

발 간 등 록 번 호

11-B090039-000875-01

Policy Research 2022-22-05



2022 K-Innovation Partnership Program with Cambodia

Capacity Building and Pilot Implementation Program on Technology Roadmap

2022 K-Innovation Partnership Program with Cambodia

Title	Capacity Building and Pilot Implementation Program on Technology Roadmap
Institute	Science and Technology Policy Institute (STEP I)
Partner Country	The Kingdom of Cambodia
In Cooperation with	Ministry of Industry, Science, Technology, and Innovation (MISTI)
Program Investigators	Ji Hyun Kim, Head, Office of SDG Innovation Research, STEP I Wangdong Kim, Senior Research Fellow, Office of SDG Innovation Research, STEP I
Project Manager	Jungwon Lee, Senior Research Fellow, Office of SDG Innovation Research, STEP I Dong Un Park, Associate Research Fellow, Office of SDG Innovation Research, STEP I
Researchers	Jung Won Lee, Senior Research Fellow, Office of SDG Innovation Research, STEP I Dong Un Park, Associate Research Fellow, Office of SDG Innovation Research, STEP I Jeehye Min, Researcher, Division of Global Innovation Strategy Research, STEP I Dr. Try Sophal-Deputy Director General / Department of Science, Technology & Innovation of Ministry of Industry, Science, Technology Dr. Srun Pagnarith-Director of STI Policy Department · Ministry of Industry Science Technology & Innovation Dr. Chen Sovann- Deputy Director of the STI Data Management Department, GDSTI, MISTI Dr. Seng Touch-Deputy director of Rural Water Department at Ministry of rural development, Cambodia Mrs. KHUN Chanmony- Head / Department of Tourism, Faculty of Social Science and Humanities, RUPP Mr. SAUT Moeun- Deputy Head / Department of Tourism, Faculty of Social Science and Humanities, RUPP Mr. HAN Haknokrith-Lecturer / Department of Tourism, Faculty of Social Science and Humanities, RUPP Mr. Chhem Siniwat - Team Leader / Director of the Centre for Inclusive Digital Economy (CIDE) at AVI Dr. Chheang Vannarith- Public Policy analyst at AVI Dr. Hossain Alamgir-Vice President (Academic Affairs & Research), Cambodia University of Technology & Science Mr. Chum Pharitho-Dean of Faculty of Engineering of Cambodia University of Technology and Science in the Kingdom of Cambodia
Project Coordinator	Jeehye Min, Researcher, Division of Global Innovation Strategy Research, STEP I

Policy Research 2022-22-05

Capacity Building and Pilot Implementation Program on Technology Roadmap

Published on 30 December 2022

Copyright © 2022 by Science and Technology Policy Institute(STEPI). All Rights Reserved.

5th~7th Flrs., Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong-si,
30147. Republic of Korea

Tel: +82-44-287-2000 Fax: +82-44-287-2068

www.stepi.re.kr

For more information, visit

stepi.re.kr/global

Printed by Mirae Media

Tel: +82-2-815-0407 Fax: +82-2-822-1269

ISBN 978-89-6112-877-3 93300



Executive Summary

The goals of the K-Innovation ODA Program are to build each partner country's capacity to formulate STI policy for implementing the project, with the specific aim of promoting the project's purposes. The K-Innovation ODA Program's professional and academic experience in designing strategies have enabled the Kingdom of Cambodia (Cambodia) project to produce both the national strategies and technology roadmaps.

This is the third year of phase 2 of the K-Innovation project with Cambodia. Phase 1 of the project was carried out from 2018 to 2019 under the title of 'Policy Consultation on the Development of a National STI Policy and Management of the STI System for Cambodia's Policymakers'. During the second year of phase 2 in 2021, it was recognized that the project had produced some meaningful outcomes despite the restrictions caused by the COVID-19 pandemic. Based on the experiences and lessons learned from the first year, the project's third year was effectively conducted in similar ways to the second year, utilizing online meetings and drawing on support from local institutions in Cambodia. In addition to the way the project is being operated, STEPI and the Ministry of Industry, Science, Technology & Innovation (MISTI) of the Royal Government of Cambodia have concluded a service contract to develop the first national technology roadmap of Cambodia.

The consultation process for the project comprised the following activities as explained below: As in the first and second years, the whole process of the 2nd phase of the project was implemented online due to the ongoing COVID-19 pandemic. Since direct face-to-face activities were discouraged in both countries, the kick-off meeting, contract signing process and ceremony, and a training workshop were held online or via e-mail.

The contracts were signed in early August 2022. MISTI selected three local institutions from among government organizations and academia based in the STI field. MISTI's main activities consisted in organizing and operating national TRM committees for three selected technologies. The main roles of each committee were as follows: 1) to elaborate and identify the scope of the selected technologies; 2) to develop TRM for each selected technology based on the guidelines established by STEPI; 3) to collect opinions from interested groups and build a consensus on the developed TRM; and 4) to finalize the TRM and submit the report. MISTI also designated the principal institutes and a project manager (PM) for each technology roadmap (TRM) for the contract with STEPI.

The training workshop was convened on August 3, 2022, and Korean experts reviewed the draft version of the roadmap shared by three local organizations and provided relevant feedback. The roadmap focuses on three categories of technologies: digital, energy, and tourism. The review workshop was held virtually (i.e. online) on November 24, 2022 to produce a final version of the roadmaps. The three reports were finalized and submitted in late December after a series of reviews of the first drafts with Korean experts. The main output of the reports was the establishment of a concrete technology development strategy that can be viewed as a national policy plan for fostering the relevant experts and government officials.

Despite the numerous difficulties caused by the Omicron variant of COVID-19 pandemic in 2022, STEPI was able to accomplish the objectives of the project. First, it succeeded in developing three additional national technology roadmaps of Cambodia, i.e. roadmaps for digital, energy, and tourism technologies, in addition to three TRMs, i.e. agricultural, educational, and health technologies, in 2021. Furthermore, Korea and Cambodia have successfully accumulated sufficient experience to explore a 'more effective and safer' ODA project during the COVID-19 pandemic era.

Finally, the overall efficiency and effectiveness of the project were carefully managed throughout the project process. MISTI and two other local organizations managed the budget for each activity in a thorough and professional manner, and also developed technology roadmaps which could serve as stepping stones to designing, implementing, and monitoring the Science, Technology and Innovation field in Cambodia.

CONTENTS

Chapter 1. Project Overview.....	3
1. Introduction	3
2. Objectives	4
3. Project Framework.....	5
4. Project Team	7
5. Main Activities of the Project.....	8
6. Project Schedule.....	10
7. Project Outputs	11

Chapter 2. The STI Policies and Technology Roadmap of Cambodia ...	15
1. Summary of Cambodia's 2020 STI Policies.....	15
2. STI Policy Roadmap and Survey	17
3. 2021 STI Roadmap of Cambodia: Agriculture, Education, and Health	20

Chapter 3. Capacity Building Process	29
1. Meetings for Project Implementation	29
2. Selection of Local Implementer and Training Workshop.....	32
3. Technology Roadmap Consultation.....	36

Chapter 4. The STI Roadmap of Cambodia	45
1. Digital.....	45
2. Energy	60
3. Tourism.....	100

Chapter 5. Conclusion	201
1. Outcomes and Expected Benefits.....	201
2. Dissemination and Future Plans	204

REFERENCES.....	206
------------------------	------------

APPENDICES	207
-------------------------	------------

List of Tables

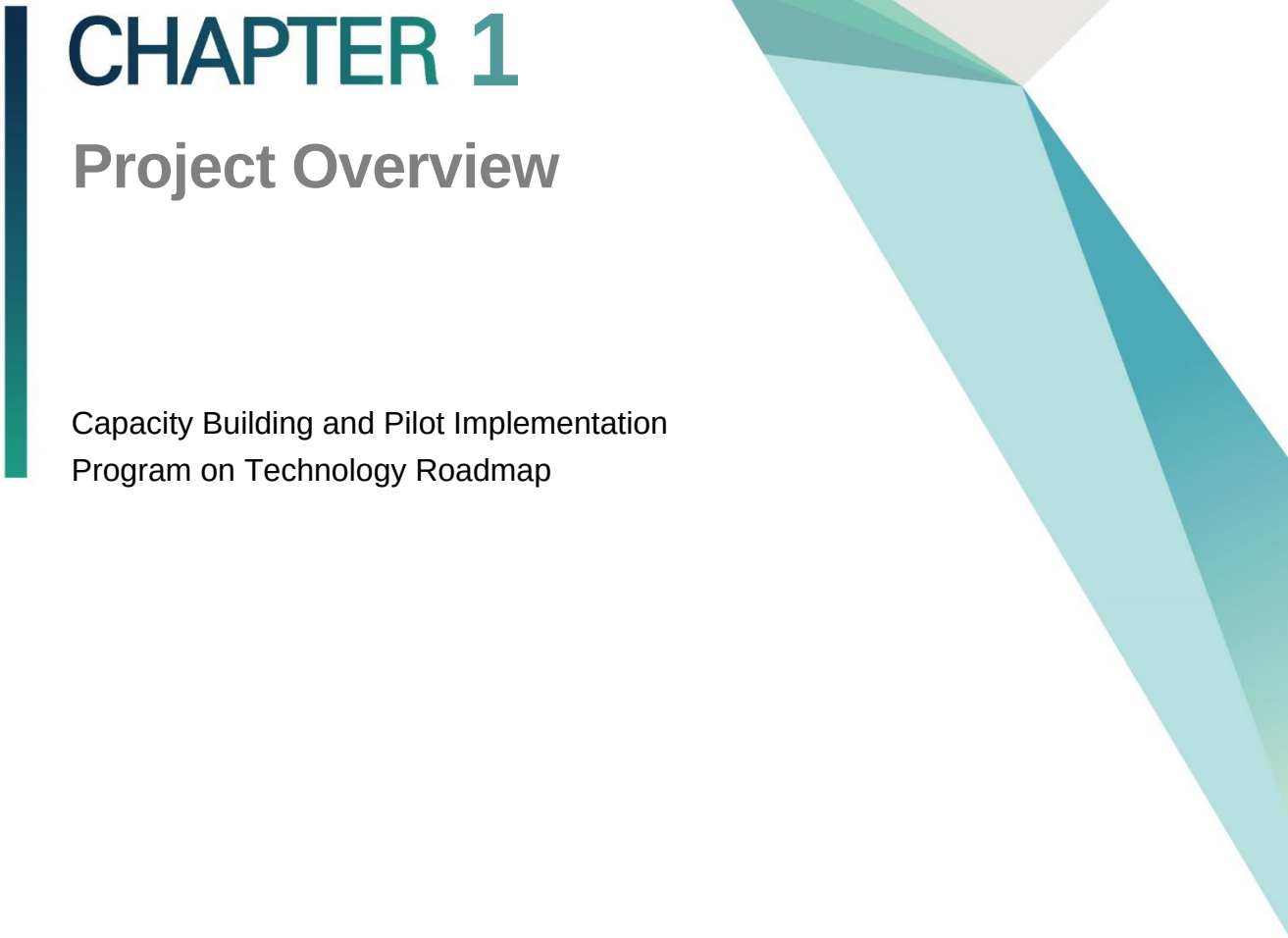
[Table 1-1] Composition of the research team.....	7
[Table 1-2] Current status of local representatives in Cambodia	7
[Table 1-3] Project Implementation Schedule.....	10
[Table 3-1] Concept of the technology roadmap and cases in Korea	40
[Table 4-1] Development and application of technologies to tourism	108
[Table 4-2] Social drivers	119
[Table 4-3] Technological drivers	120
[Table 4-4] Economic drivers.....	121
[Table 4-5] Environmental drivers	122
[Table 4-6] Political drivers.....	123
[Table 4-7] Drivers, opportunities, threats, and candidates of products and services ...	124
[Table 4-8] Candidates of products/services for the tourism technology roadmap	130
[Table 4-9] Evaluation of products/services	132
[Table 4-10] Evaluation of technologies based on two criteria for evaluating technologies for virtual tour services.....	133
[Table 4-11] Evaluation of technologies based on two criteria for integrated transportation service.....	135
[Table 4-12] Evaluation of technologies based on two criteria for sustainable tourism services	137
[Table 4-13] Evaluation of technologies based on two criteria for green tourism products.....	138
[Table 4-14] Evaluation of technologies based on two criteria for smart experience .	140
[Table 4-15] Evaluation of technologies based on two criteria for digital transaction services	142
[Table 4-16] Evaluation of technologies based on two criteria for digital communication	145
[Table 4-17] Evaluation of technologies based on two criteria for tourism security and safety service	146
[Table 4-18] Evaluation of technologies based on two criteria for automated and contactless services	148

[Table 4-19] Evaluation of technologies based on two criteria for digital marketing services	150
[Table 4-20] Evaluation of technologies based on two criteria for research and innovation.....	152
[Table 4-21] Evaluation of technologies based on two criteria for destination management.....	154

List of Figures

[Figure 1-1] Policy consulting framework for Cambodia’s national technology roadmaps.....	6
[Figure 1-2] Building process of Cambodia’s national technology roadmaps	6
[Figure 2-1] Number of Interviewees by qualification	18
[Figure 2-2] Main Technology Activities by Organization	19
[Figure 2-3] The 3 Pillars of the Health Technology Roadmap Vision	22
[Figure 2-4] One Health Approach.....	23
[Figure 3-1] Kick-off meeting.....	31
[Figure 3-2] Spread of the COVID-19 pandemic in Cambodia.....	33
[Figure 3-3] Training workshop program.....	34
[Figure 3-4] Photos of the training workshop.....	35
[Figure 3-5] Five key areas of the Cambodia STI Roadmap 2030	38
[Figure 3-6] Guidelines on worksheets for Cambodia’s NTRM (example)	41
[Figure 4-1] FDI Digital Services	51
[Figure 4-2] Growth rate of energy consumption.....	74
[Figure 4-3] Growth rate of petroleum supplies.....	74
[Figure 4-4] Cambodia’s energy consumption.....	75
[Figure 4-5] Energy consumption in various sectors.....	76
[Figure 4-6] Growth rate of TEEC.....	76
[Figure 4-7] Shares of energy production sources.....	77
[Figure 4-8] Development process of the National Energy Technology Roadmap	84
[Figure 4-9] Key technologies	88
[Figure 4-10] Chart of the Energy Technology Roadmap.....	93
[Figure 4-11] Statistics on Tourists Visiting Cambodia, 20210-20219 (Source: MoT, 2021)	103
[Figure 4-12] Portfolio analysis to identify strategic products and services	132
[Figure 4-13] Setting priorities to identify key technologies for virtual tour services..	134
[Figure 4-14] Setting priorities to identify key technologies for integrated transportation services	136

[Figure 4-15] Setting priorities to identify key technologies for sustainable tourism products.....	138
[Figure 4-16] Setting priorities to identify key technologies for green tourism products.....	140
[Figure 4-17] Setting priorities to identify key technologies for smart experience.....	142
[Figure 4-18] Setting priorities to identify key technologies for digital transaction services	144
[Figure 4-19] Setting priorities to identify key technologies for digital communication	146
[Figure 4-20] Setting priorities to identify key technologies for tourism security and safety services.....	148
[Figure 4-21] Setting priorities to identify key technologies for automated and contactless service	150
[Figure 4-22] Setting priorities to identify key technologies for digital marketing services	152
[Figure 4-23] Setting priorities to identify key technologies for research and innovation	154
[Figure 4-24] Setting priorities to identify key technologies for destination management.....	156
[Figure 5-1] Technology launch event program draft.....	202
[Figure 5-2] Participants in the technology launch event	203

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a light grey triangle pointing left, and a larger teal shape composed of several overlapping triangles pointing right.

CHAPTER 1

Project Overview

Capacity Building and Pilot Implementation
Program on Technology Roadmap



Chapter 1. Project Overview

1. Introduction

This program is a follow-up to the second phase of the “2018-2019 Policy Consulting Program for the Establishment of STI Policy in Cambodia” carried out as a part of the K-Innovation project. In the first phase of the project, the *National STI Policy 2020-2030*, the Cambodian government's first national science and technology master plan, was established and adopted as a key outcome of policy consulting.

Based on the trust developed between STEPI and the Cambodian government following the successful implementation of the first phase of the program, the Cambodian government requested and promoted the “Consulting Program for Building a National Technology Roadmap” for the implementation of the plan as a second phase project.

However, due to the outbreak and spread of the COVID-19 pandemic at the beginning of 2020, which has made visits to Cambodia and invitational training in Korea equally difficult to implement, the second phase of the program has been carried out largely through online-centered cooperation and consulting activities.

In 2021, the second year, the local partner organization was changed from GS-NSTC to the Ministry of Industry, Science, Technology, and Innovation (MISTI) following a

reorganization of the Cambodian government. As the COVID-19 pandemic continued unabated, online-centered cooperation and consulting activities were conducted based on cooperation with MISTI, the executing agency.

In the 2nd phase of the project, we transferred the methodology for creating the national technology strategy/technology roadmap to the Cambodian government while deriving three strategic technologies. Next, the national technology roadmaps were built based on the results of a 2020 survey which led to the selection of education, agriculture, and health as the three main technology strategy sectors. These were then approved by the National Council on Science, Technology & Innovation of Cambodia as the target technologies for the national technology roadmap (July 2021).

The three national technology roadmaps were built as follows: agriculture by MISTI, Cambodia's executing agency; education by Cambodia University of Technology and Science (CamTech); and health by the University of Puthisastra (UP).

2. Objectives

Policy consulting on the establishment of national technology strategies and technology roadmaps for the strategic technologies was carried out in cooperation with Cambodian policymakers and the associated researchers. Mutual economic cooperation was promoted to contribute to the development of science and technology in Cambodia and to improve the various associated systems and conditions (Paragraphs (1) and (2).2 of Article 3 (Basic Ideas and Objectives) of the Framework Act on International Development Cooperation, Korea).

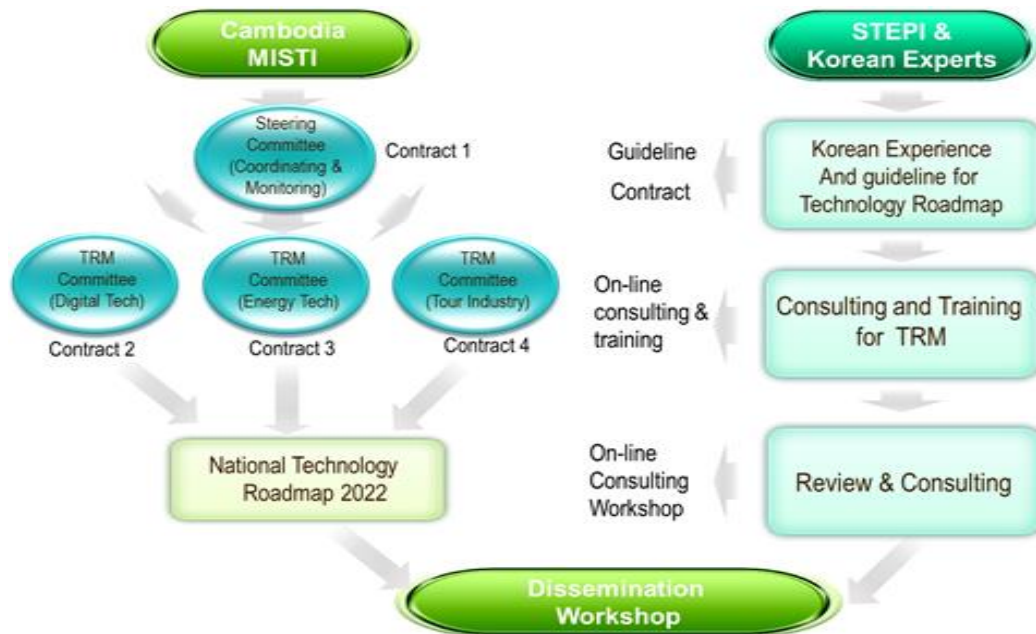
3. Project Framework

■ Consulting on Cambodia's National Technology Roadmap

As in the second year, we built NTRMs for three Cambodia's strategic technologies based on Korea's success cases and experience of building the Korean national technology roadmap in 2002 and the Korean strategic technology roadmap in 2014.

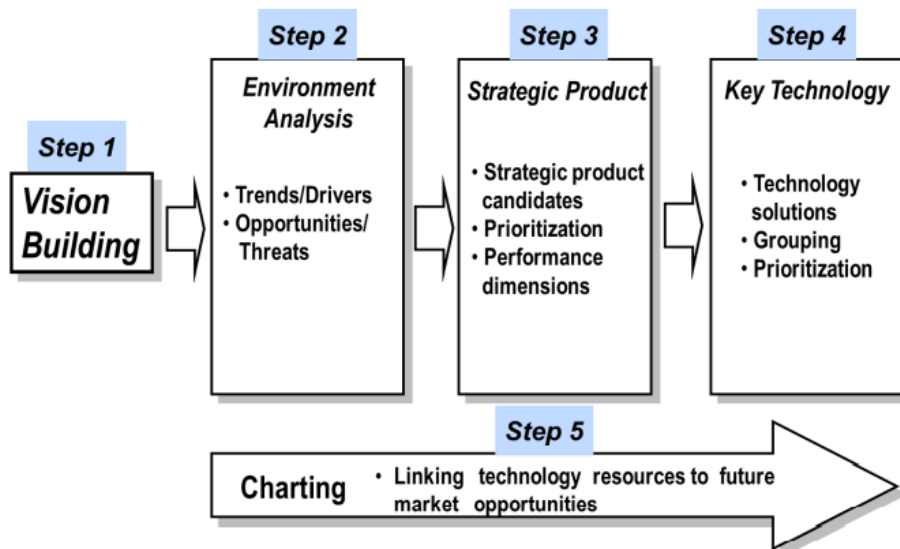
- NTRMs were produced in six areas for the purpose of enhancing Cambodia's national competitiveness over the next ten years (until 2030) based on its "National STI Policy 2020-2030." Cambodia's national technology roadmap was generated to improve Cambodia's national competitiveness over the next ten years (until 2030).
- In the 1st year of the Cambodia National Technology Roadmap Program, the roadmaps for agriculture, health, and education were produced.
- In the 2nd year of the Cambodia National Technology Roadmap Program, digital, tourism, and energy were selected as the three sectors of the technology roadmaps, with the input of a local expert committee composed of experts from industry, academia, and research.
- The Steering Committee, which is composed of a general coordinator and a coordinator for each field, and in which MISTI plays a central role, monitored the overall process and supported the development, by sector, of excellent roadmaps according to the established guidelines.

[Figure 1-1] Policy consulting framework for Cambodia's national technology roadmaps



Source: Kim Ji-hyun et al. (2021, p.206).

[Figure 1-2] Building process of Cambodia's national technology roadmaps



Source: Kim Ji-hyun et al. (2021, p.206).

4. Project Team

4.1 Research Team

■ Korean Research Team

- A research team was formed with experts in ODA projects (including the creation of technology roadmaps) from inside and outside STEPI to ensure that the program is continuously carried out.
- Given that it is a three-year program, experts who participated in the 1st year (2020) were included for the sake of continuity of the program.

[Table 1-1] Composition of the research team

Description	Institute	Position	Name	Major role
Internal expert	STEPI	Senior Research Fellow	Lee Jungwon	Overall control of the Cambodia program
	STEPI	Associate Research Fellow	Park Dong Un	Research & surveys and local authority consulting
	STEPI	Researcher	Min Ji-hye	Research & surveys and contact with local authorities
External expert	KISTEP	Senior Research Fellow	Lim Hyun	Creation of technology roadmaps

■ Cambodia Research Team

[Table 1-2] Current status of local representatives in Cambodia

Description	Organization	Position	Name	Major role
Cambodia research team	Ministry of Industry, Science, Technology and Innovation (MISTI)	Director General	Hul Seingheng	Overall local control of the Cambodia program
		Deputy Director General	Try Sophal	Program control in Cambodia
	Asian Vision Institute (AVI)	Director	Chhem William Siriwat	Overall control and working-level promotion of the digital technology roadmap
	Royal University of Phnom Penh (RUPP)	Chair	Khun Chanmony	Overall control and working-level promotion of the tourism sector roadmap

5. Main Activities of the Project

5.1 Program adjustment and operation meetings

- March 3, 2022: Holding of the program kick-off meeting between STEPI and MISTI.
- March 31, 2022: Completion of the progress report on the selection of the three technology sectors and the review of future plans.

5.2 Preparations for the selection of Cambodia's technologies and the promotion of local service projects

- March 3, 2022: Selection of service providers in the three technology sectors for service contracts and on-site surveys; drafting the progress report on the program outcomes in the second year; review of future plans.

5.3 Business consultations related to service contracts by technology field

- May 30, 2022: Division of duties and sharing of schedule between MISTI and the service providers regarding overall program control; identification of the three technology sectors to be developed for the technology roadmaps.
- June 23, 2022: Announcement of the technology roadmaps for three technology sectors as the outcomes of the 2nd year of the project.

5.4 Workshop on building technology roadmaps

- August 3, 2022: Holding of a workshop for the STEPI team, Cambodian local institutions responsible for the relevant technologies, and the committee members.
- Sharing of Korea's NTRM cases and discussion on how to apply them to Cambodia's technology roadmap, and agreement on the need for continuous consulting.

5.5 Preparing and signing a local service contract

- Diverse documents, including task instructions, program budget, registration certificate with the local government agency, bank account information, etc. were prepared in order to conclude a service contract.¹
- October 14, 2021: Signing of service contract.

5.6 Consulting workshop

- The decision was made to hold a consulting workshop in December 2022 on the program progress and outcomes, including the national technology roadmaps for the three technology sectors.

¹ In the case of the AVI, which is not a public institution, it was necessary to purchase guarantee insurance, but MISTI issued a memorandum of guarantee instead.

6. Project Schedule

The overall project schedule is as follows:

[Table 1-3] Project Implementation Schedule

Task	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Planning												
Plan Revision												
Kick-off meeting												
Select three technologies												
Select three partner institutions												
Draw up Interim Report												
Sign local service contracts												
Build the Technology Roadmaps												
Workshop on outcome dissemination												
Final Report												

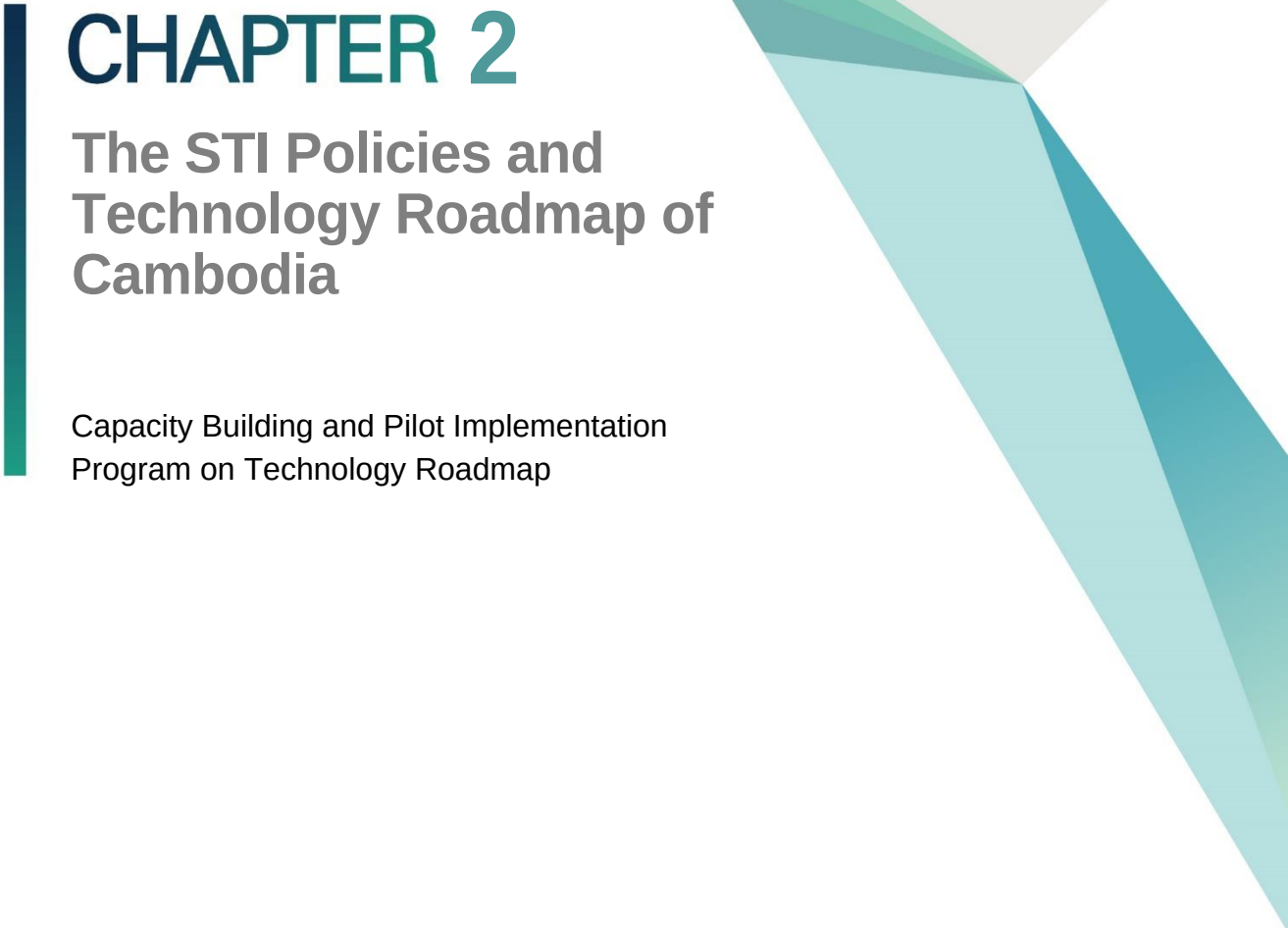
7. Project Outputs

STEPI provided the following deliverables:

- Deliverable 1: Guidelines for TRM and Sample Worksheet (August 2021)
- Deliverable 2: Online Training Workshop (August 2022)
- Deliverable 3: Feedback on the Draft and Final Report (December 2022)

MISTI provided the following deliverables:

- Deliverable 1: List of Candidate Technologies and Implementation Institution (April 2022)
- Deliverable 2: Final Report on TRM (December 2022)

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and light blue, also pointing downwards.

CHAPTER 2

The STI Policies and Technology Roadmap of Cambodia

Capacity Building and Pilot Implementation
Program on Technology Roadmap

Chapter 2. The STI Policies and Technology Roadmap of Cambodia

1. Summary of Cambodia's 2020 STI Policies

1.1 Cambodia's Sustainable Development Goals (2016-2030)

A project to establish Cambodia's Sustainable Development Goals (CSDGs) was launched in late 2015 by researching and mapping the global visions and targets while taking into consideration national priorities. The Government of Cambodia chose a total of seventeen SDGs, and included an extra goal related to the clearance of landmines and "Explosive Remnants of War" (reflecting the national priority of de-mining of the national territory) (STEPI, 2020). The final version of the CSDGs consists of the following: 18 Cambodian Sustainable Development Goals, 88 nationally important targets, and 148 (global and locally defined) indicators (STEPI, 2020).²

The CSDG Framework has four clear goals:

- Presentation of the national goals, targets, and indicators based on Cambodia's main priorities;
- Identification of the organizations' important activities to accomplish the targets;
- Identification of data sources for each indicator and the data cycle, with provisions on the working definitions and the methods for computing the indicators;
- Presentation of pathways toward achievement of the targets, setting (2015) of the national baselines and the annual (or cycle-based) target values, and their implementation at the sub-national and regional levels (CSDGs, 2016) (STEPI, 2020).

² Chapter 2 is a summary of the report written by the STEPI Team in 2021.

1.2 Cambodia's Industrial Development Plan (2015-2025)

Cambodia's industry remains weak due to the family-focused nature of many production activities, the absence of entrepreneurship, and insufficient use of technology, all of which restrains the country's ability to compete in international markets to a significant extent. The crucial characteristics of the country's industrial weakness can be summarized as follows: a weak industrial base; a misplaced middle and informal industrial structure; weak, urban-centered entrepreneurship; low value addition; and a low level of technology application (STEPI, 2020).

However, coordination among additional regulators remains problematic, and sometimes it varies relevant to their own goals. For instance, the tax system puts immoderate weight on tax collection, whereas the trade system pays attention to market openness. Such absence of coordination triggers unnecessary barriers to the implementation of industrial development objectives.

2. STI Policy Roadmap and Survey

2.1 Introduction

In order to achieve the strategies and ultimate goals of the Rectangular Strategy (RS) Phase IV and the National Strategic Development Plan 2019-2023, the Royal Government of Cambodia (RGC) needs to determine the priorities of the national strategy for technology development, which are crucial to keeping Cambodia's economy alive amid the effects of the global Covid-19 pandemic.

This survey includes text and illustrations about the results of a small survey conducted by the MoP research team investigating the current uses of technology from public and private institutions that were randomly selected for interview and respondents who filled out the forms. The survey aimed to review strategic technology candidates and their current activities in Cambodia and their suggestions; highlight the main technology areas from both the short-term and long-term perspectives, and analyze the economic impacts, technical feasibility, national safety, and security concerns according to the responses of the interviewees.

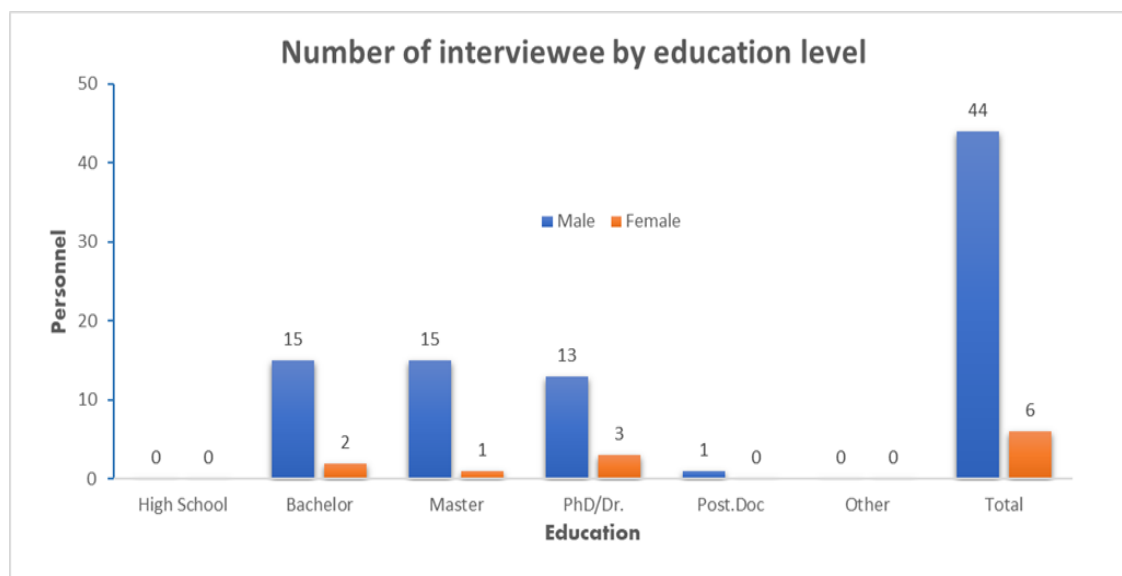
2.2 Method of Investigation

The team developed the questionnaires and data sets and organized a list of the top twelve technology areas and some technology candidates to make it easy and convenient for the interviewers and respondents to select or fill in the 10 technology areas and 10 strategic technology candidates for each question. All experts were cordially asked to answer the questions based on the given guidelines and conditions.

The GS-NSTC staff were assigned to interview experts from the selected sample size because their ideas would be key to identifying the current status of technology development and challenges in Cambodia. After carefully considering only those candidates with a high educational background and abundant experience in the field of S&T, a total of fifty respondents were selected and interviewed face-to-face.

2.3 Results of the Survey

[Figure 2-1] Number of Interviewees by qualification



The graph in Figure 2.1 above shows that fifteen of the respondents were men with master's degrees, with only one female respondent having graduated from university with a master's degree. The second group of male respondents held bachelor's degrees, and only two female respondents had bachelor's degrees. On top of that, thirteen male respondents had PhD/doctorate degrees; but, surprisingly, there were three female respondents in the sample who had PhD/doctorate degrees. Finally, one man with a post-doctoral degree was interviewed by the interviewers. In short, forty-four men and only six women completed the questionnaire survey prepared by GS-NSTC.

[Figure 2-2] Main Technology Activities by Organization

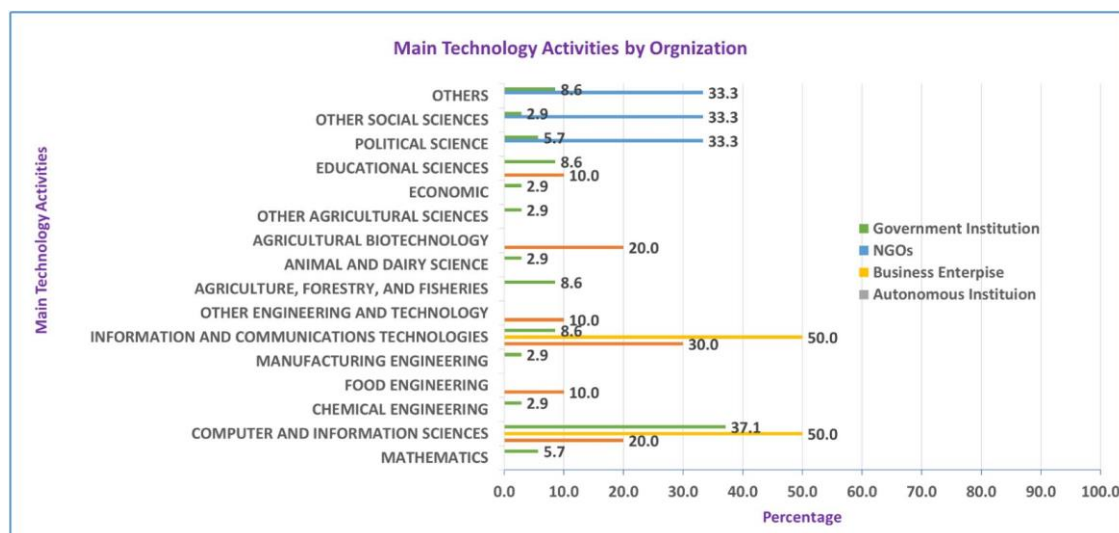


Figure 2.2 above shows the variety of the main technology activities of the respective entities - including government institutions and agencies, NGOs, business enterprises and autonomous institutions - surveyed by GS-NSTC officials. Information and communications technologies and computer and information science constituted about 50% of all the activities of the selected targets of this survey, which suggests that business enterprises are conducting these activities more actively than government institutions, NGOs, and autonomous institutions. Meanwhile, notable efforts are being made by NGOs related to technologies in other areas (33.3%), social sciences (33.3%), and political science (33.3%). On top of that, some autonomous institutions are conducting technology, including information and communications technologies (30%), computer and information sciences (20%), agricultural biotechnology (20%), other engineering and technologies (10%), food engineering (10%), and education science (10%).

Finally, as can be seen in the figure above, government institutions have also carried out a variety of technology activities, including computer and information science (37.1%), other areas (8.6%), educational science (8.6%), information and communication and technologies (8.6%), agriculture, forestry, and fisheries (8.6%), mathematics (5.7%), political science (5.7%), chemical engineering (2.9%), manufacturing engineering (2.9%), animal and dairy farming (2.9%), other types of agricultural science (2.9%), economics (2.9%), and other social sciences (2.9%).

3. 2021 STI Roadmap of Cambodia: Agriculture, Education, and Health

3.1 Agriculture

3.1.1 Building the vision and goals

■ Vision

To increase productivity and the high value-added of agricultural commodities destined for the global supply chain through technology and innovation by 2030.

■ Goals

To harness the power of Science Technology and Innovation in the agriculture sector in order to:

- Increase agricultural productivity and infrastructure food security and for exports.
- Leverage the industrial high-value-added of agricultural commodities to increase agriculture's contribution to gross value added (GVA) and exports.
- Integrate agricultural commodities into the global supply chain in order to expand Cambodia's agricultural businesses and reputation in international markets and networks.

3.2 Education

3.2.1 Building the vision and goals

The shared vision and goals were developed with the involvement of all the relevant stakeholders from academia, government, the private sector, and students. The initial vision and goals were synthesized and developed on the basis of in-depth interviews and a roundtable panel discussion, and were then put forward for further discussion during the technical consultation workshop held in December 2021 to encourage the development of comprehensive ideas and information and to promote collaboration

and a sense of ownership in the development and implementation of the national roadmap, with the ultimate goal of achieving desirable outcomes. After gathering additional inputs, suggestions, and comments from the participants, the vision and goals were then refined and sent to the relevant specialized department(s) within the Ministry of Industry, Science, Technology, and Innovation in order to seek final technical inputs before submitting them to the Committee for Educational Technology and the National Council of Science, Technology, and Innovation for endorsement and final approval. The national vision and goals for educational technology are presented below.

■ Vision

- To build the next-generation technology-enhanced learning ecosystem with the focus on improving innovation and entrepreneurship skills.

■ Goals

- Offer an open educational resource platform by creating a central repository to manage and deliver freely accessible learning resources.
- Increase time spent on personalized learning to complement the learning outcomes of the “physical” classroom.
- Increase Internet access at school and at home via computers.
- Improve learning outcomes.
- Improve innovation and entrepreneurship skills.

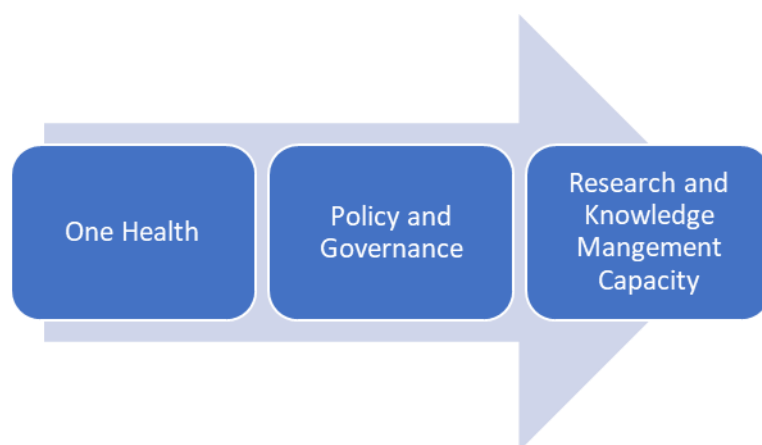
3.3 Health

3.3.1 Vision

The Cambodian Health Technology Roadmap vision aims to “Increase the health and well-being of Cambodians through research and development, innovation and technologies”. This is underpinned by the four pillars of Technology, One Health, Governance, and Research. This conception is rooted in the acknowledgement that for

an integrated One Health approach to be successful, it is necessary to adopt an inclusive, multi-disciplinary, participatory governance and policy approach to ensuring that the system is robust and responsive to the dynamic healthcare space in the kingdom. By the same token, there must also be a quality research environment to support evidence-informed policy with the appropriate applied technological interventions. Each pillar of the vision is discussed further below.

[Figure 2-3] The 3 Pillars of the Health Technology Roadmap Vision



3.3.1.1 An Integrated One Health Approach.

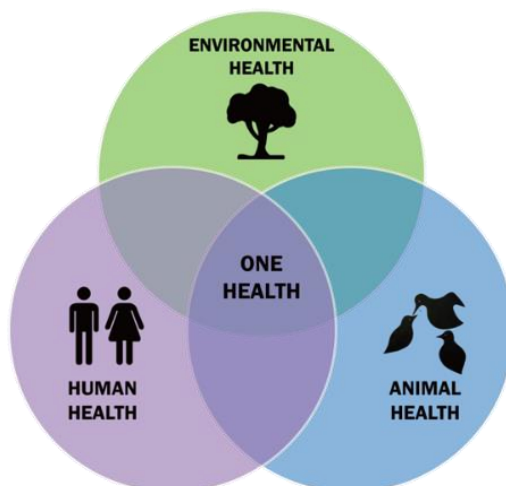
Firstly, the health technology roadmap envisions an integrated, nationwide “One Health” approach that aligns the interrelated and individual health objectives encompassing human health, environmental health, and animal health.

As outlined by the WHO,³ One Health (OH) is an approach to designing and implementing programs, policies, legislation, and research in which multiple sectors communicate and work together for the pursuit of better public health outcomes. Noting the broad classification of OH, there are areas of work in which an OH approach is particularly relevant, including food safety, the control of zoonoses, and combatting antibiotic resistance. The motivation of an OH approach is rooted in recognition of the

³ WHO, 2021. *One Health*. [online] Who.int. Available at: <https://www.who.int/news-room/q-a-detail/one-health>

interconnections between people, animals, plants, and their shared environment. Although this recognition of interplay and interrelation in our shared environment is not new, many factors have changed the interactions in our shared environment in recent years.⁴ In particular, human populations are growing and expanding into new geographical areas, consequently changing climate conditions and land use, and there has been an increase in the mass movement of people, animals, and animal products across borders. Such factors have contributed to the spread of existing and new zoonotic diseases such as rabies and Ebola.

[Figure 2-4] One Health Approach



3.3.1.2 Multidisciplinary Policy and Governance for Health Technology.

The successful planning, adoption, and execution of health technology intervention, policy, and governance requires collaboration and knowledge sharing between stakeholders from diverse and inclusive backgrounds. This pillar is crucial to ensuring that the adoption of health technologies works in tandem with national policies and strategies such that interventions are complementary to addressing priority development needs. By the same token, civil-society and research institute actors must be consulted in order to ensure a transparent process that minimizes oversights in

⁴ Cdc.gov. 2021. *One Health Basics* / *One Health* / CDC. [online] Available at: <https://www.cdc.gov/onehealth/basics/index.html>

planning and execution. At the forefront of such collaboration, in many ways, is the role of the private sector in bringing technologies to the market. Working as a collective, government, research institutions, civil society and the private sector require a collaborative process for the purpose of defining and executing a health technology roadmap.

3.3.1.3 Strengthened Research and Knowledge Sharing Capacity.

Health technology prioritization requires an enhanced local decision-making context (as outlined in Vision 2). However, in order for this to be the case there must exist a robust technical research landscape. This vision outlines the collaboration between stakeholders in this space – including hospitals, universities, the private sector, and the Ministry of Health – to develop a robust research culture in the kingdom.

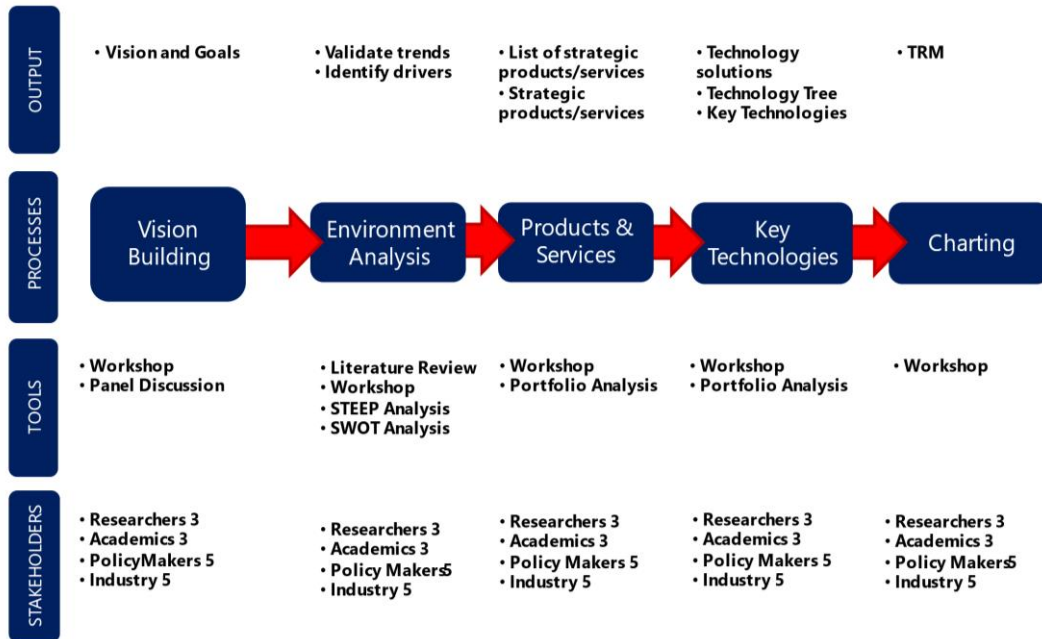
To care for patients effectively, healthcare providers and staff must be able to share not only clinical knowledge, but also knowledge of organizational processes and procedures and the most up-to-date drug and treatment information. Given that healthcare is a knowledge-driven industry, a healthcare knowledge management system is crucial to creating a more efficient flow of information between all providers and staff, as this ultimately leads to increased efficiency and productivity.

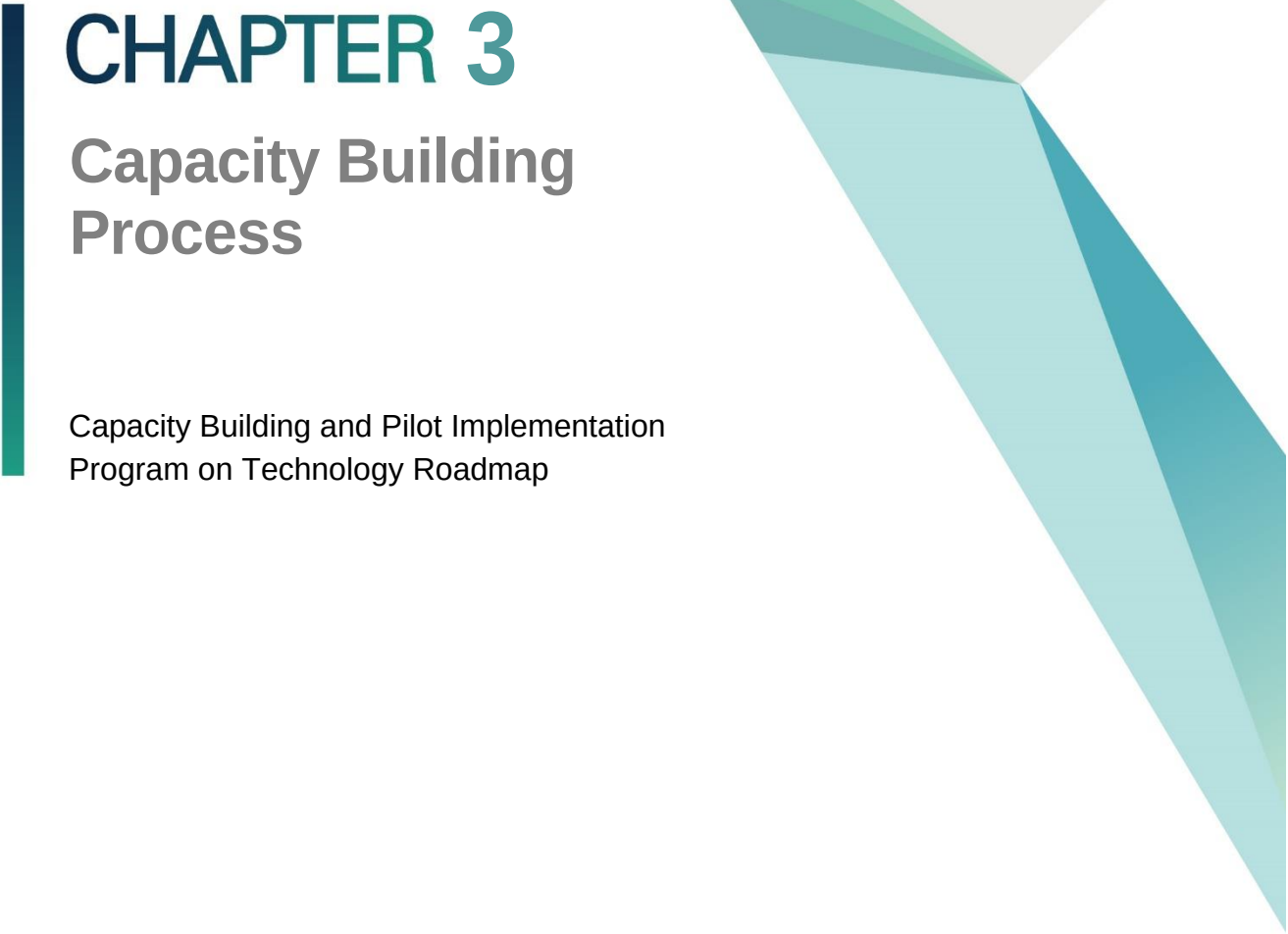
In the context of Health Technology, a robust knowledge management system is expected to deliver enhanced and effective decision making with respect to the adoption and implementation of health technology, in addition to supporting a continuous-learning environment that promotes collaboration and knowledge exchange.

3.3.2 Goals

Following the outline of the vision for the Health Technology Roadmap, it is important to map those goals and objectives which are considered crucial to achieving that vision. There are nine reinforcing goals in total, each of which is described in detail below.

Health Technology Road Map Process



A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and a series of overlapping teal and light blue triangles also pointing downwards, creating a dynamic, layered effect.

CHAPTER 3

Capacity Building Process

Capacity Building and Pilot Implementation
Program on Technology Roadmap

Chapter 3. Capacity Building Process

1. Meetings for Project Implementation ⁵

1.1 1st Video Conference (kick-off meeting)

- STEPI and MISTI shared information on the COVID-19 situation in their respective countries and held the kick-off meeting online on March 3, 2022.
- The online kickoff meeting was held as the COVID-19 situation in Korea and Cambodia had not improved since 2021.
- The items of agenda, such as allocation of the budget for the selection of technologies, were discussed in advance by STEPI and MISTI via e-mail.
- The STEPI researchers arranged a video conference to allow MISTI to carry out the overall control of the local service providers, select local service providers from among research institutions and universities, and discuss contract-related matters in detail, so that the program could proceed smoothly without any face-to-face interactions.
- The conference was held online on March 3, 2022 for the purpose of kicking off the project. It was attended by four people from STEPI (Senior Research Fellow Lee Jungwon, Director Kim Ji-hyun, Associate Research Fellow Park Dong Un, and Researcher Min Ji-hye), and three people from MISTI (Hul Seingheng, Srun Pagnarith, and Try Sophal).

*Key points for discussion:

- Sharing information on the COVID-19 situation in each country
 - As the number of confirmed cases of COVID-19 continued to increase, all the activities in the first half of the year had to be carried out online, and it seemed highly likely that the outcome dissemination workshop scheduled for the second half probably would have to be held online, too.
- Activities scheduled for 2022
 - Derive the three national strategic technologies (energy, digital, tourism technologies) and select the related organizations according to Cambodia's demand.
 - Compose each technology committee of representatives from diverse universities, research institutes, etc.
 - Plan and build each committee's technology roadmap according to STEPI's guidelines.
 - Submit a roadmap report to STEPI after establishing a technology roadmap.
- Budget support for technology roadmap activities
 - It was agreed to allocate financial support of approximately US\$15,000 (KRW 20 million) per technology.
 - It was agreed to pay 50% of the sum in advance and the remaining 50% upon complete fulfillment of the contract.
 - It was agreed that service contracts must be concluded between local service providers and STEPI.

[Figure 3-1] Kick-off meeting



1.2 2nd Video Conference

- The second Video Conference between STEPI and MISTI was held in order to discuss the contract with the local program executing agency.

*Key points for discussion:

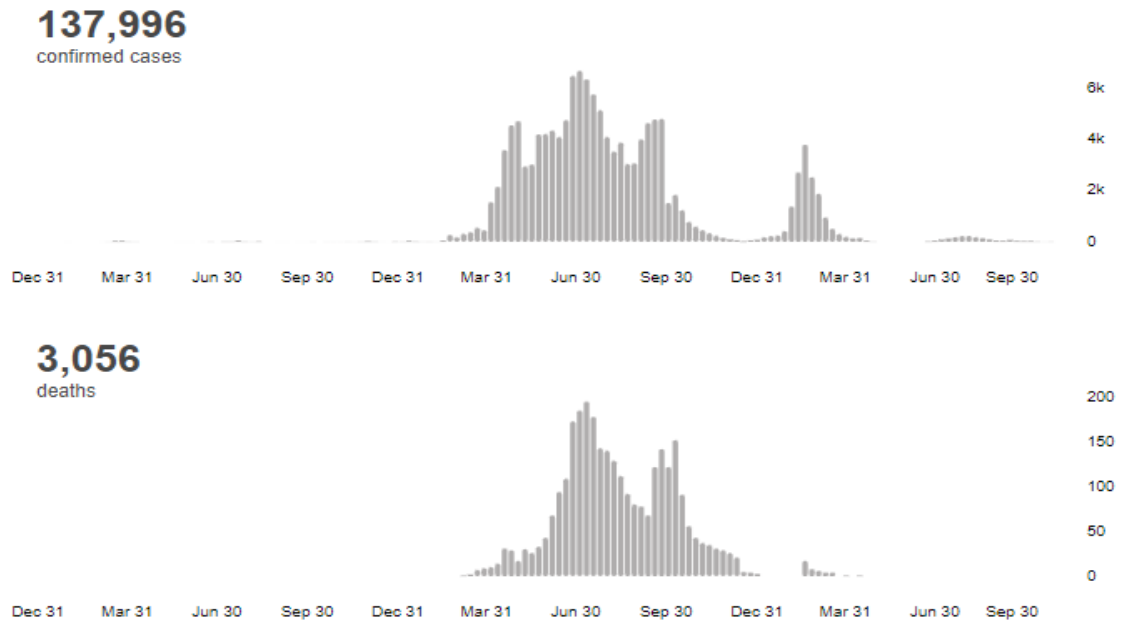
- Ways to perform tasks and contract-related matters
 - Regarding the task performance method, as in 2021, MISTI was responsible for the overall control in Cambodia, including monitoring of and support for the local institutions' performance of their tasks.
 - The technology roadmap report was to be submitted as output in time for the conclusion of the contract.
- A training workshop was scheduled after signing the contract.
 - Building the roadmap was scheduled to start after the training workshop.
 - STEPI agreed to continue providing consulting on the roadmap after the training, with the expectation that MISTI would be of great help to the organization performing the task this year, as it had carried out the same task the previous year.

2. Selection of Local Implementer and Training Workshop

2.1 Selection of Local Implementer

- The prolonged COVID-19 pandemic has disrupted the progress of the program.
 - Since January 2020 Cambodia has also found it difficult to advance the program face-to-face through on-site visits or invitations to Korea due to the ongoing COVID-19 pandemic.
 - As of November 7, 2022, Cambodia had recorded 137,996 confirmed cases and 3,056 deaths from COVID-19 (WHO, 2022).
 - As of October 20, 2022, a total of 40,463,517 COVID-19 vaccines had been administered
- MISTI, the executing agency, has also opted for telecommuting during the pandemic, except for unavoidable cases.
- Due to the COVID-19 situation, in 2022, the third year of the 2nd phase program, policy consulting and training were conducted online as in the second year.
 - Given that the Covid-19 situation made it impossible for Korean experts to visit and conduct field research, it was agreed with MISTI to proceed in the same way as in the second year.
 - MISTI selected local partners with the capacity to produce technology roadmaps in the three sectors, with the local partners conducting their tasks according to STEPI's technology roadmap guidelines and policy consulting.
 - As in the second year, all consulting was conducted via e-mail and periodic online video conferences. Training workshops designed to build a national technology roadmap were also conducted via video conference.
 - The opinions of STEPI's research team and domestic experts on the roadmap building process and the draft of the national technology roadmap report were delivered via e-mail and online meetings.

[Figure 3-2] Spread of the COVID-19 pandemic in Cambodia



Source: WHO website (2022)⁶.

2.2 Training Workshop

- The STEPI-Cambodia workshop on the technology roadmap for energy, digital and tourism technologies was held online on August 3, 2022. This event was planned to provide the guidelines for establishing a technology roadmap in 2022 and for sharing Korean cases. STEPI invited Lim Hyun, a TRM expert and senior research fellow at KISTEP.
- Regarding the “Development of a National Technology Roadmap,” Dr. Lim gave a detailed definition and explanation of a technology roadmap and outlined the processes involved in its creation, as well as sharing cases of Korea's national technology roadmap and introducing the process of carrying out the national technology roadmap tasks through a STEEP analysis. Dr. Lim also proposed the vision and goals of the roadmap in consideration of the local infrastructure, laws and regulations, and cyber threats.

⁶ <https://covid19.who.int/region/wpro/country/kh>, Search date: Nov. 8, 2022.

2022 K-Innovation Partnership Program with Cambodia

- This workshop aimed to produce the first draft of the roadmap based on the established guidelines, after which it would be revised through continuous Q&A processes.
- The Cambodia team aimed to build the roadmap by November, while STEPI would continuously monitor the roadmap building process and discuss any necessary support between the parties.

[Figure 3-3] Training workshop program



MINISTRY OF INDUSTRY SCIENCE TECHNOLOGY & INNOVATION
General Department of Science Technology and Innovation

First Meeting
Committee on Implementing the Development of
Energy, Tourism and Digital Technology Roadmaps

Ministry of Industry Science Technology & Innovation (MISTI) has collaborated with the Science and Technology Policy Institute (STEPI) of Republic of South Korea to develop technology roadmaps in the prioritized areas under the guidance of National Council of Science, Technology & Innovation (NCSTI). In 2021-2022, MISTI has coordinated, facilitated, developed and rolled out three technology roadmaps successfully including Education, Health and Agriculture Technology Roadmap. In this 2022, with the endorsement of NCSTI in its 2nd National Meeting, MISTI will continue the collaboration with STEPI to develop 3 more technology roadmaps consisting of Energy Technology Roadmap, Tourism Technology Roadmap and Digital Technology Roadmap and it is expected to complete by the end of this year. The main purposes of the establishment of these 3-technology roadmaps are to develop the key strategic and activity plans in developing and deploying technologies in the prioritized sectors of Cambodia in order to realize the government's ambitious vision to transform Cambodia to be the higher middle-income country by 2030 and high-income country by 2050, and, in addition, to respond to the changes of the socio-economic development of the country, the region and the global as a whole, especially during and post Covid-19 pandemic. To this end, to ensure the sustainably and inclusively successful development of these 3-technology roadmaps, MISTI has established the working committee in implementing the developing of the technology roadmaps participated by line ministries and relevant stakeholders in co-creation approach.

The first meeting of committee on implementing the development of Energy, Tourism and Digital technology roadmaps has main objectives as follow:

1. To share the background of the technology roadmap development project
2. To introduce the members of the working committee
3. To present the process in developing the technology roadmap and share the experiences of the previous technology roadmap development.

The expected outcomes of this project are:

1. To have a good understanding of the background of this collaborative project in developing the technology roadmaps.
2. To get to know the members in the working committee and sub-committee and the responsibility of each member.

3. To well understand the process in developing the technology roadmap and be ready for implementing the project to achieve the expected result successfully.

AGENDA

First Meeting Committee on Implementing the Development of Energy, Tourism and Digital Technology Roadmaps

03rd August 2022
08:00 AM - 11:00 AM

Venue: General Department of Science, Technology & Innovation (GD/STI)
#1157 National Road 2, Sangkat Chak Angrae Leu, Khan Meanchey, Phnom Penh
And via teleconference for Science and Technology Policy Institute (STEPI)

Hour	Activities	Moderator
07:30-08:00 AM	Arrival of Distinguished Guests	Guests
08:00-08:05 AM	Announcement	GD/STI
08:05-08:30 AM	Remarks of STEPI Representative Welcoming and Opening Remarks, MISTI	STEPI Representative H.E. Prof. Dr. CHHEM Keith Rethy , Minister Delegate Attached to the Prime Minister, Secretary of State of MISTI, and the Chairman of Steering Committee
08:30-08:50 AM	Presentation on Background and Objective of the 3-Technology Roadmap	H.E. Dr. HUL Seingheng , Director General, General Department of STI, MISTI, and Vice Chairman of Steering Committee
08:50-09:30 AM	Ice breaker & Introduction of the Members and Responsibility of Committee and Sub-Committee of the 3-Technology Roadmap	Dr. TRY Sopha , Deputy Director- General of GD/STI, MISTI, and Permanent Member of Steering Committee
09:30-09:45 AM	Introduction of Consultant Team on Developing Technology Roadmap	Chairman of Sub-Committee Consultant Teams on Developing Technology Roadmap
09:45-10:30 AM	Coffee Break	
10:30-10:50 AM	Presentation on Principle, Process, and Technical Aspects in Developing the Technology Roadmap	STEPI Expert GD/STI
10:50-11:00 AM	Q & A	Participants

[Figure 3-4] Photos of the training workshop



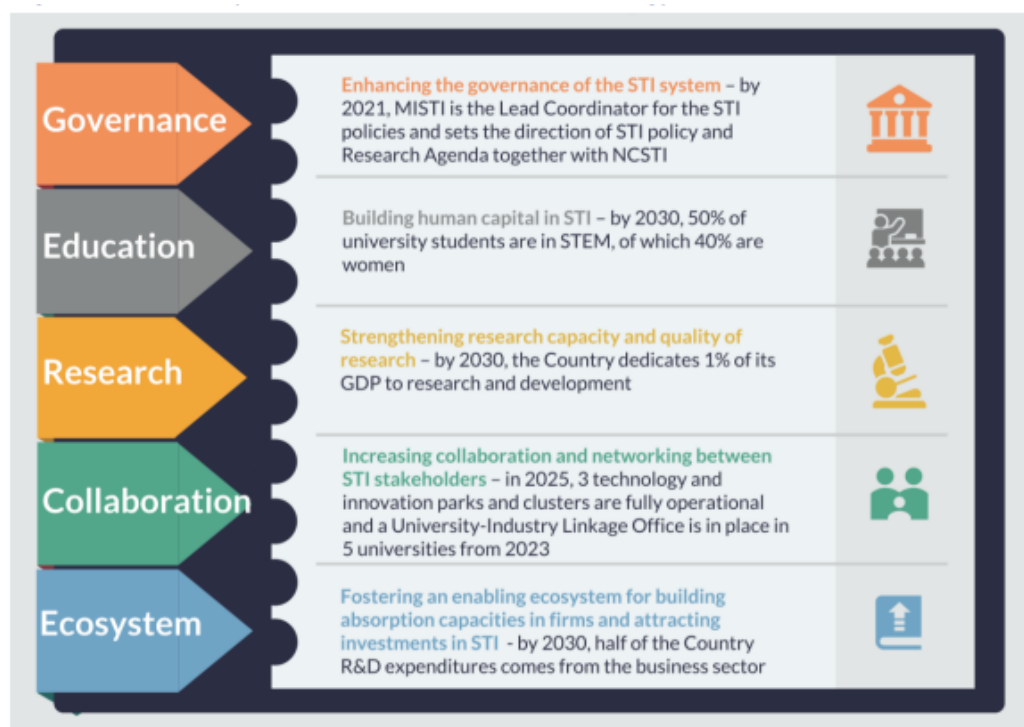
3. Technology Roadmap Consultation

3.1 Consulting on Cambodia's science and technology system

- Understanding the current state of science and technology in Cambodia
 - Along with the National Strategic Development Plan, Cambodia has established and is implementing the “Rectangular Strategy Stage 4,” a national development strategy.
 - The Rectangular Strategy, a national development strategy launched in 1998, is updated every five years.
 - Stage 4 features the following core strategies under the vision of growth, employment, equity, and efficiency.
 - Focusing on the role of science and technology in the realization of “Rectangular Strategy Stage 4,” the Cambodian government established the “National STI Policy 2020-2030,” Cambodia’s first national plan for science and technology.
 - To establish the National STI Policy 2020-2030, Cambodia used the Cambodian policy consulting program (2018-2019), which was conducted as the K-Innovation phase 1 project.
 - The main policy goals of the National STI Policy 2020-2030 are to develop and nurture ethically motivated human resources for science and technology innovation, with the focus on achieving a positive gender balance, to promote efficient and effective national R&D, and to strengthen the capacity to absorb foreign technologies.
- After establishing the national science and technology basic plan, Cambodia recently reorganized governmental governance related to science and technology to better realize the plan.
 - Previously, there was no official government organization in charge of STI, so the GS-NSTC was put in charge of the related tasks.

- Absorbing the GS-NSTC, the Ministry of Industry and Handicrafts was reorganized into the Ministry of Industry, Science, Technology and Innovation (MISTI) and made exclusively responsible for STI (April 2020).
- To that end, MISTI established the General Department of Science, Technology & Innovation (GDSTI).
- It also established the National Institute of Science, Technology & Innovation (NISTI), a specialized research institute designed to support the realization of STI policies.
- The National Science and Technology Council (NSTC) was reorganized into the National Council on Science, Technology and Innovation (NCSTI).
- Along with UN ESCAP, Cambodia pushed for joint development of the STI (Science, Technology, and Innovation) Roadmap, which was adopted in 2021.
 - The STI Roadmap serves as a policy roadmap for implementing Cambodia's "National Science, Technology and Innovation Policy 2020-2030."
 - The Roadmap is composed of the following five key areas: governance (enhancing the governance of the STI system); education (building human capital in STI); research (strengthening research capacity and the quality of research); collaboration (increasing collaboration and networking between STI stakeholders).

[Figure 3-5] Five key areas of the Cambodia STI Roadmap 2030



Source: Cambodia MISTI website⁷.

- Consulting on Cambodia's STI policies
 - Developing countries such as Cambodia show country specificity in science and technology innovation systems, too, according to each country's stage of development and its own political, economic, and social characteristics.
 - Belonging to the lower middle-income country group in terms of per capita income, Cambodia is currently in the process of establishing a national STI system.
 - Therefore, it is necessary to find the most suitable model and development path for Cambodia's current development stage and its STI capacity.
 - In this regard, Cambodia's "National STI Policy 2020-2030," the STI Roadmap 2030, and the reorganization of government governance for science and technology innovation policies are all considered timely.

⁷ <https://www.misti.gov.kh/public/file/202108261629990117.pdf>, search date: Nov. 9, 2022

- The following figure shows the results of the diagnosis of the current state of science and technology innovation in Cambodia using a policy diagnosis tool that is applied to developing countries of the UNIDO (United Nations Industrial Development Organization).
- Cambodia ranks in a low position overall in most indicators, including infrastructure, institutions, networking, and capability, as well as the creation of demand in domestic and foreign markets.
- However, Cambodia is in a high position in terms of its power capacity, Internet penetration, and IPR system based on patent applications by overseas Cambodian nationals.
- In terms of the development stage of the science and technology innovation system, Cambodia has passed the initial introductory stage and now entered the system establishment stage. Accordingly, “execution” should be emphasized rather than “planning” in science and technology innovation policies.
- In the early stage of development, support for MISTI through strong leadership is essential in terms of role sharing and cooperation with the relevant ministries and agencies, and of efficiency of budget allocation and policy execution.
- Consulting on Cambodia’s National Technology Roadmap
 - Concept of the technology roadmap and introduction of Korean cases
 - As in the first year, we briefly introduced the concept of the technology roadmap and cases in Korea to the executing agency and newly selected service providers.
 - We explained the concept of the technology roadmap and introduced the details of various cases including their backgrounds, procedures, promotional organization, and contents.
 - We emphasized the necessity of knowledge in terms of enhancing the technology strategy and planning capabilities of the roadmap building process.
 - We specified the backgrounds and procedures of Korea's 2002 National Technology Roadmap and 2014 Strategic Technology Roadmap

- We emphasized the necessity of a technology roadmap to present the vision and strategic direction of science and technology development at the national level.

[Table 3-1] Concept of the technology roadmap and cases in Korea

Description	Major role
Concept	- Technology planning methods to establish demand-based technology strategies. - Functions for identifying, selecting and developing technological alternatives that meet product demand.
Korean cases	- 2002 National Technology Roadmap - 2014 Strategic Technology Roadmap

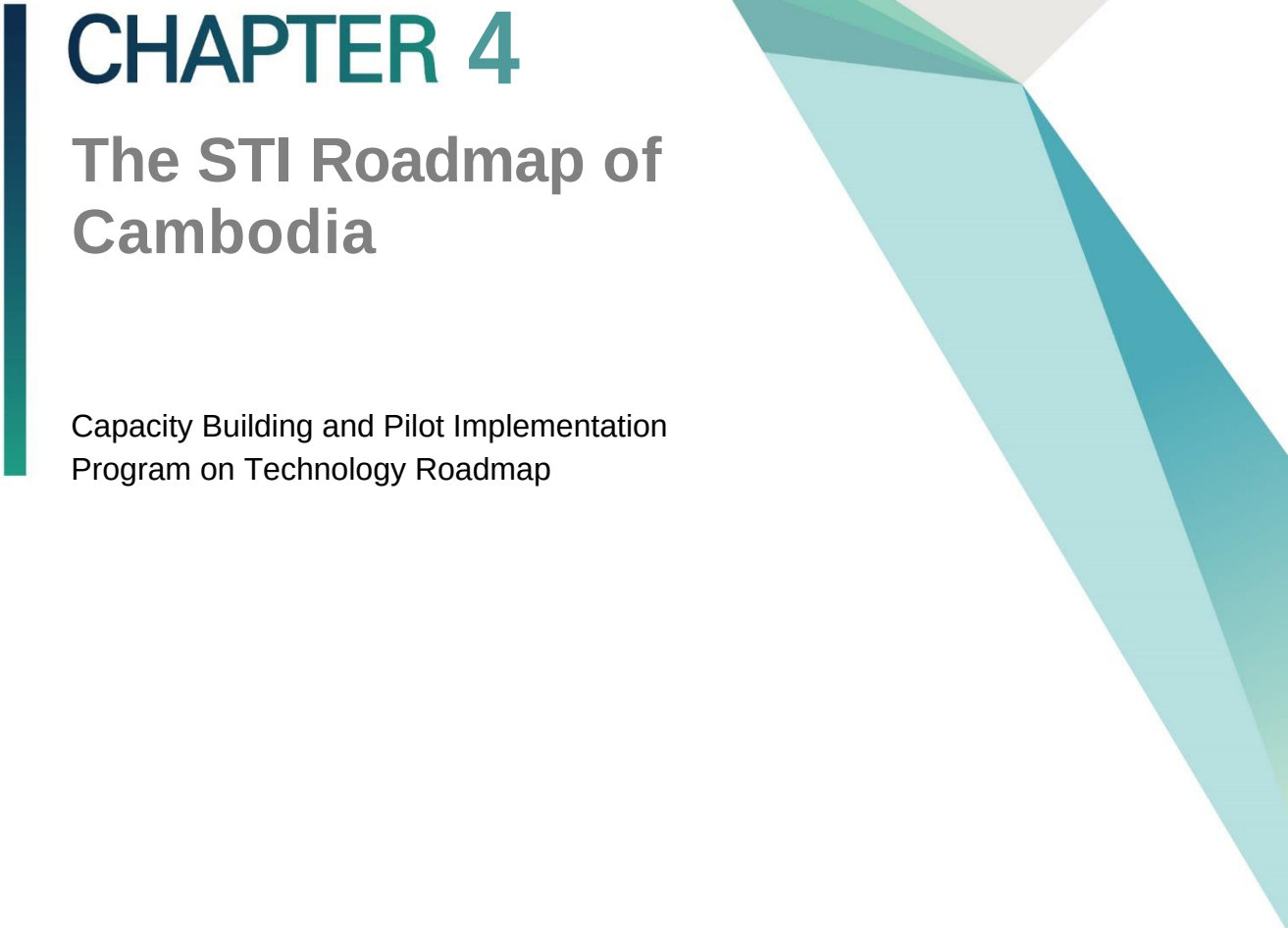
Source: STEPI Research Team.

- Consulting at the preliminary stage of the Cambodia National Technology Roadmap
- It was agreed that MISTI, the executing agency, should obtain approval from Cambodia's National Council of Science, Technology and Innovation (NCSTI) for the roadmap target areas and promote national technology roadmaps in the fields of energy, digital, and tourism technologies.
- STEPI prepared and provided guidelines and worksheets for Cambodia's national technology roadmap.
- Contents of Cambodia NTRM Guidelines
 1. Introduction: Overview, Objectives, Scope
 2. Methods and Process
 - Step 1. Vision Building
 - Step 2. Environmental Analysis
 - Step 3. Strategic Products
 - Step 4. Key Technologies
 - Step 5. Charting

[Figure 3-6] Guidelines on worksheets for Cambodia's NTRM (example)

Source: Kim Ji-hyun et al. (2021, p.230).

- Consulting on the composition of Cambodia's NTRM Drafting Committee
- Consultation on the composition of the Technology Roadmap Committee, emphasizing the importance of selecting experts to serve as committee members, and the need for diversity (various stakeholders, especially from the private sector).
- Consultation on the roles of the executing agency and the Technology Roadmap Committees in the fields of energy, digital, and tourism, as well as the appointment of coordinators to each committee and the general coordinator
- Accordingly, MISTI launched the National Technology Roadmap Committee, with the leading organization for each of the three technology fields selected with the approval of the Minister of MISTI.
- Energy sector and overall control of local institutions: MISTI (PM: Try Sophal)
- Digital sector: Leading organization - AVI (PM: Chhem William Siriwat)
- Tourism sector: Leading organization - RUPP (PM: Khun Chanmony)

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and light blue, also pointing downwards.

CHAPTER 4

The STI Roadmap of Cambodia

Capacity Building and Pilot Implementation
Program on Technology Roadmap

Chapter 4. The STI Roadmap of Cambodia

1. Digital



Digital Technology Roadmap

Draft Report

Asian Vision Institute

January 2023

1.1 Introduction

A digital technology roadmap is a visual document that reflects the digitalization plan of a country, including the series of goals, strategies, initiatives, and enhancements that should help to transform it into a high performing, digitally driven country.

This report presents the Digital Technology Roadmap of Cambodia, which consists of five main steps, namely, 1) Vision Building, 2) Environmental Analysis, 3) Strategic Products, 4) Key Technologies, and 5) Charting, in order to demonstrate potential areas for further actions. To achieve this goal, the Asian Vision Institute (AVI) organized the half-day Workshop No. 1 on Vision Building (i.e. building the vision and goals of the roadmap) and Environmental Analysis (i.e. collecting and analyzing the various drivers), along with the associated opportunities and threats, which was held on October 4, 2022.

Workshop No. 1 was organized to consolidate the inputs provided by the Working Committee on the 'Vision Building' and 'Environmental Analysis' components of the Digital Technology Roadmap. Furthermore, the discussion includes the strategy for preparing the Strategic Products, Key Technologies, and Charting for Workshop No. 2.

Twenty-five experts in digital technologies from the public and private sectors and academia attended the workshop. Three groups of twenty-five participants with multi-disciplinary backgrounds discussed the two core issues raised by Workshop No. 1.

At the beginning of the workshop, MISTI's representative and the AVI coordinator of the Digital Technology Roadmap gave a briefing on the goals of the workshop. After the briefing, the three groups of participants engaged in interactive discussions with the support of the three workshop leads, and then exchanged views and knowledge for several hours and identified the key elements of the digital transformation goals. Finally, the groups discussed various complicated and uncertain global issues in order to identify, evaluate, and select strategic digital technology alternatives that could help achieve the country's desired growth.

The three groups mainly focused on the three core elements of the digital technology

roadmaps. These are briefly presented below:

(i) Vision Building: The group discussed several points related to the roadmap vision and identified the goals to achieve this vision.

■ **Vision:**

"To leverage digital technology in order to improve the productivity and efficiency of Cambodia's digital economy, society, and government."

■ **Goals:**

- 1) Strengthen human resources' digital literacy and IT skills.
- 2) Invest in technological infrastructure at the national level.
- 3) Increase connectivity between people, organizations, data, and devices.

(ii) Environmental Analysis: This issue was discussed with the goal of identifying the drivers of the following contexts.

- Who is the driving force in Cambodian society?
- Which technologies are the key drivers?
- Which of the key drivers could boost the country's economic growth?
- Key environment drivers?
- Political driver that includes policies, stability, transformational leadership, etc.?

■ **Drivers:**

- Youth
- Employment
- Mobile and Internet reach
- E-commerce
- Tech start-ups and MSMEs

- Foreign investment
- Land and natural resources
- Policies and legal frameworks
- Social
- Digital business
- Technology
- Political

After exploration and discussions, the groups presented and shared their achievements with diverse views during the last part of the workshop. The consensus outcomes with priorities were considered to co-relate with Workshop No. 2. Overall, the workshop went very well. All of the participants engaged in the discussion, shared their knowledge and views, and demonstrated the actions of the digital technology roadmap.

Workshop No. 2 was held on November 18, 2022. During this workshop, the members of the sub-committees were once again convened to provide their inputs on Strategic Products and Key Technologies, based on the findings of Workshop No. 1.

1.2 Demand, Supply, and Trends

Services have become the backbone of the world economy. In 2019, services accounted for about two-thirds of both world gross domestic product (GDP) and Foreign Direct Investment (FDI) and provided the majority of all employment (WTO, 2020). The value of international trade in services is forecast to increase from \$6.1 trillion in 2019 to \$8.0 trillion by 2025, equivalent to one-third of the entire value of global financial flows over this period (Oxford Economics and Western Union Business Solutions, 2020). The WTO (2019) projects that the share of services in world trade will grow by 50% by 2040.

Asia is the second-largest trader of digital services, with this trading segment growing

faster than in other parts of the world. In 2020, amid the COVID-19 pandemic, the year-on-year growth of trade in digital services in Asia and the Pacific region remained resilient (at a 1% increase), while other services (global tourism, travel, and distribution sectors) plummeted by 38% (Liberatore, Avendano, and Cho, 2021). In recent years, the sectors which have made the greatest contribution to Asia's growth in digital services trade are other business services and telecommunications, and computer and information services. Some Southeast Asian economies have also been driving expansion in digital services trade by posting rapid growth in digital exports—as in Cambodia, the Lao People's Democratic Republic (Lao PDR), and the Philippines. Meanwhile, emerging areas of further growth include e-commerce, digital transactions, demand for automation, and the remote delivery of services (Chanda 2021).

The Asian Development Bank (ADB) estimated that Cambodia's technologies and digital businesses generated revenues of \$470 million for 2019 report in Asian Economic Integration report 2021. There was an uptick in cashless payments in Cambodia, especially during the global Covid-19 pandemic, as well as the growth of fintech adaptation and a push by the government for its citizens to go cashless for health and safety reasons. When broken down by sector, online travel accounted for 37.9%, followed by e-commerce at 27.6%, advertising technology at 12.7%, digital media at 10.2%, e-services at 7.8%, and transportation at 3.8%. Moreover, according to the National Bank of Cambodia (NBC), the market demand in Cambodia stands at approximately US\$29.6 billion collectively, with digital financial apps only accounting for about US\$5.7 billion.

Currently, Cambodia has a population of 16.38 million people, of whom 24.5% live in urban centers, while 75.5% live in rural areas. In terms of age demographics, people aged 15-35 account for 40.13% of the total population, and 65.4% of the population are of working age, i.e. between 15 to 65 years old. This demographic dividend is one of the main drivers behind the advance of digital technology and the digital economic transformation of Cambodian society due to the adoption of technologies for both socio-economic development and self-development purposes in keeping with general global trends. In addition, Cambodia belongs to ASEAN, a region consisting of large

economies, and is an attractive location for the development of regional and global value chains.

As of January 2021, the number of mobile phone subscriptions to Cambodia's six operators was 21.18 million, showing an increase of 0.8% year-on-year. This figure was equivalent to 125.8% of the total population of Cambodia. Furthermore, the total number of internet users for all devices was 8.86 million in January 2021, representing an increase of 1.1 million between 2020 and 2021, with an internet penetration rate of 52.6% of the total population. Generally speaking, Cambodian internet users prefer to use mobile phones to access the Internet, which is reflected in the statistics obtained from Digital 2021: Cambodia from Datareport.com. Among all devices, mobile phones accounted for 52.0% of all web traffic, followed by laptops and desktops at 45.9%, and tablet computers at 2.2%.

Developments in ICT, and digitalization in particular, have been responsible for the growing importance of trade in services. Digitalization cut costs dramatically and lowers barriers to entry, facilitating the trading of an ever wider range of services. Given the prevalence of social distancing and travel restrictions during the COVID-19 pandemic, the adoption of digital technology accelerated even further as companies expanded their online presence and consumers adopted new habits - especially in the health, education, telecommunication, and audio-visual services sectors (WTO, 2020).

In the Asia and Pacific region, digital services are an important component of total foreign investment. Even in Asia's digital services, where the reinforcing relationship of the Global Value Chain (GVC) and FDI is apparent, Cambodia showed both a high FDI presence and a high GVC participation rate in the digital service sectors. Growing FDI in these sectors could improve the region's trade and integration into international production networks through various channels. Advancements in information and communication technologies and digital platforms have reduced distance-based barriers to the trade of goods and services and allowed many firms and businesses to integrate into the global supply networks (ADB 2021a).

[Figure 4-1] FDI Digital Services

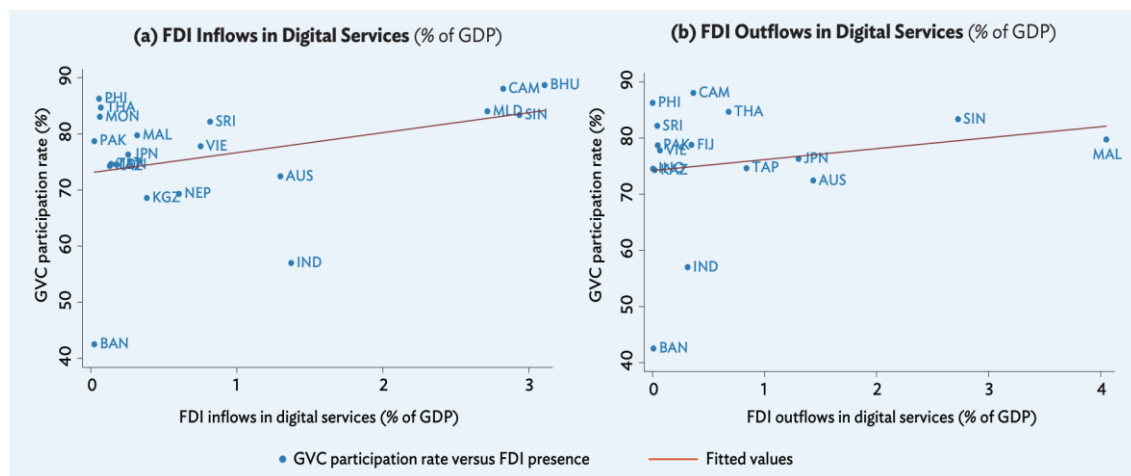


Figure: Scatter Plots on GVC participation and FDI—Asia and the Pacific, 2020: AUS = Australia; BAN = Bangladesh; BHU = Bhutan; BRU = Brunei Darussalam; CAM = Cambodia; FDI = foreign direct investment; FIJ = Fiji; GDP = gross domestic product; GVC = global value chain; HKG = Hong Kong, China; IND = India; INO = Indonesia; JPN = Japan; KAZ = Kazakhstan; KGZ = Kyrgyz Republic; KOR = Republic of Korea; LAO = Lao People's Democratic Republic; MAL = Malaysia; MLD = Maldives; MON = Mongolia; NEP = Nepal; PAK = Pakistan; PHI = Philippines; PRC = People's Republic of China; SIN = Singapore; SRI = Sri Lanka; TAP = Taipei, China; THA = Thailand; and VIE = Viet Nam.

Source Asian Economic Integration Report 2022.

1.3 Vision, Policy, and Strategy

1.3.1 Vision

Cambodia nurtures the ambition to establish itself as a digital economy, society, and government by 2035 in tandem with its vision of becoming a higher middle-income country by 2030 and a high-income country by 2050.

1.3.2 Policy

The digital economy is regarded as one of the key drivers of economic growth. Recently, digital adoption has been accelerated by the COVID-19 pandemic, while policy frameworks have been introduced to speed up digital transformation. So far, three key policy instruments have been established that are critical to digital technology development and digital transformation, namely, the Digital Economy and Society Framework 2021-2035, the Digital Government Policy 2022-2035, and the Science, Technology and Innovation Roadmap 2030.

Cambodia's Digital Economy and Society Policy Framework 2021-2035 set policy measures to (a) develop the digital infrastructure with the focus on digital connectivity, financial technology infrastructure and digital payment systems, and logistics and last-mile delivery; and to (b) build trust and confidence in digital systems with the focus on establishing responsive and effective legal frameworks and strengthening digital security management.

Cambodia's Digital Government Policy 2022-2035 aims to achieve the following ten goals: (a) establish and improve the digital connectivity infrastructure; (b) establish the digital payment system infrastructure for public services; (c) strengthen the digital security infrastructure; (d) develop the postal service infrastructure; (e) organize digital government governance; (f) digitalize government and public services; (g) develop digital human capital; (h) promote digital research and innovation; (i) promote collaborations with digital technology companies; and (j) promote digital start-ups. In this regard, the Minister of Post and Telecommunications leads the Digital Government Committee.

The STI Roadmap 2030 aims to achieve the following goals: (a) enhance the governance of the STI system and identify MISTI as the leading coordinating body for STI policies and implementation; (b) build human capital in STI by increasing the number of university students in STEM fields to at least 50 per cent; (c) improve scientific research by investing 1 per cent of GDP in research activities; (d) strengthen partnership building between key stakeholders; and (e) foster an ecosystem that facilitates innovation, improves the absorption capacities of firms, attracts investment in STI-related industries, and promotes technology transfer.

To implement the policy frameworks effectively, it is essential to identify policy challenges and gaps with regard to digital technology applications, develop policy support mechanisms and an ecosystem, support strategic products and key technologies that meet the needs of rapidly evolving industries, and better align and build synergies with the existing policies and strategies relevant to the digital economy, society, and government.

1.3.3 Strategy

To further build and strengthen the ecosystem to support digital transformation, it is necessary to (a) foster inter-ministerial collaboration (MISTI, MEF, MPTC, etc.) so as to complement the policy frameworks related to the digital economy, society, and government; (b) promote public-private partnerships for policy consultation and design in order to support private sector investment and growth in terms of digital technology development; (c) strengthen university-industry linkage to address the practical demands of the workforce and research and development; (d) develop and harmonize the regulatory and policy frameworks in order to strengthen cybersecurity and protect critical infrastructures; and (e) promote an inclusive digital economy by promoting digital literacy and universal education on STEM subjects, enhancing the digital capacity of MSMEs, and providing special support to women and girls.

1.4 Development of the National Digital Technology Roadmap

1.4.1 Building the Vision

As, arguably, the most fundamental component in the development of the national digital technology roadmap, the sub-committee members were divided into four diverse groups with the aim of balancing perspectives from the public sector, private sector, and academia.

This intentional categorization helped to form well-rounded visions and goals for the digital technology roadmap, based on contrasting views and expertise.

Visions	Goals
1. Establish a digitally literate society and economy by 2035 or 2030	- Digitalize public services - Digitalize rural areas. - Attract a greater contribution from the private sector - Provide cyber security and data protection - Establish a digital curriculum for all schools
2. Increase productivity and connectivity	- Build infrastructure and connectivity - Increase digital literacy - Promote research & development - Promote entrepreneurship and innovation
3. Establish a digital technology that is relevant locally and competitive internationally; that is dynamic and adaptive to the changing world; and that is focused on creating a positive impact for the country and its people (relevant and competitive, dynamic and adaptive, bringing positive impact)	- Evolution: positive evolution of digital citizens, digital government, and digital business. - Inclusion: inclusion of everyone in the digital era: elderly, rural, people with disabilities and females. - Knowledge hub: focus on skills and access to technology to support tech advancement for everyone.

1.4.2 Environmental Analysis

Having established the vision and goals, the groups proceeded to identify the drivers of digital technology based on the results of a STEEP (Social, Technological, Economic, Environment, Political) analysis.

STEEP	Drivers
Social	- Youth - Employment - Skills development - Sufficient skills and understanding of benefits of technology - Efforts to increase female participation - Efforts to advance knowledge of digital skills in rural areas - Offering of life-long learning opportunities to all groups, including elderly people - Inclusion of people with disabilities
Technological	- Mobile and internet reach - E-commerce - Artificial Intelligence - Internet - Research and development - Infrastructure and access for everyone - Closing the gap between rural and urban areas in accessing technology - Ensuring access to affordable devices
Economic	- Tech startups - Foreign investment - Digital business - E-banking - E-payment - Access to funding - Access to funding to support innovations and new technology solutions - More possibilities to connect with investors - Funding opportunities from government and international organizations

STEEP	Drivers
Environmental	- Land and natural resources - Environmentally-friendly
Political	- Policies and legal framework (e.g. DES, STI roadmap, etc.) - Policy and legal framework - Stable political situation and supportive political environment that encourages digital technology - Clear and transparent legislation - Benefits for investors and implementers of tech solutions, such as tax support, legal support and others - Digital government tools and opportunities that ensure easy and fast services for citizens

Next, the sub-committee members proposed potential opportunities and threats that could stem from the drivers, taking into account the associated advantages and disadvantages.

Drivers	Opportunities/Threats
A. Youth	O = Higher youth population (2/3 under 35) T = Limited technology training/awareness of youth
B. Employment	O = High demand with high salary in the market T = Availability of skilled workforce
C. Mobile and internet reach	O = Many service providers T = Lack of infrastructure and poor quality of services
D. E-commerce	O = Higher adoption starting from the banking sector T = Consumer protection
E. Tech startup & MSMEs	O = Higher adoption and innovation T = Lack of investment and sustainability
F. Foreign investment	O = Open economy, more attractive to investors, cheap labor force T = Sanctions (EBA withdrawal)
G. Land and natural resources	O = Availability of abundant resources T = Climate change with low resilience of local people

Drivers	Opportunities/Threats
H. Policies and legal frameworks	O = Many endorsements and drafting from all related ministries T = Awareness to public, implementation, and investment
I. Social	O = Young people are quicker learners and are able to use the technology T = Security
J. Digital business	O = Larger Number of Users T = Customer Security
K. Technology	O = Technology aspect, founding T = Protection, ethics of AI use, too rapid development of technology
L. Political	O = Strong Collaboration (ASEAN, and Mekong Country) T = How to align all the regulations?

1.4.3 Strategic Products

In Workshop No. 2, the sub-committee members began by proposing strategic products and services that would be relevant to the development of digital technology. Once the various groups had provided their suggestions, all the sub-committee members proceeded to vote for the most important strategic products and services, based on the following three evaluation criteria: economic impact, strategic importance, and potential for success. The highest-scoring strategic products and services were as follows: **1) access to digital devices, 2) internet infrastructure, 3) online business registration platform, 4) focus on STEM subjects in the education system, 5) re-skilling/upskilling agency for professionals, 6) raising public awareness, 7) national data center, and 8) center of excellence.**

2022 K-Innovation Partnership Program with Cambodia

Products/Services	Code	Evaluation Criteria			Sum
		Economic Impact	Strategic Importance	Potential for Success	
Access to Digital Devices (including computer labs and makerspaces in schools)	1	57	58	52	167
Internet Infrastructure (fiber optic cables, routers, switches, satellites, etc.)	9	59	58	50	167
Online Business Registration Platform (transparent and clear closure process for companies)	6	56	55	52	163
STEM Focus in Education System (teacher training and student extra-curricular activities)	3	56	57	49	162
Re-skilling/Up-skilling Agency for Professionals (to connect training providers with businesses to support digital skills)	4	54	54	48	156
Raising Public Awareness on Digital Technology (via news, social media, events, etc.)	5	51	53	48	152
National Data Center (store public and private sector data securely)	12	50	54	46	150
Center of Excellence (regional technology transfer and talent pool)	13	50	51	47	148
Decentralized Financial Systems (blockchain-based)	11	53	49	44	146
Startup Support Programs (incubation and accelerator programs)	7	51	48	45	144
Free Online Educational Resources in Khmer (online access for fundamental digital skills)	2	50	49	39	138
Consumer Protection Enforcement (one point of contact online)	8	49	48	41	138
Hardware Development Center (PCB Assembly)	10	45	45	42	132

1.4.4 Key Technologies

Based on the selected strategic products and services, the groups then proceeded to propose key technologies that would be essential for the development of the aforementioned products and services. The highest-scoring key technologies were as follows: **1) cybersecurity, 2) 5G, 3) automation systems, 4) big data, 5) AI and machine learning, 6) computer software, 7) WiFi nos. 6 and 7, and 8) cloud computing.**

Vision: To leverage on digital technology to improve productivity and efficiency, for Cambodia's digital economy, society, and government					
Score scaling from 1 to 5 1= Not at all Important 2= Slightly Important 3= Moderately Important 4= Very Important 5= Extremely Important					
Key Technology	Criteria				Sum
	Strategic Importance	Feasibility			
		Short-term 2024	Middle-term 2027	Long-term 2030	
Cybersecurity	45	42	44	45	176
5G	44	41	42	44	171
Automation Systems	39	35	43	44	161
Big Data	42	31	38	42	153
AI & Machine Learning	40	31	39	42	152
Computer Software	40	38	37	37	152
Wifi 6/7	40	33	40	38	151
Cloud Computing	38	33	37	39	147
Robotics	35	31	37	42	145
IoT	40	32	36	36	144
Natural Language Processing	35	32	37	38	142
Blockchain	35	31	34	39	139
AR/VR	32	29	32	35	128
Chatbot	28	29	33	32	122
Parellel Processing	31	26	33	32	122
Quantum Computing	30	21	26	30	107

1.5 Conclusion

(To be continued...)

- All of the key components (Vision Building, Environmental Analysis, Strategic Products, Key Technologies, Charting) have been finalized based on the inputs provided by the sub-committee members
- Requiring further elaboration: conclusion, final roadmap, and chart

2. Energy



Energy Technology Roadmap

General Department of Science, Technology and Innovation
Dec. 31, 2022

2.1 Introduction

2.1.1 Background

The United Nations Agenda 2030 highlights access to energy in its Sustainable Development Goal (SDG) No. 7, which aims to provide “affordable, reliable, sustainable and modern energy for everyone by the year 2030” (UN, 2015). SDG 7 is considered a medium-term goal that will contribute to the achievement of other goals. As stated in SDG 7, renewable energy includes solar energy, hydropower, wind turbines, bioenergy (waste-energy), geothermal energy, hydrogen and fuel cell energy, nuclear power and ocean energy. The availability of energy has transformed the course of humanity over the last few centuries (Mika et al., 2021).

Based on the report of the International Energy Agency (IEA), some 770 million people around the world did not have access to electricity as of 2019 (IEA, 2019). The accelerating progression of this trend is largely the result of the expansion of electricity grids and the use of fossil fuels (IEA, 2015). Electricity generation is still heavily reliant on fossil fuels, nuclear energy or large-scale hydropower, making the sustainability aspects of energy sector development somewhat problematic (Mika, et al., 2021). Currently, various sources of energy are being used around the world, including fossil fuels, nuclear power, hydropower and other renewable technologies. Indeed, the energy system has transformed dramatically since the days of the Industrial Revolution. Demand for energy is growing massively across many countries around the world, as societies become increasingly affluent and populations increase (Hannah Ritchie). Based on the World Count (TWC), a global energy consumption observatory, global energy consumption will reach 740 million terajoules (additional growth of 30%) by 2040. The Association of Southeast Asian Nations (ASEAN) was considered the seventh-largest economy and the fifth-largest investment destination in 2016. As one of the largest economic coalitions, ASEAN has a cumulative GDP of USD2.5+ trillion. This vigorous economic growth has increased energy demand by 70% compared to 2000, and the region currently accounts for 5% of total global energy demand. Therefore, given that the region is experiencing ever increasing energy demand, each country will need to make an effort to meet the projected target (Yang Liu, 2020). To

generate electricity as well as energy, location is one of the primary factors to consider when contemplating energy generation. Naturally enough, some countries and regions are better suited to certain types of renewables depending on their natural resources. In 2019, five out of the ten ASEAN countries (Indonesia, Thailand, Malaysia, Singapore, and the Philippines) issued green bonds (Bloomberg Terminal, 2019), which could provide a unique opportunity to mobilize finance for energy efficiency and renewable energy projects in those countries. At the same time, the total primary energy supply in Cambodia was about 4,671 kilotons of oil equivalent (ktoe) in 2015. Therefore, to reach SDG 7, Cambodia will need to have a clear strategy, such as this energy roadmap, which could improve energy sufficiency and sustainability with cutting-edge technologies by 2030. Thus, the goal of this roadmap is to (1) increase renewable energy capacity, (2) increase energy generation, and (3) promote the use of green energy across all sectors in order to reduce the country's overall CO₂ emissions.

2.1.2 Country Context

2.1.2.1 Demographics, Energy Products

Located in the southern part of the East Asia Peninsula, Cambodia has a total land area of 181,035 square kilometers and a population of 15,552,211 (males: 48%; females: 51.5%). its population growth rate is 1.4%.⁸ During the period 1998-2019, Cambodia enjoyed an average economic growth rate of 7.7%, making it one of the fastest-growing economies in the world⁹, although it nonetheless remains among the lowest in Asia, with poverty falling substantially from 47.8% in 2007 to 14.0% in 2014, and poverty rates declining by 1.6% points per year from 2009-2019/2020 (World Bank, 2022). However, the World Bank Open Data (2021)¹⁰ reported that Cambodia's economic growth has been decreasing at an average rate of 3% due to the Covid-19 pandemic and the global economic downturn, and GDP per capita is \$15,691. Cambodia moved to the lower-middle income status in 2015, and plans to become a

⁸ (NIS, 2019)

⁹ (The World Bank, 2022)

¹⁰ (World Bank , 2021)

higher middle and high-income country by 2030 and 2050, respectively.

Cambodia primarily depends on diesel power plants, biomass (agriculture and firewood), thermal power plants (coal), hydropower plants, and other types of renewable energy as its sources of energy. The Electricity Authority of Cambodia's (EAC) annual report for 2015-2017 recorded that Cambodia's main source of energy was "diesel" (91%), with "coal" (3.20%) accounting for only a small portion in 2010. During the period 2010-2017, however, the consumption of diesel dropped sharply to 10.69% as Cambodia switched to hydropower (60.67%), coal (53.80%) and biomass (0.89%) as its main sources of energy (Mika, et al., 2021). Even though these trends have improved the energy mix in Cambodia, the Asian Development Bank (ADB)¹¹ has reported that around 5 million citizens were unable to access to grid electricity and still relied on old methods to generate electricity, such as batteries, firewood, and other traditional fuels. Historically, Cambodia could access to electricity less than 1% in 1991, 56% in 2014 and more than 90% in 2019 (World Bank, 2019); and there were few differences (90%) between urban and rural access (World Bank, 2018). Cambodia generated 73.45% of its domestic energy demand and imported the rest (26.55%) from neighboring countries (Thailand, Lao PDR, and Vietnam), including non-renewable energy (48.83%) and renewable energy (51.17%) from local generation. In brief, the country's non-renewable energy sources consisted of coal (41.20%) and fuel oil (7.63%), while its renewable energy consisted of hydropower (44.17%), solar power (6.36%) and biomass power (0.64%) in 2021¹².

Hydropower, agriculture biomass, wind energy and solar energy are significant sources of renewable energy in Cambodia. First, hydropower projects have been expanded to include nine hydropower dams in Cambodia with a capacity of 979MW installment since 2002; and China has invested US\$ 1.6 billion in the construction of hydropower. It has been reported that the government has approved the construction of more hydropower facilities as Cambodia is projected to require a hydro energy generation

¹¹ (ADB, Cambodia Energy Sector Assessment, Strategy and Roadmap, 2018).

¹² (EAC E., 2021)

capacity of 4,000MW by 2040. Second, agriculture biomass also produced 54.29 million kWh of energy from seven biomass generation plants (five plants in 2016, and two plants in 2017). Third, another renewable source of energy is wind energy. Cambodia can generate around 500 MW from wind based on the speed recorded (5km/s) in the coastal provinces and the southern part of Lake Tonle Sap. The last source of energy is solar energy. It has been found that “the country has an average solar irradiance of 1,400-1,800 kWh per square meter per year throughout the country, and an average irradiance of 6-9 hours in the daytime.” The government plan to scale up 3.5 % share of total energy by 2030, which will come from solar farms (Marabona, 2019).

2.1.2.2 Infrastructure, Sources, Usage Data, Supply Chain

■ Energy Consumption

Cambodia’s Total Final Energy Consumption (TFEC) grew stably at a rate of around 7.2% per year in the eight years after 2010. Biomass still played a vital role in generating energy for household and industrial uses and “electricity production”. In 2018, the biomass share of TFEC was 25.5%, compared with 40.9% in 2010; and the rate of oil and electricity trade increased to 8.1% and 18.3% annually, respectively, as reported by the Ministry of Mines and Energy of Cambodia in 2020. From 2000 to 2008, Cambodia relied heavily on biomass, followed by oil.

For cooking purposes, charcoal and gas are mainly used in urban areas, followed by electricity, due to better access and less concern about the price of electricity (Mika, et al., 2021), whereas in rural areas people use firewood and plants as their primary energy source (ADB, 2018). ADB (2018) reported that Cambodians utilized liquefied petroleum gas (31%), charcoal (5%), and electricity (2%) for cooking. This suggests that in rural areas it is easy to access firewood for cooking, followed by gas.

The Ministry of Mines and Energy (2022) has concluded that “The country’s GDP increased by 7.6% per year in 2000-2019, which means that the energy elasticity of GDP to the TFEC (2000–2019) was 0.95. If biomass is excluded from the TFEC, the

energy elasticity would be around 1.4 since the TFEC without biomass has been growing 10.5% faster than GDP. In this regard, an aggressive energy efficiency and conservation program must be in place to improve elasticity in the future.”

■ Energy Demand

In 2040, the industrial sector will require 2.41 Mtoe of energy, followed by the residential and commercial sectors at 5.67 Mtoe, (Kimura and Han, 2019). The main consumers of TFEC will be the industrial, transportation, and other sectors. This is followed by energy demand in the transportation sector, which is projected to grow at an average annual rate of 5.6%, or 4.78 times, from 1.75 Mtoe in 2019 to 10.12 Mtoe in 2050. Demand in other sectors is expected to grow at an annual rate of around 4.4%, rising from 1.81 Mtoe in 2019 to 7.15 Mtoe in 2050. In 2022, the MME projected that the consumption of coal, electricity and oil would grow at an average annual rate of 10.3%, 7.6% and 5.1%, respectively.

■ Energy Supply

As local energy generation and energy resources cannot meet the growing demand for energy, Cambodia has decided to import electricity, coal and diesel from other countries. For example, in 2021, Cambodia’s annual power consumption amounted to 9,255 GWh, of which 3,345 GWh was imported from Vietnam (1,136 GWh), Thailand (306 GWh), and the Lao People’s Democratic Republic (1,904 GWh) (EAC, 2021). Cambodia’s supplied electricity sales, commercial, residential, industrial and government sectors were 31%, 28%, 23%, 16% and 4%, respectively. These figures indicate that Cambodia needs to increase the power supply to meet the growing demand in the above sectors.

2.1.2.3 Energy issues facing the country and the related support programs and policies

Cambodia faced many challenges including the following: 1) electricity cost, reliability and coverage in rural areas fell behind neighboring countries; 2) long distance from the

long connection costs, and low incomes prevented some rural households from accessing electricity; 3) poor economic competitiveness (Marabona, 2019), and 4) poor energy security (during the dry season, need to import energy, etc.).

To ensure a successful transition to sustainable energy, the Royal Government of Cambodia has endorsed policy documents and action plans to develop the energy sector, as outlined below:

- National Strategic Development Plan (NSDP), 2014–2018
- Industrial Development Policy (IDP), 2015–2025
- Power Sector Strategy 1999-2016
- Master Plan of the Ministry of Mines and Energy
- Renewable Electricity Action Plan (REAP) 2002–2012
- Rural Electrification by Renewable Energy Policy 2006
- Renewable Energy Development Program

2.1.3 Sector Context

■ Energy Institutions

The energy sector in Cambodia is governed by the MME and the Electricity Authority of Cambodia (EAC) in accordance with the 2001 electricity law and the subsequent amendments of 2007 and 2015 (EAC, 2018). According to the law, the MME is responsible for developing energy policies, strategies, plans, and technical standards, while the EAC acts as the electricity regulator with the authority to issue rules, regulations, and procedures on power market operations; award licenses; and set tariffs (ADB, 2018)

■ Energy Supply and Demand

Cambodia's total primary energy supply (TPES) increased from 3,350 kilotons of oil equivalent (ktoe) in 2010 to 4,761 ktoe in 2015, at an average annual growth rate of 7.3 percent. The sources comprised fuel wood and other biomass, which accounted for an estimated 44.4% of TPES, oil and petroleum products at 38.5%, coal at 10.7%, hydropower at 3.6%, and electricity imports at 2.8%. The completion of new coal and hydropower facilities between 2010 and 2015 greatly boosted the share of coal and hydropower in Cambodia's energy supply. During this period, the share of coal rose from 0.7% to 10.7%, while that of hydropower rose from 0.1% to 3.6%.

The total final energy consumption (TFEC) in Cambodia was about 3.4 million tons of oil equivalent in 2015, growing at an average annual rate of 6.9% during the period 2010-2015. The total energy consumption consisted of 50.5% petroleum products, 36.0% biomass, 13.1% electricity, and 0.4% coal. Even though the TFEC grew steadily during these years, it was still lower than the Asian average (about 480 kilograms of oil equivalent per capita per year in 2016 compared to an average of 740 kilograms among the Asian economies according to the statistics of the International Energy Agency).

According to the ADB, by 2030, the country's energy requirements will have doubled due to continuous urbanization, economic growth, and population increase. As the nation becomes increasingly motorized and as the government encourages the formation of industrial zones for manufacturing and agro-processing, demand for diesel and gasoline will be the primary drivers of oil consumption. As kerosene is gradually phased out, it is anticipated that more liquid petroleum gas will be used in the residential sector.

■ Energy Resources

As fossil fuel (coal and oil) resources in Cambodia are either small in quantity or untapped (limited exploration, mining, drilling, and processing), the country imports all the fossil fuels that it consumes. However, Cambodia's oil and gas potential, though

not proven, seems promising due to similarities to adjacent areas in Vietnam and Thailand, where these resources are being produced. According to the General Department of Petroleum of the MME, the nineteen-block onshore petroleum resources contain three times more potential than the six-block offshore area.

Renewable energy consumption in Cambodia accounted for 65% of the TFEC, of which 46% came from traditional biomass such as wood, charcoal, and dung; 15% from modern biomass such as biogas produced from human and animal waste; and 3% from hydropower (Bank, 2018).

■ Issues in the Energy Sector

Alongside rapid economic growth, increased electrification rates, and growing energy demand, Cambodia faces many challenges such as limited capacity in energy planning, energy security, affordability, and environmental sustainability. The Asian Development Bank's (ADB) *Assessment of Cambodia's Strategy and Road Map for the Energy Sector* (ADB, 2018) calls for a more cohesive strategy that links policies and physical infrastructure plans, and the well-balanced integration of new renewable power generation within the existing energy mix. The document also addresses the need to develop the capacity of the energy sector's main governing bodies (MME, EAC) in cooperating with the private sector (public-private partnerships) in order to scale up clean energy generation. Another energy challenge is the high price of electricity resulting from high tariffs, which has not only left many Cambodians unserved, but has also had the effect of restricting economic competitiveness and investment.

2.1.4 Approach and Scope

In response to the vision and objectives of the National STI Policy, along with the determination of the Royal Government of Cambodia (RGC) to transform Cambodia into a high middle-income country by 2030 and a high-income country by 2050, efforts to improve the energy sector are considerably more important than for other sectors of the nation. Boosting the country's energy supply with environmentally-friendly energy can only be achieved by leveraging science, technology and innovation and by

identifying and resolving issues and needs. This energy roadmap was developed using a quantitative approach, including an assessment of Cambodia's energy sector, the collection of data on the country's energy situation, surveys and consultation with the main bodies involved in the energy sector.

The need for advanced clean energy technologies is more necessary than ever before. “Energy technology” refers to a “combination of hardware, techniques, skills, methods, and processes used in the production of energy and the provision of energy services” (IEA, 2020), which relates to how we go about creating, converting, storing, transferring, and utilizing energy. Although the emerging technologies may take years to become effective in terms of their technical performance, Cambodia should pay close attention to the current global trend of transition to renewable energy sources, including the following:

- *Solar technologies* convert sunlight into electrical energy either through photovoltaic (PV) panels or through mirrors that concentrate solar radiation.
- *Wind turbines* transform the kinetic energy of the wind into electricity via blades and a generator.
- *Hydropower* is power generated from the natural flow of water.
- *Geothermal energy*, a virtually untapped energy resource derived from the earth's heat, is more vital today than ever—it supplies clean, renewable power around the clock, emits few or no greenhouse gases, and requires only a very small environmental footprint to develop.
- *Bioenergy technology* includes all technologies which derive energy from biomass. Biomass is an organic renewable energy source that includes such materials as agricultural and forest residues, energy crops, and algae.
- *Hydrogen & Fuel Cells* produce electricity from a number of domestic fuels, including hydrogen and renewables, and can provide power for virtually any application, ranging from cars and buses to commercial buildings.

Numerous technical- and political-level workshops and meetings were held to produce

this strategic document. On October 13, 2022 the General Department of Science, Technology and Innovation hosted the first consultation workshop, attended by high-level energy officials and experts from the relevant ministries under the RGC, in order to brainstorm ideas and acquire strategic information.

This technology roadmap introduces the enabling factors and major challenges facing the development of the energy sector.

2.2 Analysis of Cambodia's Energy Supply and Demand

This section looks into the country's energy funding, how it constitutes the national supply of and demand for energy, as well as the implications.

2.2.1 Energy Resources

Though debatable, the country has a substantial energy potential consisting of both non-renewable and renewable energy resources. Although it currently imports all the fossil fuels, such as coal, oil and gas, that it consumes, as yet unquantified amounts of these non-renewable energy resources also exist in the country. Other domestic yet renewable energy resources include biomass, hydropower, solar energy, and wind energy. The following subsections describe each energy resource in detail.

2.2.1.1 Non-Renewable Energy Resources

Coal: The availability of coal deposits has long been claimed, but there has been insufficient exploration to confirm the exact location and volume of this mineral. While there can be more coal in un-surveyed areas, preliminary estimates of the deposits are over 150 million tons in northern Cambodia. These small deposits could supply enough coal for a 400-megawatt (MW) coal power plant for thirty years (JDI, 2009). The coal is of the lignite type (ERIA, 2022), and is similar in quality to Thai coal, which has a calorific value of 3,000 kilocalories per kilogram and is suitable as boiler or kiln fuel (JDI, 2009). However, no commercial-scale coal mining has ever taken place in the country to date (ADB, 2018).

Oil and gas: Seismic data show that the country has more oil and gas reserve potential in onshore areas than offshore areas. There are nineteen onshore petroleum blocks (I-XIX), six offshore blocks in national waters (blocks A-F), and four more blocks in the offshore Overlapping Claims Area (OCA) with Thailand for oil and gas exploration. To what extent these areas contain oil and gas has yet to be clearly determined. However, it is estimated that the 27,000 km² OCA may contain around 11 trillion cubic feet of natural gas and 3.6 billion barrels of oil. It is also estimated that Block A could produce about 30 million barrels of oil over a period of nine years. But fluctuations in global oil prices, the complexity of geological factors, maritime disputes, and the long duration of contractual negotiations could severely constrain the development of all the onshore and offshore petroleum blocks. Over the years, the government has issued concession licenses to some companies for oil and gas exploration and extraction activities in a number of those blocks, but their licenses have been terminated due to their lack of progress. Thus, the country has oil and gas reserves that remain largely untapped, leaving aside the fact that it has no oil refineries. In May 2017, a US\$ 2.3 billion oil refinery construction project was begun on 365 hectares (ha) of land straddling the provinces of Kampot and Sihanoukville. This will be the country's first-ever refinery, with a projected annual production capacity of 5 million tons of oil upon its completion (ADB, 2018).

2.2.1.2 Renewable Energy Resources

Biomass: The country has plenty of biomass energy resources deriving from rubber plantation forests, tropical trees such as *Gliricidia* and *Acacia*, and agricultural residues such as rice husks, rice straw, corn cobs, waste from palm oil extraction, cassava stalks, sugarcane bagasse, and cashew nut shells, as well as animal waste. The estimated total theoretical potential of agricultural residues alone is around 15,000 gigawatt hours (GWh) per year, 80 percent of which could be converted from rice residues (See Annex 1). By 2017, seven biomass generation power plants were in operation in Cambodia, representing roughly 39 MW of the installed capacity. The production of biofuels for transport also uses biomass. A joint-venture with a Korean company, MH Bio-Energy Group (ADB, 2018), produces nearly 36 million liters of bioethanol per year from

100,000 tons of dry cassava flour or 400,000 tons of cassava. Other major sources of biofuels include plantations that produce jatropha (1,000 ha), palm oil (4,000-10,000 ha), and sugarcane (20,000 ha). In addition, the country can theoretically produce as much as 2.4 million cubic meters of biogas per day (See Annex 2). The fact that most agricultural holdings nationwide are small-scale, however, means that the technical potential could be much less than the theoretical (ADB, 2015).

Hydropower: There are sixty-three possible locations across the country for the development of small and large hydropower projects (MME, 2019), fifty percent of which are located in the Mekong River basin, forty percent on its tributaries, and ten percent in the southwestern coastal highlands. Hydropower generation varies according to seasonal changes such that it is higher in the rainy season than in the dry season. The country's technical hydropower potential is estimated to be about 10,000 MW. Currently, only 980 MW of this capacity has been developed, 400 MW is under construction, and 90 MW is in the feasibility-study stage (ADB, 2018).

Solar energy: The country has a considerably high level of solar irradiation, rendering it rich in solar energy resources. The daily average solar irradiation is around 5 kilowatt hours (kWh), with daily average sunshine of 6-9 hours (MME, 2019). The Global Horizontal Irradiation (GHI) is valued at 1,644-2,009 kWh per square meter per year, while the Direct Normal Irradiation (DNI) ranges from 1,168 to 1,607 kWh per square meter per year. Most parts of the country have a GHI level of 1,826 kWh per square meter per year or more, and a DNI level of 1,314 kWh per square meter per year or more (See Annexes 3 and 4). The total suitable land area for solar photovoltaics development is 134,500 square kilometers, which corresponds to a maximum technical solar energy potential of around 8.1 megawatt-peak (MWp) or 12 terawatt hours (TWh) per year and a maximum economic solar energy potential of around 7.2 MWp or 10.8 TWh per year (see Annex 5) (ADB, 2015).

Wind energy: The country has limited wind energy potential as only 3-5 percent of its total land area has wind resources of medium intensity viable for utility-scale wind turbines. The annual average wind speed is between 6-9 meters per second in the southern regions of Tonle Sap Lake, the southwestern mountainous areas, and along

the coast (ADB, 2018). The theoretical wind potential is 65 gigawatts, with a potential production capacity of 154 TWh per year (See Annex 5). The technical wind potential, on the other hand, is 18 MW at the lower limit (5 percent of grid capacity) and 72 MW at the upper limit (20 percent of grid capacity) (ADB, 2015).

2.2.2 Primary Energy Supply

The country's domestic energy only includes hydropower and biomass as its other energy resources remain unutilized. To meet the local energy needs, it imports sub-bituminous coal primarily from Indonesia (GGGI, 2018); petroleum from Singapore, Thailand, and Vietnam; and electricity from Laos, Thailand, and Vietnam. Its import dependency ratio¹³ was 68 percent in 2019, rising from 35 percent in 2000. Greater reliance on energy imports could pose a greater threat to its energy supply security (ERIA, 2022).

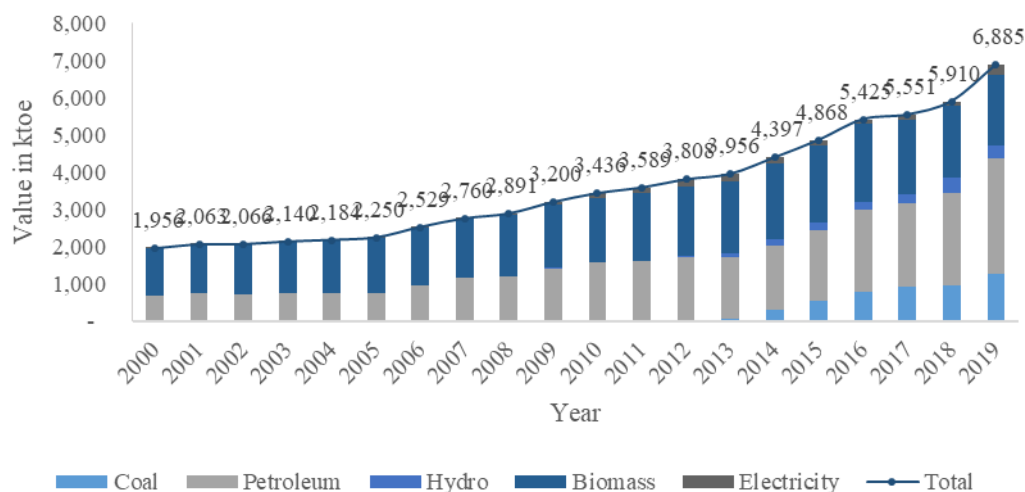
The total primary energy supply (TPES)¹⁴ increased from 1,956 kilotons of oil equivalent (ktoe) in 2000 to 6,885 ktoe in 2019. In the same period, the supply of petroleum increased from 670 ktoe to 3,104 ktoe, that of hydropower from 3 ktoe to 346 ktoe, that of biomass from 1,279 ktoe to 1,902 ktoe, and that of imported electricity from 4 ktoe to 263 ktoe. Coal began to be imported in 2008, rising from 14 ktoe in that year to 1,270 ktoe in 2019. Petroleum now makes up around 45 percent of TPES, replacing biomass as the most dominant energy source. The steeper slope of the TPES curve over time indicates that supply is increasing faster than before. Its growth rate was almost two times higher in 2010-2019 than in 2000-2009 (See Figure 4-2).

This trend toward growth of the primary energy supply can be explained by the growing need for energy to support the rapid economic development of Cambodia, which is seeing a greater influx of foreign direct investments in the construction of commercial buildings, infrastructures, and economic zones (ERIA, 2022).

¹³ The import dependency ratio measures a country's reliance on energy imports. It is equal to energy imports divided by the sum of energy production and energy imports.

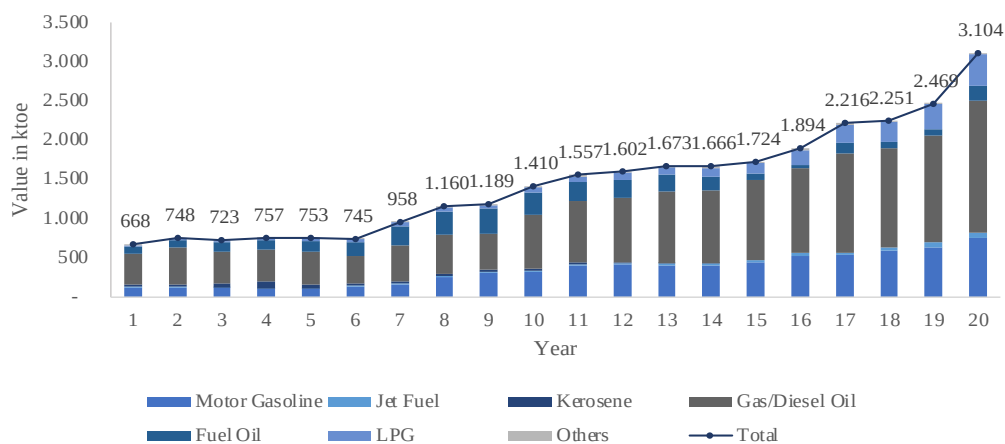
¹⁴ From the energy balance table, the total primary energy supply shows the amount of energy available within a country. It is equal to the sum of indigenous production, imports, exports, international marine bunkers, international aviation bunkers, and stock changes.

[Figure 4-2] Growth rate of energy consumption



The petroleum products supplied to Cambodia include motor gasoline, jet fuel, kerosene, gas or diesel, fuel oil, liquefied petroleum gas (LPG), and others. In 2000, gas or diesel oil accounted for the largest share of supply (59%), followed by motor gasoline (19%), fuel oil (14%), kerosene (5%), LPG (2%), jet fuel (1%), and others. The supply of kerosene became obsolete from 2014 onwards. In 2019, gas or diesel oil still accounted for the largest share of the total petroleum supply (54%), while motor gasoline, LPG, fuel oil, and jet fuel accounted for 24%, 13%, 6%, and 2%, respectively (See Figure 4-3).

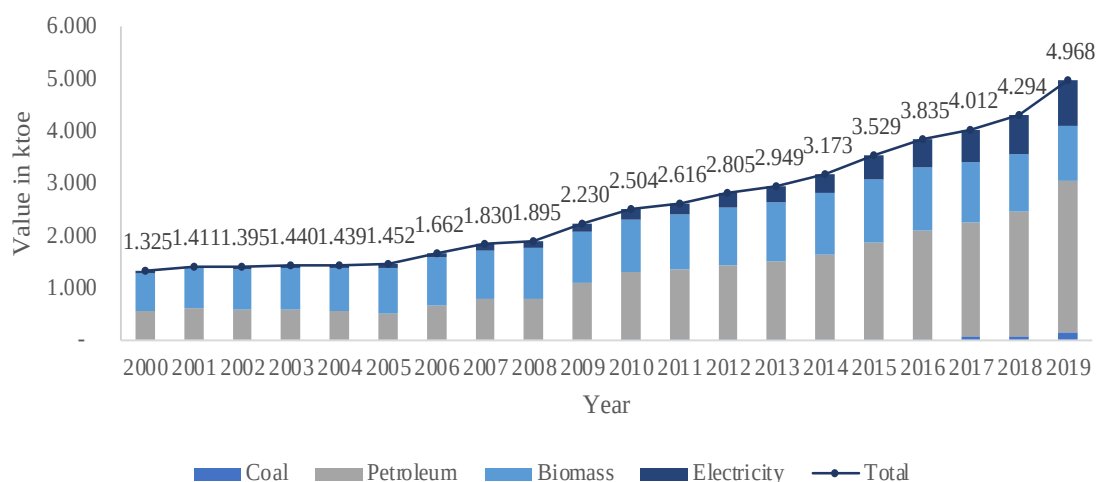
[Figure 4-3] Growth rate of petroleum supplies



2.2.3 Final Energy Consumption

The country's total final energy consumption (TFEC)¹⁵ stood at 4,968 ktoe in 2019, showing a dramatic, almost fourfold increase from the 1,325 ktoe consumed in 2000. Petroleum accounted for the largest share at 2,899 ktoe, followed by biomass at 2,053 ktoe, electricity at 879 ktoe, and coal at 137 ktoe. In the last decade, coal consumption grew at the fastest rate, i.e. 55% per year on average, followed by electricity (18%), petroleum (10%), and biomass (2%) (See Figure 4-4).

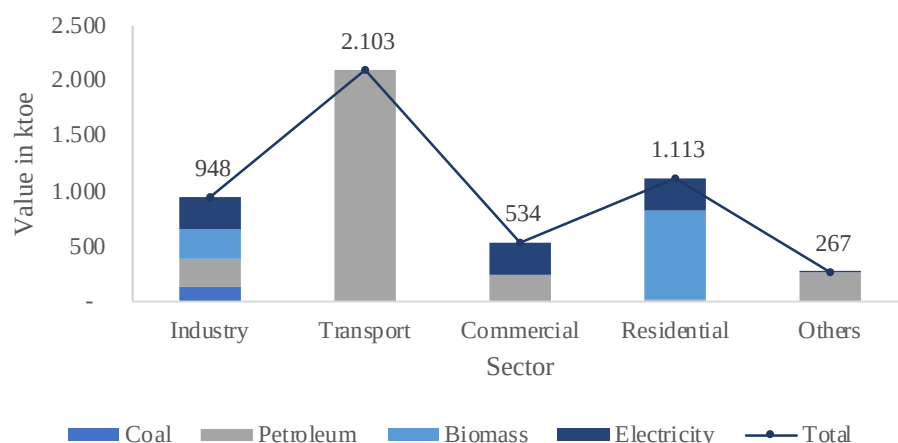
[Figure 4-4] Cambodia's energy consumption



In 2019, the transport sector showed the highest final energy consumption (2,103 ktoe), followed by the residential (1,113 ktoe), industrial (948 ktoe), commercial (534 ktoe), and other sectors (267 ktoe). The residential sector was the major consumer of biomass, whereas the consumption of petroleum was mainly concentrated in the transport sector, and that of electricity was almost equally distributed across the industrial, commercial, and residential sectors. As for coal, it was only consumed in the industrial sector (See Figure 4-5).

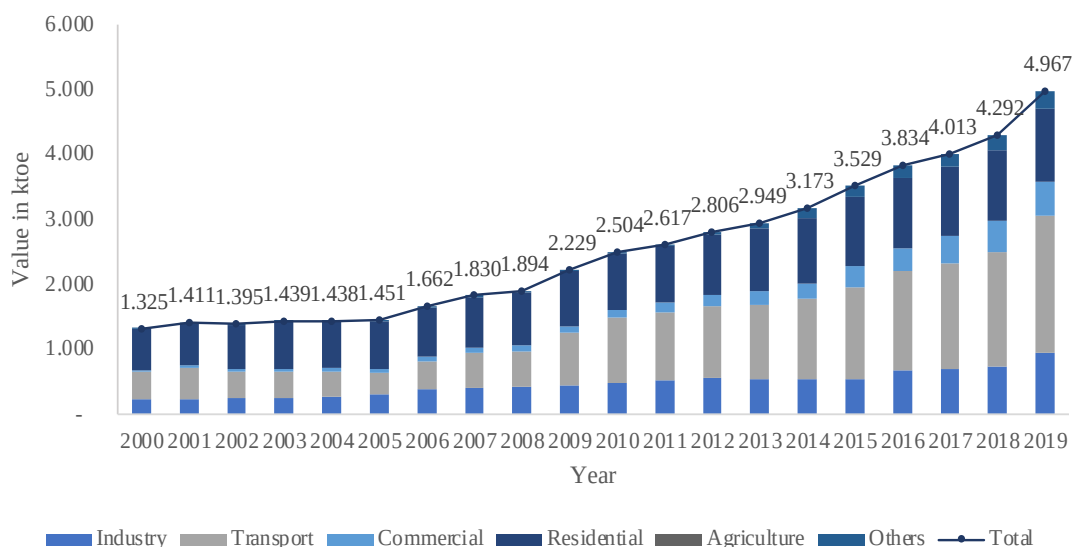
¹⁵ From the energy balance table, total final energy consumption shows how energy is consumed by the final sectors.

[Figure 4-5] Energy consumption in various sectors



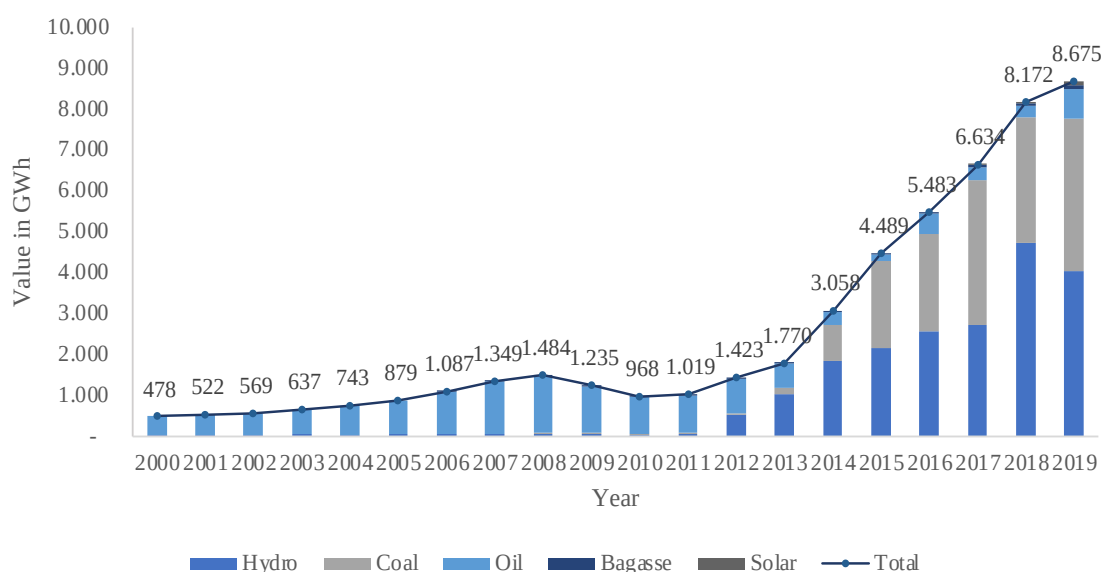
The rise of the TEEC during the period 2000-2019 was largely driven by the expansion of the industrial, transportation and commercial sectors. The average annual final energy consumption growth rates of the industrial, transport, commercial and residential sectors were 8%, 10%, 17%, and 3%, respectively (See Figure 4-6).

[Figure 4-6] Growth rate of TEEC



Besides electricity imports, the country generates electricity from a variety of energy sources, including hydropower, coal, oil, bagasse, and solar power. The total volume of electricity generation rose dramatically from 478 GWh in 2000 to 8,675 GWh in 2019. Oil was the primary source of power generation up until 2013, when more hydro and coal power plants started to operate. By 2019, oil's share of power generation had fallen to just 8%, while hydropower and coal accounted for 46% and 43%, respectively, and bagasse and solar each had a share of 1% (See Figure 4-7).

[Figure 4-7] Shares of energy production sources



2.3 Cambodia's Policy and Regulatory Framework

Energy consumption in Cambodia has been increasing rapidly in recent years. According to the Report on Energy Demand and Supply of Cambodia 2010–2018 (2019)¹⁶, one of the energy research reports produced by the Economic Research Institute for ASEAN and East Asia (ERIA), total final energy consumption (TFEC) grew at 7.2% per annum in the period 2010-2018. However, its elasticity over gross domestic

¹⁶ Ministry of Mines and Energy, Cambodia and ERIA (2019), Energy Demand and Supply of Cambodia 2010–2018. Jakarta: ERIA. Available at: <https://www.eria.org/publications/energy-demand-and-supply-of-cambodia-2010-2018/>

product (GDP) was largely attributable to the 7.2% growth rate of GDP recorded in the same period. The TFEC includes biomass, which accounts for a significant share of the TFEC in Cambodia (biomass's share of the TFEC was still 25.5% in 2018). During the period 2010-2018, the growth rates of commercial energy, such as oil and electricity, were 8.1% and 18.3% per year, respectively, i.e. much higher than the TFEC. In the future, biomass's 25.5% share of the TFEC will surely be replaced by commercial energy sources such as oil and electricity. Therefore, Cambodia needed to formulate and implement energy efficiency and conservation (EEC) policies and programs to limit and check its commercial energy consumption. Unfortunately, no such EEC program had been formulated and, consequently, the General Department of Energy (GDE), the Ministry of Mines and Energy and the ERIA had to come up with the EEC Master Plan for Cambodia. The following five EEC policies and programs were focused on the following: (i) energy service company (ESCO), (ii) fostering of energy managers, (iii) standard and labelling system, (iv) education and campaigns, and (v) preparation of energy efficiency indicators (EEIs). ERIA has also prepared a 5-year road map (2020–2025) for each EEC policy and program mentioned above.

Similarly, the government's Industrial Development Policy (IDP) recognizes that insufficient coordination and investment in physical infrastructure, including electricity and clean water, are holding back the development of manufacturing and other value-added industries. In line with the IDP's 2018 target, the government has reduced the price of electricity for specific industrial zones and expanded the transmission networks to improve the reliability of supply. The IDP has also called for a review of the forecast for long-term electricity demand and energy supply options so as to better align them with the country's goals for economic and industrial development. Cambodia does not yet have a comprehensive energy strategy; rather, the government has adopted, or is preparing to adopt, various sector-oriented strategies and action plans. The MME's PDP, which was prepared in 2007 for the period 2008-2020 and then updated in 2015 for the period 2015-2030¹⁷, includes plans to invest in the expansion

¹⁷ Chugoku Electric Power Co., Inc. 2015. The Project for Revision of Cambodia's Power Development Master Plan. Presentation prepared for the Government of Cambodia, Phnom Penh, September, Unpublished.

of power generation and transmission. With regard to the need to expand power distribution and rural electrification, the MME, in cooperation with the Electricity Authority of Cambodia (EAC) and REF, sets the goals.

The Royal Government of Cambodia established the Rectangular Strategy Phase IV for National Growth, which was implemented during the sixth term of the National Assembly, 2013–2018¹⁸. The National Strategic Development Plan (NSDP), 2018–2023, the primary document guiding, was endorsed by the National Assembly on June 26, 2014¹⁹. The NSDP aims to meet the Cambodia Millennium Development Goals,²⁰ integrate Cambodia into the economic community of the Association of Southeast Asian Nations, reduce poverty to about 13% by 2018, and achieve the long-term goal of making Cambodia an upper middle-income country by 2030. Regarding the energy sector, Cambodia's National Energy Sector Development Policy, established in 1994²¹, stipulates that the government's main goal shall be to establish an adequate, affordable, and sustainable energy supply to support national economic development.

The NSDP, 2014–2018, emphasizes that although progress has been made in such areas as expanding the electricity supply, reducing tariffs, and strengthening the relevant institutions, further development of the energy sector is essential to increase national competitiveness and promote sustained economic growth. The NSDP includes the following eight key energy policy objectives:

- Expand the capacity to generate low-cost and high-tech electricity, especially from new and clean energy sources, and continue to develop relay networks at all levels in order to strengthen energy security and ensure high-efficiency, high-quality, safe,

¹⁸ Government of Cambodia (2013) Rectangular Strategy Phase III, 2013–2018, Phnom Penh. The government has drafted the new Rectangular Strategy Phase IV, 2018–2023, which will guide the development of the NSDP, 2019–2023.

¹⁹ Government of Cambodia (2014) National Strategic Development Plan, 2014–2018, Phnom Penh.

²⁰ The United Nations Sustainable Development Goals came into effect in January 2016, replacing the Millennium Development Goals, which were to be achieved by 2015. Cambodia is currently developing localized Cambodia Sustainable Development Goals, which will be integrated into future plans and policies, including the NSDP, 2019–2023.

²¹ Government of Cambodia (2015) Cambodia Industrial Development Policy, 2015–2025: Market Orientation and Enabling Environment for Industrial Development, Phnom Penh.

reliable, and affordable electricity supply and distribution and respond to development needs.

- Provide further incentives for the private sector to invest in the electricity generation, transmission and distribution infrastructure, with the focus on boosting technical and economic efficiency and mitigating environmental and social impacts.
- Increase the implementation of the electricity strategy to meet the goal of ensuring that all villages in Cambodia have access to electricity from the national grid or other sources by 2020.
- Provide additional support for the Rural Electrification Fund (REF) to help achieve equitable electricity access for the entire population with funds from Electricité du Cambodge (EDC) and other government budget allocations, as well as funding support from development partners.
- Establish rational measures for the use of electricity by reducing energy costs during part-time hours to serve production and irrigation systems, in order to increase agricultural productivity and accelerate the development of industry and handicrafts.
- Step up exploration and commercialization of the oil and gas sector, which has enormous potential for ensuring energy security and will provide valuable resources for Cambodia's economic development in the long term.
- Carry out institutional capacity building, and human resource/energy sector planning and management.
- Promote active involvement in energy cooperation under the Greater Mekong Sub-region GMS regional framework.

Apart from generally encouraging the use of renewable energy and setting aspirational goals for rural electrification through renewable sources, Cambodia has not yet set a clear national renewable energy target. However, a National Policy, Strategy, and Action Plan for Energy Efficiency in Cambodia was developed by consultants funded under the EU Energy Initiative Partnership Dialogue Facility, and it was expected to be

adopted by the end of 2018²². A strategy and action plan for the promotion of renewable energy, including a target percentage of renewable energy in the energy mix, was to be undertaken as part of the MME'S CCAP, 2016–2018²³. In addition, Cambodia's Nationally Determined Contribution targeted a 24% reduction in emissions from renewable energy and energy efficiency initiatives²⁴. Preparation of the Nationally Appropriate Mitigation Actions required to meet this target and establish an emissions management approach for the sector were also part of the MME's CCAP.

Moving forward, the government will need to formulate a consolidated energy sector strategy and implementation plan that specifies and assigns sector priorities and timelines for the agencies responsible for different initiatives and activities. Consolidating the various thematic strategies and plans should facilitate more organized and efficient development of the energy sector, including the development of physical infrastructures and the required legal, regulatory, and institutional frameworks.

In a like manner, Cambodia has embraced the new 2030 Agenda for Sustainable Development. Eradicating poverty in all its forms and dimensions is at the core of the new 2030 Agenda adopted in September 2015 by 193 world leaders at the UN Sustainable Development Summit held in New York.

The benefits in terms of energy efficiency of Sustainable Development Goal 7 are as follows:

- Access to energy: Energy efficiency can increase the services delivered by each kilowatt of electricity and improve energy access.
- Air quality: Energy efficiency can reduce both indoor and outdoor concentrations of air pollutants.

²² A. M. Bassi and S. Rey (2017) National Policy, Strategy, and Action Plan for Energy Efficiency in Cambodia, Eschborn, Germany: EU Energy Initiative Partnership Dialogue Facility. This is an update to the 2013 draft version of the same name.

²³ Government of Cambodia, MME (2015) Climate Change Action Plan for Mines and Energy Sectors, 2016–2018, Phnom Penh.

²⁴ Government of Cambodia (2015) Cambodia's Intended Nationally Determined Contribution, Phnom Penh.

- Asset values: Energy efficiency can increase asset values for homeowners, businesses and utilities.
- Economic benefits: Cost-effective energy efficiency improvements can have positive macroeconomic impacts, thus boosting economic activity.
- Emissions savings: Energy efficiency reduces GHG emissions, including direct emissions from fossil fuels, and indirect emissions from electricity generation.
- Employment: Energy efficiency can induce job creation, improve productivity and decrease employee absenteeism.
- Energy prices: Energy efficiency can lower energy prices by reducing the need to build new power generation or transmission capacity and by reducing pressure on energy resources.
- Energy savings: Energy efficiency reduces the amount of energy used to provide a service.
- Energy security: Energy efficiency can reduce the reliance on energy imports and reduce the risks of supply interruptions.
- Health and wellbeing: Energy efficiency supports physical and mental health with healthy temperature, humidity and noise levels and good air quality.
- Household savings: Energy efficiency can ensure a higher disposable income by lowering energy bills and other household costs.
- Productivity: Energy efficiency leads to productivity gains by lowering maintenance issues and optimizing processes.
- Public budgets: Energy efficiency delivers financial benefits to public budgets through increased income and decreased expenses.

2.4 Development of the National Energy Technology Roadmap

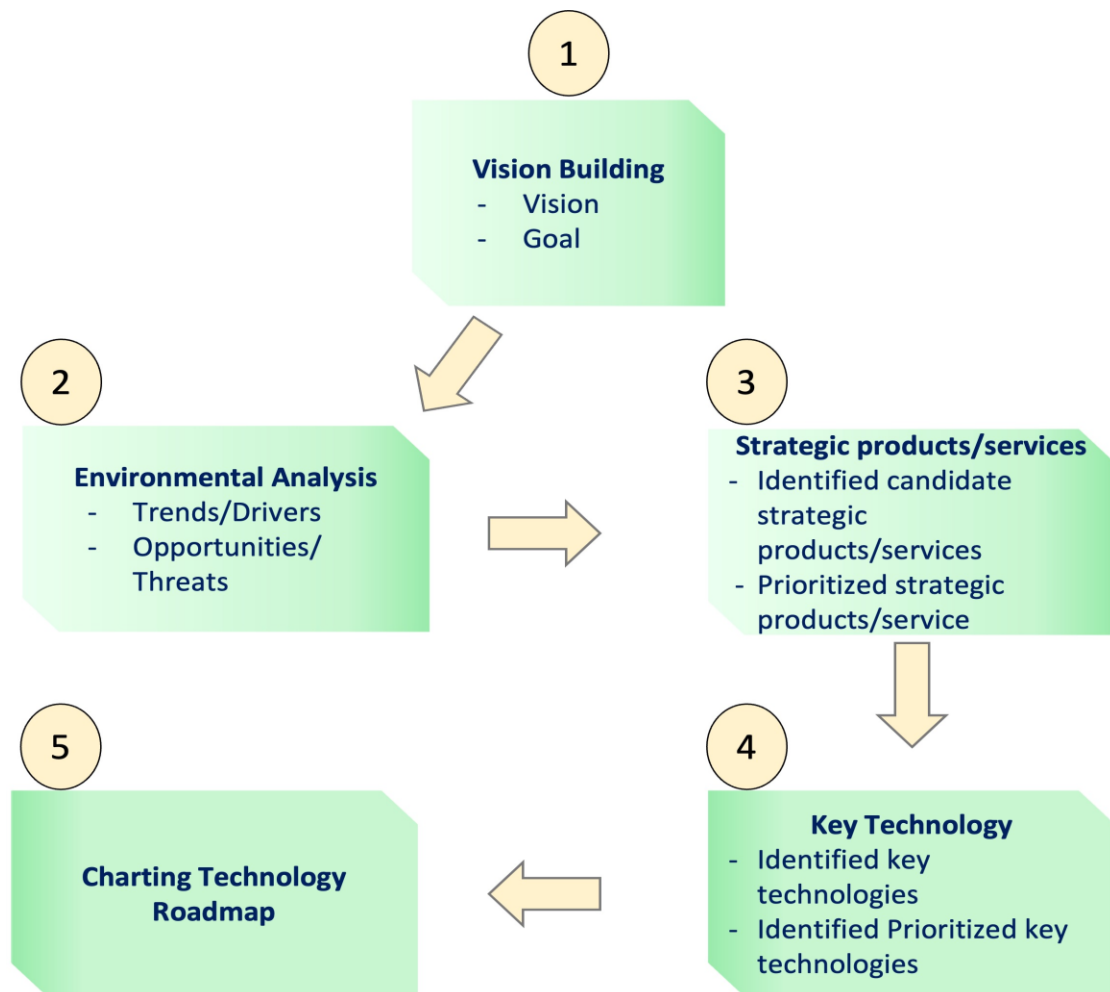
This section describes the process of developing the Energy Technology Roadmap (Energy Tech Roadmap), a document designed to serve as a guideline for the development of key technologies to be implemented by 2030, in order to achieve the co-development consensus-building vision. Additionally, this roadmap gives government, private sector, and academic institutions investment priority and sets the guiding principles for the technologies listed in this roadmap. According to background knowledge, the primary objectives of the Energy Tech Roadmap are as follows:

- Establish a technology roadmap to support the implementation strategy for the National STI Policy 2020-2030.
- Provide a strategic direction for resolving the difficulties facing energy technology.

The following steps make up the process of development of the Energy Tech Roadmap:

- 1- The first step consists in defining the vision and goals for Cambodia's energy sector over the next eight years, as well as the essential technologies required to develop the sector globally, and more particularly aims to contribute to and help accelerate national economic growth.
- 2- The second step consists of an environmental analysis that takes into consideration social, technological, economic, environmental and political concerns, and then, depending on the drivers, an evaluation of the opportunities and threats in the energy sector.
- 3- The third step aims to determine potential functions, strategic products and services. Then, the expert committee scores the priority products and services to determine the final selection.
- 4- The fourth stage consists in identifying key technologies. The expert committees then rank the most important technologies and choose the priority key technologies.
- 5- Finally, a technology roadmap is created with timeframes ranging from the short to the long term.

[Figure 4-8] Development process of the National Energy Technology Roadmap



2.5 Development of the Energy Technology Roadmap

2.5.1 Vision and goals

This Energy Technology Roadmap has been developed to achieve the following vision:
“To become an energy self-sufficient nation by deploying technology and innovation, and leveraging renewable energy sources for sustainable development by 2030.”

In order to achieve this vision, the roadmap sets out the following goals:

1. Increase the domestic energy generation capacity with the support of science, technology and innovation.
2. Increase renewable energy’s share of national energy consumption and its storage capacity for sustainable development.
3. Leverage energy efficiency for domestic and industrial purposes.

2.5.2 STEEP analysis

To realize the vision and goals set out in the above section, a STEEP analysis was conducted to effectively identify the main drivers as well as the strategic products necessary for the development of the roadmap. The STEEP analysis looked into Social, Technological, Economic, Environmental and Political aspects, and subsequently identified the opportunities and threats based on Cambodia’s context. Table 1 depicts the STEEP analysis and opportunities and threats in Cambodia.

Vision	“To become an energy self-sufficient nation by deploying technology and innovation and leveraging renewable energy sources for sustainable development by 2030”
Goals	1. Increase domestic energy generation capacity with the support of science, technology and innovation. 2. Increase renewable energy’s share of national consumption and its storage capacity for sustainable development. 3. Leverage energy efficiency for domestic and industrial purposes.

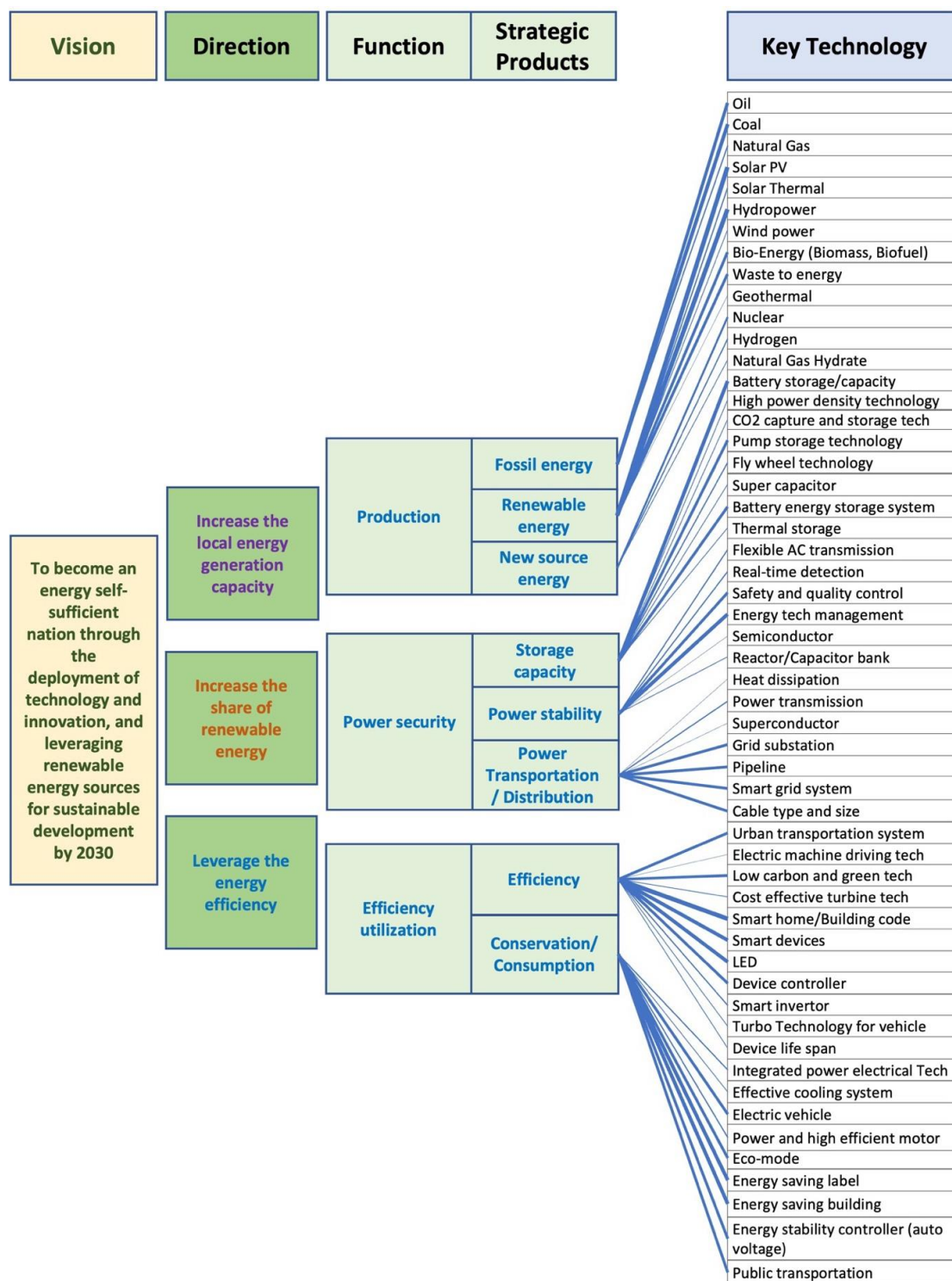
Drivers	Social: - Young population - Population growth - Increasing energy use - Community involvement or public acceptance - Limited supplies - Covid-19 - Create & increase jobs	Technological: - Clean energy - Solar energy/sunlight - Digital control - Biomass - Remote sensing - Automation - Energy storage systems - R&D on new inventions - Wind energy - Internet of Things - Big data analysis - Artificial Intelligence (AI) - Robotics - Mobile technology - Sensor technology - Increasing battery storage prices - Smart homes - Tech startups - Technology Transfer - Hydropower - Grid development	Economical: - Growing economy - Productivity increase - Large Enterprise - Community development and fund - Investment attraction - Financial support - Fuel prices - Oil crisis - Competition with foreign firms - Price instability - Investment in the energy industry	Environmental: - Environmental pollution - Climate change - Resource dependence - Production environment impact - Natural disasters (response system) - Vulnerability - Environmental concerns	Political: - Political stable - Support (technical and financial) from the government - Political and geographical crises - War causing price instability - Partnerships and cooperation - Political will and commitment
Opportunities & Threats	Opportunities - Renewable energy sources (hydropower, solar energy, biomass energy) - Fast pace of technological development - Smart home - Political will and commitment - Attraction of investment in the energy industry - Partnerships and cooperation - Financial support - Community development and funding - Capacity building - Grid development		Threats - Climate change - Political and geographical crises - Oil crisis - Increasing battery storage prices - Production environment impact - Natural disaster (response system) - Limited supplies - Resource dependence - Vulnerability - Community involvement or public acceptance - Energy security - Environmental concerns - War causing price instability - Covid-19 (3) - Competition with foreign firms - Price instability - Fuel prices		

2.6 Key Technology

Figure 8 shows the structure of the Energy Tech Roadmap, including its vision, direction, functions, strategic products and key technologies. The Roadmap is oriented in three main directions that are necessary to support the vision and achieve the goals by 2030: increase domestic energy generation capacity with the support of science, technology and innovation; increase renewable energy's share of national consumption and its storage capacity for sustainable development; and leverage energy efficiency for domestic and industrial purposes. Strategic products and services were determined within these three directions by the committee members, along with the drivers identified by the STEEP analysis, eight of which were chosen as priorities. The three directions include increasing domestic energy generation, increasing renewable energy's share of national energy consumption, and leveraging energy efficiency, as well as eight strategic products and services related to fossil energy, renewable energy, new sources of energy, storage capacity, power stability, power transportation/distribution, efficiency, and conservation/consumption. As shown in Figure 8, each key technology serves to the strategic product; also, the thicker the line, the more important the strategic product's key technology.

The fifty-four important technologies serve to support the potential strategic products and services, as shown in Figure 8. To emphasize its importance, each key technology is given a score based on its strategic importance, economic impact, potential for success in terms of the vision, and its viability in the short, medium, and long terms. Given that they are regarded as priority key technologies, the selected key technologies are included in Figure 9 (Charting).

[Figure 4-9] Key technologies



2.7 Charting the Technology Roadmap

Figure 4.10 below shows the chart of the Energy Technology Roadmap, which depicts the key technologies illustrated in Figure 4.9 above. To fully realize this vision, concerted efforts must be focused on making changes to the energy structure with the focus on six main categories, i.e. Energy Production, Power Security and Stability, Power Storage, Energy Transportation and Distribution, Energy Efficiency and, last but by no means least, Energy Conservation. In the short term, Cambodia will prioritize energy self-sufficiency in order to achieve national energy security and ensure that reliable energy coverage covers every inch of the country. In the medium term, more attention should be paid to renewable energy sources such as solar, hydro, and wind power, all of which are clean, environmentally-friendly forms of energy.

Up to the present time, fossil fuel have remained the main source of energy production. The use of petroleum, natural gas and coal as energy sources dates back several centuries. Likewise, Cambodia has also been heavily reliant on fossil fuels for its energy consumption. Hence, in the short term, energy from oil will still play an important role as an energy source for consumption. However, concerns about energy consumption centered on fossil fuels have increased in recent decades - especially concerns over the global depletion of fossil fuels in the near future, and the possibility of an increasingly serious energy crisis threatening the economic well-being of every country in the world. In addition, the large-scale development and utilization of fossil fuels poses severe problems for the environment, one of which is climate change coupled with ecological deterioration, which in turn directly affects human health and the sustainability of humankind. To this end, clean and renewable, pollution-free energy sources have been introduced and adopted as strategic energy plans by many countries around the world in a drive to replace conventional fossil fuel-based energy sources. As with the favorable condition, Cambodia should embrace the hydropower in the short and medium term, while solar power should be adopted under the long-term energy strategy in order to achieve Cambodia's sustainable development goal and

transform it into a carbon-neutral country.

To achieve the realized energy vision, it is crucial to ensure the stability and security of energy operation, particularly for long-distance transmission. Cambodia must constantly deploy and develop technologies to achieve the optimally effective power grid for energy management, as well as safety and quality control in the short to medium term. As regards the medium to long term, it is necessary to focus on real-time perception and detection technologies as they can monitor the power grid situation, including abnormalities, and transfer data instantly via digital and information technologies to centralized and decentralized control centers or directly to the users. Flexible transmission technology can also improve the stability and capacity of energy transmission over long distances, thereby enhancing the quality and stability of the energy supply.

Power storage is another aspect that must be taken into account. Because renewable sources are scattered across many areas and their energy generation capacity is not available at all times (for example, solar energy can only be produced in the daytime), storage technology is vital to ensure the stability and security of renewable energy sources. Battery energy capacity is also strategically important for storing surplus energy and can be used as an alternative source of energy generation when needed. It is also necessary to adopt high-power density cell capacity in the short to medium term, while pump storage technology could also be used in the hydropower system.

Effective transportation and distribution must be improved to ensure energy security and stability because long-distance transmission can pose problems, such as energy leakages and losses, and renewable energy sources like solar and hydro power require transportation from a suitable area to cities and urban areas. Grid substations and a smart grid system must be put in place in the short and medium term and in the medium to long term, respectively. Meanwhile, power transmission and super-conductive power technologies could be used to enhance energy and power

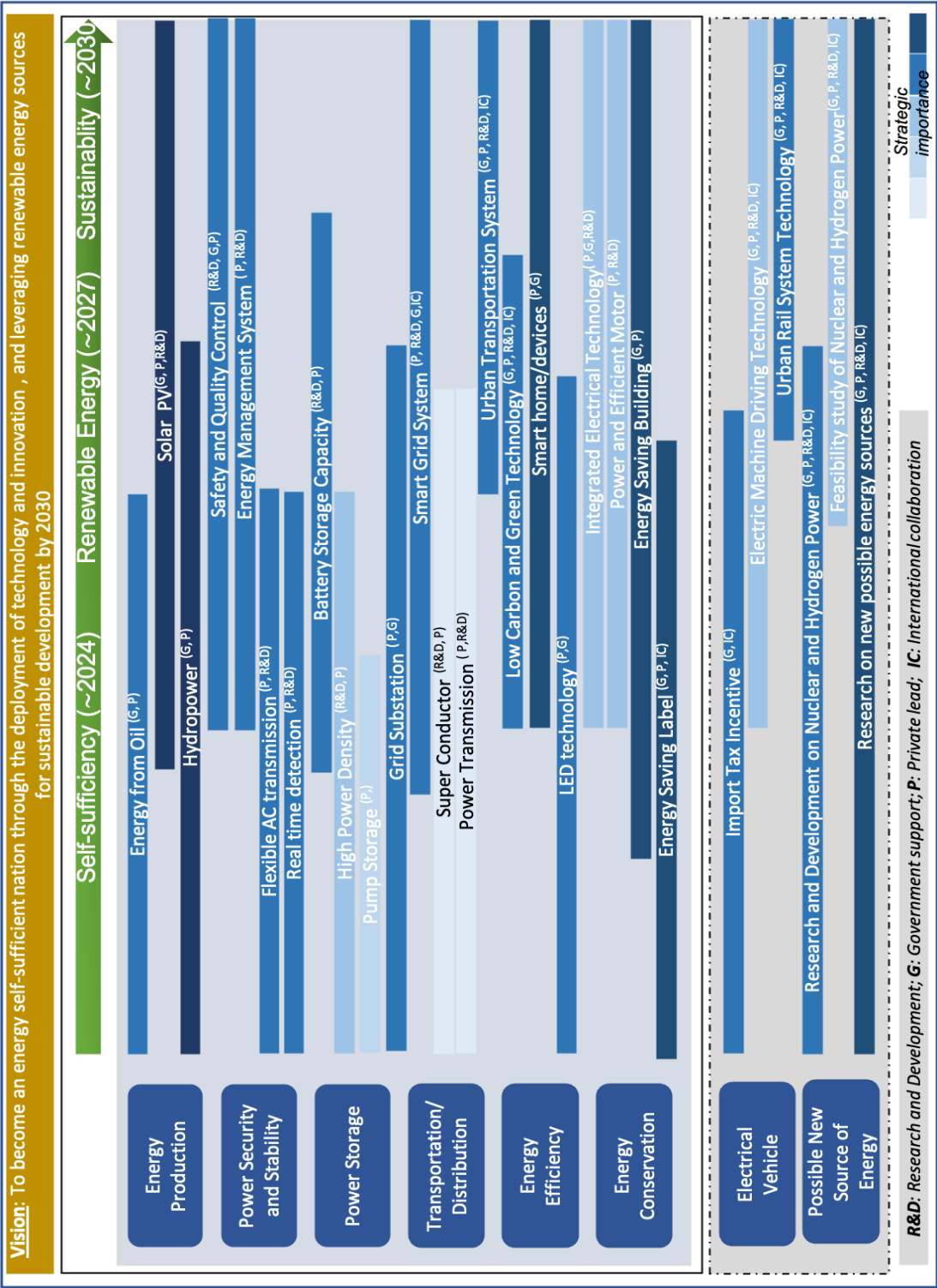
transportation and distribution.

Energy efficiency is another aspect that needs to be included in the strategic plan for achieving energy security and sustainability. Smart home and smart devices powered by sensor, IOT, information, AI and other technologies could help enhance the efficiency of energy utilization in homes and public buildings as they minimize energy wastage due to unnecessary usage, and can deliver energy and power on time and be immediately disconnected when unnecessary. Technologies like LED (Light Emitted Diode) could save a lot of energy and provide better quality in housing and industrial consumption. Low carbon and green technologies can not only improve the efficiency of energy consumption, but they also are environmentally-friendly and can help realize the goals of sustainable development and carbon neutrality. Urban transportation could also help reduce energy consumption on the one hand, and reduce carbon emissions on the other. As such, these technologies have been deployed in the strategic roadmap.

Energy conservation cannot be disconnected from energy efficiency. To achieve sustainable development, integrated electrical technology should be deployed to replace fossil fuel consumption entirely, and power and high effect moto technology could replicate and even go far beyond the capacity of traditional fossil fuels given the speed of development of the relevant technologies in recent years. Energy saving buildings are another effective strategy that could be implemented to conserve energy. Energy saving labels are also very effective, and are used worldwide for energy conservation as they can provide accurate information to consumers about their energy usage. Energy labels coupled with governmental incentivizing schemes are one of the most - if not the most - effective ways of promoting energy conservation in a country.

Two additional directions for the energy roadmap have been proposed due to the current global mega trend, namely, Electric Vehicles (EV) and New Possible Sources of Energy. Electric vehicles are becoming very popular as a means of reducing the reliance on, if not completely replacing, fossil fuel-powered vehicles as they not only bring great savings, but are also environmentally-friendly. Import tax incentives, together with efforts to promote electric machine driving technology, could guarantee the success of electric vehicles. Searching for possible new sources of energy could solve the energy problems facing Cambodia today and accelerate the accomplishment of the energy strategy and vision. Finally, nuclear and hydrogen power are clean new sources of energy production that should not be completely ruled out without first undertaking a thorough feasibility study.

[Figure 4-10] Chart of the Energy Technology Roadmap



2.8 Conclusion

Energy is a mandatory resource that is required for human survival and societal advancement. Energy is not only used for economic development, but it is also crucial from a political and strategic perspective. No country, without internal energy security, can develop fully if it relies on energy sources from other countries. Thus, the development of energy production has a direct impact on the development and security of a nation. A nation's economic and social development require increasingly huge amounts of energy each year as its civilization develops. The lack of energy hampers urban and economic growth as well as population expansion.

Due to the possibility of a severe energy crisis and concerns about the depletion of fossil fuels, the energy structure has been modified over time: Until relatively recently it was dominated by fossil fuels such as petroleum, natural gas and coal, but it has now switched to renewable energy sources including solar, wind, ocean, biomass, and geothermal energy, and potential new energy sources such as nuclear and hydrogen. This pattern could lead to a significant evolution of energy technology. As such, science and technology are key factors in the advancement of energy technology that will in turn bring about radical changes in socioeconomic growth. Throughout the history of humanity economic development has been associated with energy transitions ranging from human labor to steam engines to electricity to what we see around us today. Human civilization has gradually transformed from an agrarian society into an industrial civilization thanks to an energy evolution based on scientific and technological progress.

Cambodia must have a clear strategic plan such as this roadmap in order to utilize science, technology and innovation to ensure energy security for its prosperous economic development in the future as well as sustainable and inclusive growth. This energy technology roadmap has identified the technologies that Cambodia needs to adopt for its future development in order to realize the vision of becoming an energy self-sufficient nation by deploying technology and innovation and leveraging renewable energy sources for sustainable development by 2030, based on an

evaluation of the current condition, strengths, weaknesses, opportunities and threats aimed at identifying the drivers, and in turn, determining the strategic products and functions within a clear and specific timeframe.

Lastly, but perhaps most importantly, listed below are a number of policy implications that need to be considered in order to achieve energy security:

- Government policies and incentivized schemes are mandatory to meet the need for energy security and sustainable development.
- Progress in Science, Technology and Innovation (STI) is crucial for national development in all sectors, particularly the energy sector. Hence, investing in STI is indispensable for Cambodia to become an energy self-sufficient and sustainable nation.
- Talents play a central role in this energy roadmap as well as in other energy strategic plans yet to be realized. Investing in the development of human capital is of great importance for the long-term achievement of the energy vision and goals.
- It is essential to invest in and prioritize research and development on energy related areas as it can strengthen scientific-based and technological capabilities and bring out innovations.
- The training and education system for energy science and technology capability must be promoted in order to keep up with the dominant trends of the global energy sector.
- Collaboration is a key factor in promoting knowledge creation and technology transfer, as well as solutions to tackle society's need for energy security and sustainable development.
- Studies on energy strategies and policies should be continued in order to exploit the national innovation capabilities.
- Good governance and coordination among key stakeholders in the ecosystem is the key to success and should be compulsory.

To conclude, a monitoring and evaluation framework should be put in place to achieve optimal results, and this roadmap should be reviewed and upgraded every three to five years to ensure that the direction of energy technology development leads to the achievement of Cambodia's vision and goals.

References

- ADB (2018) Assessment of Cambodia's Strategy and Road Map for the Energy Sector, Manila: ADB.
- Economic Research Institute for ASEAN and East Asia (ERIA) (2022) *Cambodia Energy Statistics 2010-2019*.
- ADB (2018) Assessment of Cambodia's Strategy and Road Map for the Energy Sector, Mandaluyong: ASIAN DEVELOPMENT BANK.
- ADB (2018) CAMBODIA ENERGY SECTOR ASSESSMENT, STRATEGY, AND MAP.
- ADB, A. (2018) Assessment of Cambodia's Strategy and Road Map for the Energy Sector, Mandaluyong: ASIAN DEVELOPMENT BANK.
- Bank, T. W. (2018) *Tracking SDG7: The Energy Progress Report*. Retrieved from <http://documents1.worldbank.org/curated/en/495461525783464109/pdf/126026-WP-PUBLIC-P167379-tracking-sdg7-the-energy-progress-report-full-report.pdf>.
- EAC (2021) Salient Features of Power Development in the Kingdom of Cambodia until December 2021.
- EAC, E. (2021) *Salient Features of Power Development in the Kingdom of Cambodia until December 2021*, Phnom Penh: Electricity Authority of Cambodia, Ministry of Mines and Energy.
- Development Bank (ADB) (2015) Renewable Energy Developments and Potential in the Greater Mekong Subregion, Manila: ADB.
- Global Green Growth Institute (GGGI) (2018) Green Growth Potential Assessment: Cambodia Country Report, Seoul: GGGI.
- Development Cambodia: <https://opendevdevelopmentcambodia.net/km/topics/coal/>
- Hannah Ritchie, R. M. (n.d.) *Our World Data*. Retrieved from <https://ourworldindata.org/energy-production-consumption>.
- I.E.A. (2015) Energy Access Outlook for 2017: From Poverty to Prosperity. Retrieved from https://www.iea.org/publications/freepublications/publication/WEO2017SpecialReport_EnergyAccessOutlook.pdf.
- I.E.A. (2017) Energy Access Outlook for 2017: From Poverty to Prosperity, World Energy Outlook Special.
- I.E.A. (2019) Access to affordable, reliable, sustainable and modern energy.

- Japan Development Institute (JDI) (2009) Integrated Coal Mine and Power Generation Plant in Cambodia.
- Korkeakoski Mika, M. M.-o. (2021) Situational analysis of energy use and consumption in Cambodia: household access to energy, *Environment, Development and Sustainability*.
- Marabona, Y. (2019) Status and Challenges of Rural Electrification in Cambodia and Renewable Energy Options, Phnom Penh: Parliamentary Institute of Cambodia.
- Mika, K., Minna, M., Noora, V., Jyrki, L., Jari, K.-o., Anna, A., . . . Nicholas, H. (2021) Situational analysis of energy use and consumption, *Springer*, <https://doi.org/10.1007/s10668-021-01443-8>.
- Ministry of Mines and Energy. (2019) *Cambodia Basic Energy Plan*, Jakarta: Economic Research Institute for ASEAN and East Asia (ERIA).
- NIS, N. (2019) Phnom Penh: NIS.
- Open Development Cambodia (ODI)(Nov. 4, 2015). *Coal*.
- Statistics, N. I. (2019) Phnom Penh: National Institute of Statistics.
- The Global Economic*. (2019) Retrieved from Cambodia: Electricity Consumption: https://www.theglobaleconomy.com/Cambodia/electricity_consumption/ retrieved on 20 October 2022 at 10:14 A.M.
- The World Bank. (Oct. 19, 2022) *The World Bank IBRD, IDA*. Retrieved from The World Bank in Cambodia : <https://www.worldbank.org/en/country/cambodia/overview>.
- THE WORLD COUNT*. (n.d.) Retrieved from <https://www.theworldcounts.com/challenges/energy/global-energy-consumption>.
- U.N. (2015) Transforming Our World: the 2030 Agenda for Sustainability.
- U.N. (n.d.) AFFORDABLE AND CLEAN ENERGY: WHY IT MATTERS .
- World Bank, (2021, October 12) *World Bank Data*. Retrieved from <https://data.worldbank.org/country/cambodia>:
<https://data.worldbank.org/country/cambodia>.
- Yang Liu, R. N. (2020) ENERGY EF.

Annex 1: Workshop materials

Technology Group	Sources/Key technologies	Evaluation Criteria		
		Strategic Importance	Economic Impact	Potential for Success
Fossil energy	Oil			
	Coal			
	Natural Gas			
Renewable energy	Solar PV			
	Hydropower			
	Wind power			
	Biomass			
	Geo and solar thermal			
	Cost effective turbine tech			
New Source energy	Nuclear			
	Hydrogen			
	Natural Gas Hydrate			
Storage Capacity	Battery Capacity			
	High power density technology			
	CO2 capture and storage tech			
Power Security /Stability	Flexible AC transmission			
	Real-time detection			
	Safety and quality control			
Power Transportation/ Distribution	Heat dissipation			
	Power transmission			
	Superconductor			
Efficiency	Urban transportation system			
	Electric machine driving technology			
	Low Carbon and green technology			
Conservation/ Consumption	Integrated power electrical tech			
	Effective cooling system			
	Electric vehicle			
	Power and high efficient motor			

Technology Group	Sources/Key technologies	Important level			Acquisition Method
		Short term	Medium term	Long term	
Fossil energy	Oil				
	Coal				
	Natural Gas				
Renewable energy	Solar PV				
	Hydropower				
	Wind power				
	Biomass				
	Geo and solar thermal				
	Cost effective turbine tech				
New Source energy	Nuclear				
	Hydrogen				
	Natural Gas Hydrate				
Storage Capacity	Battery Capacity				
	High power density technology				
	CO2 capture and storage tech				
Power Security /Stability	Flexible AC transmission				
	Real-time detection				
	Safety and quality control				
Power Transportation/ Distribution	Heat dissipation				
	Power transmission				
	Superconductor				
Efficiency	Urban transportation system				
	Electric machine driving technology				
	Low Carbon and green technology				
Conservation/ Consumption	Integrated power electrical tech				
	Effective cooling system				
	Electric vehicle				
	Power and <u>high</u> efficient motor				

3. Tourism

Technology Roadmap for Tourism 2030

3.1 Executive Summary

Technologies play essential roles in society, economy, politics, environment, and technologies. Furthermore, certain technologies became even more important during the Covid-19 pandemic. Tourism is regarded as a sector that is quick to adopt new technology, and certain technologies have contributed to the management and recovery of the tourism industry during and after the Covid-19 crisis. Meanwhile, the Royal Government of Cambodia (RGC) is paying a lot of attention to technology in the development of the country. The RGC has initiated various technology policies and roadmaps for 2030 in different sectors. As a crucial sector in Cambodia's economic development, the RGC has paid particular attention to tourism, and encouraged the development of the "Tourism Technology Roadmap 2030".

This roadmap covers the period 2022 to 2030. Tourism encompasses many sectors, including accommodation, transportation, attractions, tour operators and travel agencies, and entertainment, so the technologies used in these sectors are included in the roadmap. The roadmap has been developed by reviewing the relevant existing literature, interviewing the relevant stakeholders, and holding workshops with subcommittees and relevant stakeholders from various ministries and the private sector, and the data have been subjected to a content analysis.

The vision of the Tourism Technology Roadmap 2030 is to transform Cambodia into

one of the top tourism destinations in the region through the adoption of innovative technologies and to develop a competitive, sustainable, and inclusive tourism industry by 2030. As such, it has the five following five goals: to improve the technology literacy and skills of tourism operators and users; to strengthen support for the country's technology infrastructure; to improve cyber security; to enforce the utilization of advanced technologies to improve the operation of the tourism business in Cambodia; and to promote the adoption of technologies that will transform the tourism industry.

A series of drivers has been identified and examined in terms of society, economy, politics, environment, and technology, and on the basis of these drivers, prioritized tourism products and services and key technologies have been identified.

The roadmap prioritizes the following twelve tourism products and services: virtual tour services, integrated transportation services, green tourism products, innovative tourism experiences, digital transaction services, digital communication, tourism security and safety services, automated and contactless services, digital marketing services, research and innovation, and destination management. However, virtual tour services are regarded as having the lowest priority, while digital transaction services and digital marketing services are the most prioritized ones. Thus, digital transaction and marketing services are highly prioritized and strongly recommended as the focus of tourism development.

Thirty key technologies have been identified to support the development of the prioritized tourism products and services. These technologies are classified into such areas as platforms, apps, devices, networks, biometric security and image recognition, digital map, artificial intelligence, sensing, machine learning, audio and visual, and renewable energy. The key technologies are rated based on their importance and the feasibility of each product and service, and are then identified as 'critical', 'less important', and 'strategic decision needed'.

The roadmap is divided into three periods: short-term (1 to 3 years), medium-term (3-7 years), and long-term (8-10 years). In other words, the three periods are classified

into 2022-2024, 2025-2027, and 2028-2030, and the prioritized products/services coupled with key technologies are prioritized based on these periods accordingly.

Supporting infrastructure should be provided to support the implementation of this roadmap. The infrastructures include human resources, digital payment infrastructure, security infrastructure, collaboration among stakeholders, research and innovation infrastructure, and institutional management and monitoring.

3.2 Introduction

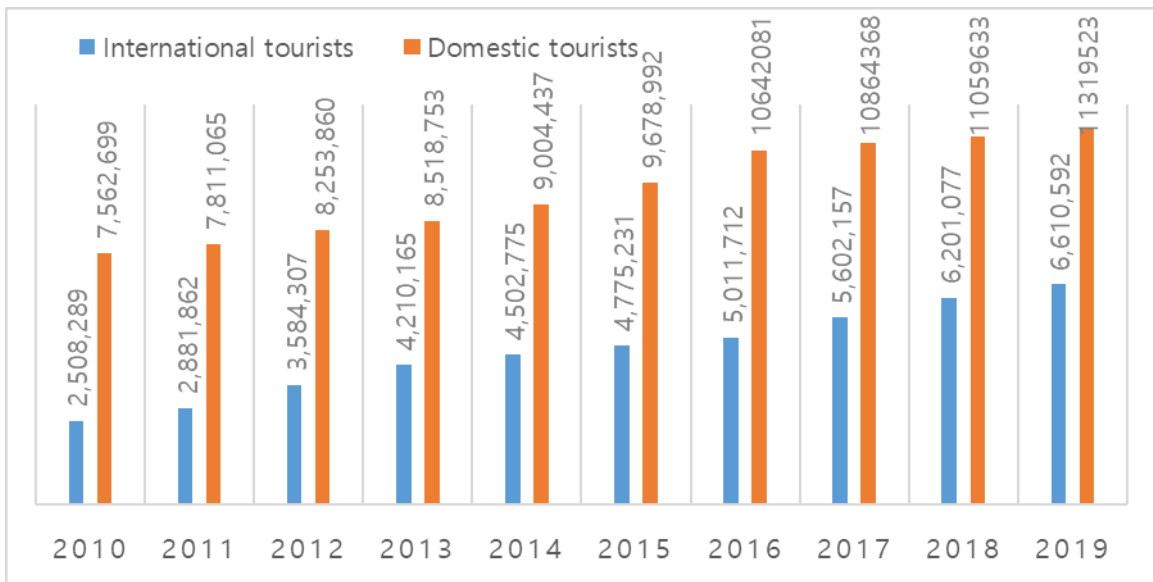
3.2.1 Development of Tourism in Cambodia

Tourism in Cambodia has become remarkably developed in recent times, with the focus on five main destinations. Phnom Penh and its surrounding areas are known as the capital, shopping, and cultural tourism area. The coastal zone, including Kep, Kampot, Preah Sihanouk Ville, and Koh Kong provinces, has been developed for coastal tourism. Cambodia's coasts have been listed as some of the world's best bays. The Siem Reap and Tonle sap areas are known for cultural heritage tourism and ecotourism. Meanwhile, the northeastern area, including Monduliri, Ratanakiri, Kratie, and Stung Treng, is known as an ecotourism area. As such, Cambodia's tourism focuses on nature and culture based on its potential resources. The Royal Government of Cambodia regards tourism as "green gold". In 2019, tourism contributed 12.1 % to the country's GDP and provided 630,000 direct jobs (Royal Government of Cambodia [RGC], 2021). However, the tourism sector was severely affected by the Covid-19 pandemic, and it was predicted that the sector's contribution to national GDP would fall to under 10%. At the same time, 2,838 tourism establishments were closed, and around 510,000 tourism employees were estimated to be at risk of losing their jobs (Ministry of Tourism [MoT], 2021).

The statistics on tourists visiting Cambodia increased yearly before the Covid-19 pandemic. The number of international tourists increased from 2,508,289 in 2010 to 6,610,592 in 2019, at an average growth rate of 12 %, while the number of domestic

tourist arrivals rose from 7,562,699 to 11,319,523, at an average growth rate of 4.62 % in the same period (MoT, 2019, 2021). The average length of stay of international tourists was 6.2 days, with an average expenditure of USD 95.50 per day and a receipt of USD 4,919 million in 2019 (MoT, 2019). However, the statistics on tourists visiting ecotourism sites recorded the lowest figure compared to all other areas. Meanwhile, infrastructure and facilities have been developed to respond to such demand. According to the MoT, Cambodia had 731 hotels with 39,328 rooms, 2,437 restaurants, 682 tour operators and travel agents, 5,377 guides, and 43 airlines as of 2017.

[Figure 4-11] Statistics on Tourists Visiting Cambodia, 20210-20219 (Source: MoT, 2021)



The outbreak of the Covid-19 pandemic in early 2020 has seriously affected tourism in Cambodia. In 2020, the statistics for international and domestic tourists dropped by around 80% and 20%, respectively, compared to 2019 (MoT, 2021). The statistics continued to fall in 2021 due to the restrictions and lockdown regulations related to Covid-19. However, it was predicted that Cambodia's tourism industry would start growing again once the Covid-19 pandemic became more manageable.

With the progress of technology, more digital devices have been used in tourism. The

tourism industry was an early adopter of digitalization business around the world²⁵. Hotels and airlines, for instance, introduced information communication technology (ICT) to their online booking services along with digital payment services including visa cards, credit cards, and other apps. Tourists, on the other hand, use their devices before, during, and after they visit Cambodia. For instance, they use smartphones, iPads/tablets to search for information on Cambodian tourist destinations, accommodation, food and beverages, and transportation before their trip. Then they take photos of their trips, post them on social media, check in to their destination, and use google maps to find the locations of diverse facilities. After their trip, they use digital devices to review their trips and photos, post them on social media, and give feedback on the services of the tourism establishments they have used. This trend is more prevalent among the younger generations, particularly generation Z, which is known to be tech-savvy. For online activities, for instance, university students in Phnom Penh have accounts on Facebook (35.40 %), YouTube (22.30%), Instagram (25.30%), Twitter (7.3%), Linked (4.4%), Blog (1.8%) and others (3.5%) (Leav, 2019). Another study found that Cambodians aged from 21 to 41 use Credit cards (17%), Visa cards (36.9%), Master Card (40%), and Debit cards (3%) to purchase products online (Vireak, 2021).

3.2.2 Government Technology Policy

The Royal Government of Cambodia is increasingly focused on technology and digital development. Its main objectives include laying the foundations for digital adoption in the period 2021-2025; digital adoption and digital transformation in the period 2026-2030; and digital transformation in the period 2031-2035. Cambodia's comprehensive digital economy and society policy framework 2021-2035 is focused on achieving the following five goals: to develop the digital infrastructure; to build reliability and confidence in digital systems; to create digital citizens; to build digital government; and to enable digital businesses. The policy framework has been developed to build digital skills for the public and private sectors and the general public. Cambodia's digital

²⁵ <https://www.unwto.org/digital-transformation>

government policy 2022-2035 aims to build an intelligent government that utilizes digital infrastructure and technology as an ecosystem for reforming and modernizing the national system of governance in a transparent and creditable way in order to create a comprehensive digital economy and society. As such, it has the following four strategic goals: to promote the development of the digital governance infrastructure; to establish digital governance; to develop digital public services and improve digital capacity and innovation; and to promote public-private relationships and cooperation.

Furthermore, the RGC has developed numerous roadmaps. First, Cambodia's Science, Technology and Innovation (STI) Roadmap 2030 has five main pillars or goals, including (1) enhancing the governance of the national innovation system, (2) developing human resources in STI, (3) reinforcing research capacity and quality among public institutions and universities, (4) promoting collaboration and networking among stakeholders in STI, and (5) promoting the ecosystem. The second is Cambodia's Digital Education Roadmap, which aims to enhance education with technology, and includes the vision of "building the next-generation technology-enhanced learning ecosystem with the focus on improving innovation and entrepreneurship skills. Third, the Agri-Tech Roadmap, aims to "increase the productivity of agricultural commodities and high value-added products and services for the global supply chain through technology and innovation by 2030". Fourth, the Health-Tech Roadmap aims to "increase the health and wellbeing of the Cambodian people through research and development, innovation and technologies". These roadmaps will enhance each sector's technology, digital knowledge, and skills by 2030. All of these policies and roadmaps demonstrate that the RGC is committed to vigorously reinforcing Cambodia's knowledge and skills in the technology and digital areas.

On the other hand, the tourism sector is also growing along with technology and digital. Like other sectors, tourism has received the attention of the RGC with technology and digital application. The MoT has focused on applying Industry 4.0 to tourism, and providing digital training for tourism-related businesses and members of the public who are interested in acquiring digital skills. The roadmap for the recovery of Cambodian tourism during and after the Covid-19 crisis also highlights the need to enhance the digital tourism industry and digital skills. Digital marketing is emphasized

in the Tourism Development Master Plan 2021-2035 for Siem Reap and Mondulkiri. The negative effects of Covid-19 on tourism have led the RGC to pay more attention to the role of digital technology in tourism.

3.3 Tourism and Technology

3.3.1 Digital Tourism

The concept of digital tourism has emerged in the last two decades. Digital tourism uses technology to enhance the tourists' visit experience (Benyon, Quigley, O'Keefe, & Riva, 2014). Their experience can be enhanced through the presence of the product through technology. Benyon et al. (2014) added that the more effectively a given technology can present a product in response to tourists' intentions, transform the touristic experience into an active one, and make sense to tourists, the more valuable the experience for the tourists. The smartphone is one of the digital devices that is driving digital tourism, since many tourists use smartphones on their tours, which influences their plans about destinations and visits (Jamal & Habib, 2020, as cited in Mikhailov & Kashevnik, 2020). Tourists can also use smartphones to check information about fares, transit schedules, transit routes, and travel times (Jamal & Habib, 2022).

Digital technology is now widely applied in tourism operations. ICT plays an essential role in digital tourism, and has already been applied in the tourism sector for some time. In the context of the nation's cultural heritage, digitalization can help it to attract new audiences, preserve the cultural heritage in the digital WorldWorld, offer massive interconnectivity and immersion, create new opportunities for creative industries, and redirect tourists to new destinations (Tourism 4.0, 2022). Tourism businesses, on the other hand, apply digital technology to marketing (Alford & Jones, 2020). Among the many technologies that have already been applied to digital tourism are 3D printing, virtual assistants, big data, the Internet of Things, blockchain, A.I., chatbots, 5G technology, robots, autonomous security robots, recognition technology, augmented reality, virtual reality, and voice technology (Cambodia 4.0 Center, 2021).

3.3.2 Technology Application in Tourism

Over the last few decades, it has been observed that tourism development has significantly benefited from the development and introduction of innovative technologies to the market (Gössling, 2021). The transformation of technology in the tourism industry can be seen throughout different periods of time from the 1970s to the present, and the industry has been transformed accordingly. The first stage, from the 1970s to 1980s, saw the development of Computer Reservation Systems (CRSs) and Global Distribution Systems (GDSs) which changed the modes of operation of different sectors of the tourism industry, most particularly the hospitality industry, i.e. the reservation of accommodation and food and beverages, and the air transportation industry, i.e. the booking of flight tickets (Buhalis & Law, *Progress in Information Technology and Tourism Management: 20 years on and ten years after the Internet—the state of e-Tourism research*, 2008). Technology has significantly improved the efficiency and effectiveness of business operations, particularly through the sharing of information between different departments in the hotel industry (Koo, Gretzel, Hunter, & Chung, 2015).

During the second stage, i.e. the late 1990s to 2000, the beginning of Internet technology and the development of the World Wide Web transformed every aspect of the tourism business and visitor experiences (Buhalis & Inversini, 2014). These technology developments modernized and revolutionized communication in the tourism industry, changing the marketing and management strategies and the approaches of tourism business operators and destinations (Buhalis, 2000; Drosos, Chalikias, Skodoulis, Kalantonis, & Papagrigoriou, 2017).

During the latest stage, from 2000 to the present, smart technologies have substantially and rapidly transformed tourism business transactions, marketing and promotion strategies, and tourist experiences. The emergence of smart technologies has transformed the tourism industry and advanced it to another stage, as well as bringing about changes in both the supply and demand sides. According to (Xiang & Fesenmaier, 2017), the key technologies adopted in the tourism industry since 2000 include the use of the internet

and websites for tourism marketing and promotion, despite the limitations of static websites that contain information on businesses and destinations. In addition, the widespread availability of e-Cash and Internet Banking has facilitated business transactions so that tourists can purchase tourism products and services anytime and anywhere they like. Email and mobile phones have effectively and efficiently replaced the traditional ways of communication in the tourism business ecosystems. On the demand side, travelers now have access to a wealth of information about destinations and attraction-type products and services to support their decision-making. Government organizations such as DMOs also benefit from adopting new technologies, such as applying smart tourism destination governance by exploiting the benefits of technology diffusion and embedding novel technologies in the main elements of destination management, including developing support infrastructures and stakeholder facilitation and activation for the sustainable development of tourism destinations (Gretzel, 2022). However, there are many challenges facing a destination seeking to adopt new technologies, including the requirements for significant changes in approaches to the governance and management of tourism destinations (Ivars-Baidal, Celdrán-Bernabeu, Mazón, & Perles-Ivars, 2019; Gretzel, 2022).

[Table 4-1] Development and application of technologies to tourism

Period	Key technology
1970-1980	CRS, GDS
1980-1990	Personal computer, Emails
1990-2000	Internet, websites, software (e.g. financial and office software, e-mail, mobile phones, e-Cash, Intranet, Internet banking World Wide Web, video conferences
2000-2010	Smart phones Social Media, Web 2.0, text messages, database and computer software, online booking systems, CRS, mobile apps, online video streaming, instant messaging
2010 to present	AR/VR, mobile apps, cloud computing and online data storage, virtual assistants, wearable technologies, social media, Google analytics, blogs and review websites, collaborative online environments, Web 2.0, chatbots and instant support, digital platforms

(Adapted from Gössling, 2021; Xiang & Fesenmaier, 2017)

Technology transformation continuously influences the operation of all major sectors in the tourism industry, such as accommodation, food and beverages, transportation, and destination attractions. The significant roles of technology in the tourism industry can be divided into different categories ranging from facilitating transaction processes to product or service marketing, supporting visitors' decision-making, improving access to the consumption of visitor experiences, and improving the sustainability and governance of public administration for sustainable tourism development.

3.3.2.1 Internet of Things

The Internet of Things (IoT) refers to the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet." (Also see Wise & Heidari for a definition, 2019). It can allow new types of real-time data, traffic-aware car navigation, smart homes, smart cities, and smart wearable devices and clothes. The IoT is one of the essential forces behind the expansion of big data (Mauro, Greco, & Grimaldi, 2015). In the tourism sector, it provides opportunities for greater access to tourism information, social media interaction, content marketing, big data and wearables (Wise & Heidari, 2019). The IoT allows tourists to access a massive variety of information on tourism destination products and services, including accommodation, food and beverages, communication, and transportation. Thus, the IoT is vital for both current and future tourism, along with the rapid growth of advanced technologies and the Internet.

3.3.2.2 Big Data

The term "big data" is concerned with the high volume, variety, velocity, and variety of information (Alaei et al., 2017; Mauro, Greco, & Grimaldi, 2015). Mauro et al. (2015) clarified its conceptualization concerning information, technology, information and impact, i. e. big data consists of the amount and variety of information, which requires specific technologies to transform such information into economic value. The more information is stored with better-processing technologies, the bigger and better the

data are. With this conceptualization, big data is vital for the tourism industry since tourism is an information-intensive service sector (Song, 2012) in which information about products and services always needs to be offered, updated and made available online for tourists. Beyond this, big data is applicable to research in the tourism sector (Alaei et al., 2017). Researchers have used data from big data for research projects. Thus, big data is essential for the development of the tourism industry.

3.3.2.3 Blockchain

Blockchain has the power to considerably transform the tourism industry (Balasubramanian, Sethi, Ajayan, & Paris, 2022). The blockchain is applicable to tourism management through digitalization, automation, disintermediation, and intelligent environments. The four blockchain applications allow tourists to utilize (information?) for their pre-trip (e.g. planning and booking, during the trip, e.g. tourist baggage tracking, tourist identification and payment), and post-trip (e.g. reviewing their last trip and planning their next one). Blockchain technologies can change the physical ecosystem in the digital ecosystem; automate tourism processes globally to increase industry efficiency, accuracy and productivity; enhance the disintermediation level, e.g., push tourists to make their travel decision; and make the tourism industry more intelligent (Balasubramanian et al, 2022). Thus, blockchain is one of the essential technologies for the development of the future tourism with advanced technologies.

3.3.2.4 Artificial Intelligence (AI) and robotic technologies

Artificial Intelligence (AI) and robotic technologies are employed in many tourism and hospitality enterprises (Ivanov & Webster, 2021; Samala, Katkam, Bellamkonda, & Rodriguez, 2022). Customers favor robots for service delivery in the tourism and hospitality sectors, and seek to reduce their payment for services fully offered by robots (Ivanov & Webster, 2021). Robotic technologies can be applied in such sectors as hotels, and food and beverages. Similarly, AI has been utilized in the tourism industry for such purposes as enhancing personalization, providing customer recommendations, and improving response speed accuracy (Samala et al., 2022).

Additionally, AI could push tourism business' marketing to gain competitiveness for automating the business activity process. These technologies can also be applied in airports, museums, and tourist attractions, as well as for service delivery (Samala et al., 2020). Hence, AI and robotics could be potential technologies for future tourism, especially for tourists who prefer services provided by modern technologies at a lower cost.

3.3.2.5 Other Digital Technologies

Numerous digital technologies have been applied in restaurants, including automation systems, QR menus, humanoid robots, the internet of objects, and 3D food printers (Altaş, 2020). Furthermore, the application of sensor-based technologies is essential for the hospitality and hotel business sectors (Thakur, 2022). These technologies also have the potential to support the future development of the tourism industry. Renewable energy sources such as solar panels, wind turbines and batteries can all contribute to tourism, especially in terms of green and sustainable tourism.

3.4 Scope and Approach

3.4.1 Scope

The roadmap is limited to the application of technologies to tourism from 2022 to 2030 in Cambodia, and does not cover the use of technologies in other sectors beyond 2030. However, because tourism encompasses a wide number of sectors, including accommodation, transportation, tourist attractions, tour operators and travel agencies, and entertainment, the technologies used in these sectors can be included in the roadmap.

3.4.2 Approach

The Tourism Technology Roadmap has been developed with a qualitative approach. First, the literature relevant to tourism, technology trends, especially in Cambodia, and digital tourism was collected. Then, the stakeholders relevant to tourism were

identified for interviews on the use of technology, and the interview questions were developed (See appendix 1). Interviews were conducted with one restaurant, one travel agency, two hotels, two museums, one airline, and one café in Phnom Penh. These organizations handled the interview request letters issued by the Ministry of Industry, Technology, Science, and Innovation. However, only the museums, café, airlines, and Travel Agency confirmed their consent to the interviews. The results of the interviews could provide valuable inputs on currently used technologies and their plans to use them.

Next, the consultative workshop was held. The results of the interviews have been used both in the development of the tourism technology roadmap and in the workshop. The results of the interviews were partially used as a guide for the consultative workshop in building the vision, developing the goals, identifying current and potential technologies applicable to tourism, and determining the drivers of the adoption of technologies for tourism. A workshop is an essential tool for gathering inputs on technologies for tourism in the present and future, i.e. toward 2030. Two workshops were organized, one in the morning and another in the afternoon. The first workshop focused on building a vision and developing goals, while the second one focused on identifying products (technologies) and the drivers of products. Consultative workshops were organized with the participants, including sub-committees composed of representatives from the relevant ministries, the private sector, and tourism-related associations.

Content analysis was applied to analyze and synthesize the data from interviews and workshops in order to develop a vision and the related goals. However, alignment by table matrix applied was used when developing the vision and goals. First, it was used to ensure that the vision of the Tourism Technology Roadmap was aligned with the Digital Economy and Society Policy Framework 2021-2035 (See Appendix 2). Second, the alignment was used to match the developed goals of the Tourism Technology Roadmap 2030 with its vision so as to ensure that the proposed goals fell within the scope of the vision. Third, alignment was carried out to match the goals of the Tourism Technology Roadmap 2030 with the goals of the Digital Economy and Society Policy

Framework 2021-2035. The drafted report will be submitted to the subcommittee for review, and any necessary revisions will be carried out accordingly.

3.5 Roadmap Development Process

The development of the roadmap consisted of five steps. The first step consists of building the vision of the Tourism Technology Roadmap 2030, and initiating the goals in accordance with the vision elaborated in this step. The second step consists in analyzing the overall environment, which includes the social, technological, environmental, economic, and political environments. The third step consists in selecting the strategic products and services required to fulfil the vision. Product priority will also be determined in this step. The fourth step is to determine the technologies for product/service development. The fifth and final step consists in creating a chart to link the strategic products and services and technologies to the target market.

3.6 Vision and Goal

3.6.1 Vision

Cambodia's Tourism Technology Roadmap 2030 has the vision of making Cambodia **"one of the top tourism destinations in the region by adopting innovative and transformative technologies in order to develop a competitive, sustainable and inclusive tourism industry by 2030"**.

3.6.2 Goals

To achieve this vision, the following goals are proposed:

(1) Improve the technology literacy and skills of tourism operators and users

The stakeholders in the tourism industry, including the private and public sectors and non-governmental organizations, need to improve their digital/technological literacy and skills in order to better understand how to use advanced technologies and skills.

They can improve such literacy and skills through both formal and non-formal education. Therefore, more courses on advanced technology and digital literacy and skills should be developed and made widely available to them.

(2) Strengthen technology infrastructure support

The supporting infrastructure should be more developed and monitored to adopt and transform technology for the tourism industry in 2030. The infrastructure could include hardware, software, and networks. For instance, phone networks, coverage, and the internet should be more developed. Infrastructure development will provide both the supply and demand sides with greater convenience in using technology.

(3) Improve cyber security

Deeper knowledge of technology and the availability of infrastructural support can lead tourism operators to adopt technology, provide that security related to technology and technology usage is guaranteed. Improving the security for technology users, for instance, data security and safety, could build trust among technology users, inducing them to adopt new technologies free from worry.

(4) Enforce the utilization of advanced technologies to improve tourism business operations in Cambodia

As advanced technologies are essential for business operations, business operators tend to adopt new technologies in order to gain a competitive advantage for their business. However, enforcing and encouraging the use of advanced technologies among tourism business operators, especially by the government, could help speed up the adoption and transformation of technologies for tourism-related businesses by 2030.

(5) Enable technology adoption and transformation in the tourism industry

As mentioned in the foregoing no. 4, business operators usually adopt new technologies to benefit their business. Therefore, enabling the adoption and transformation of technologies will help to promote and foster their use.

3.7 STEEP Analysis

STEEP analysis is conducted to identify the drivers that could push the adoption of new technologies by 2030. It covers the social, technological, economic, environmental, and political environments.

3.7.1 Social Environment

Social factors have been reported to affect the adoption of new technologies. Such factors include population growth, the young generation, the Covid-19 pandemic, better and higher education and literacy skills, globalization, religion, the durability of life, changes in lifestyles, public participation, safety and security, good hospitality, access to Internet learning online, life convenience, technology demand, and nationalization.

Globalization and regionalization bring both opportunities and threats. Regarding opportunities, globalization and regionalization allow people from all over the world to exchange ideas and values, which facilitates the circulation of massive amounts of knowledge and information among them. People now understand each other better than before, regardless of language, religion or cultural differences, which has significantly boosted the tourism industry. Globalization has increased global mobility and enabled the free flow of information, making it easier for travel enthusiasts to travel around the world and allowing them to discover hidden travel destinations, as well as learning all about different cultures before making travel decisions. In contrast, the rise of globalization and regionalization have also produced some negative effects, such as terrorism and the erosion of cultures. Globalization allows everything, including weapons and criminals, to move freely between different countries, and terror groups can now send their terror messages much more easily and faster. The richness of a given culture determines whether visitors will be attracted to it. However, with globalization and regionalization sweeping across countries and continents, many people are losing their cultural identity and adapting to foreign traditions.

3.7.2 Technological Environment

Technology itself can influence the adoption of new technologies in Cambodia. The reasons include new technologies, research and development, patent, intellectual properties, copyrights, innovation and creativity, media, branding, more convenience, advanced theology, and promotion. Research and development, for instance, requires the use of technology.

3.7.3 Economic Environment

The economy is another area that requires the adoption of new technology. Influential factors include global and regional integration, high ROI in technology businesses, high labour costs, public-private partnerships, privatization, competition, stable growth, green companies, purchasing power, indigenous reliance on tourism, and safe foods.

The economic factor also plays an important role in improving technology transformation and adoption within the tourism industry. It is undeniable that both global and local economic circumstances can significantly affect the opportunity to improve technology adoption. The integration of Cambodia's economy with other countries, whether globally or regionally, in terms of economic cooperation could provide opportunities for tourism businesses and organizations to adopt new technologies that offer them a high return on investment. Other factors that could also increase technology adoption include the rise of public-private partnerships and privatization.

3.7.4 Environment

Environmental changes could lead to the adoption of new technologies to deal with such change. Environmental factors include climate change, disaster, natural resource depletion/abundance, geological benefits, eco-tourism/community-based tourism, good temperatures, biodiversity, pollution, forecasting, irrigation systems, and environmental protection. Advanced technology can help us to predict and manage more effectively climate change, environmental change, natural resources, and

tourism. For instance, GIS remote sensing can be used to map and manage natural resource locations, disasters, and ecotourism sites. As such, it could be applied to ecotourism site management in Cambodia. Technology can also assist with the management of environmental pollution and the treatment of waste generated by the tourism industry, thereby contributing to environmental protection. Therefore, adopting advanced technology in the future is essential for environmental management.

3.7.5 Political Environment

Politics have been reported to influence the adoption of new technologies. Such factors include government policy, legal frameworks, tourism taxes and incentives, the political agenda, political stability, human rights, collaboration, Covid-19, governmental support, e-government, political will, wise political movement, and the “right” political stand.

A number of different external factors are known to influence digital transformation and adoption in tourism development. Among such factors, political factors might be the most important, even though different countries have different development contexts. Based on the results of a discussion among experts in Cambodia, the major political drivers supporting and constraining technological transformation and adoption were found to include the government’s support policy and the legal framework. This means that the transformation/adoption of technology requires the government to develop policies that will provide a guide and create an environment that is conducive to the adoption of the tourism business ecosystem. One of the key drivers is political stability, as this would lead to the stable development of the country as well as the tourism industry. Additionally, it is necessary to provide tourism tax incentives for key technologies that could support the sustainable development of the tourism industry. However, certain political factors might obstruct the adoption of technology in the tourism industry either directly or indirectly. For example, the human rights issue.

3.8 Key Drivers and Strategic Products

All the drivers of change have been identified and assessed using three main criteria, namely, level of policy relevance, level of urgency, and level of importance. As a result, the key drivers that bring both opportunities and threats to the development of tourism in Cambodia were identified and classified into the following five broad categories:

- Social
- Technological
- Economic
- Environmental
- Political

3.8.1 Social Drivers

The key social drivers influencing the future of tourism development in Cambodia would include the large young population, higher literacy rate, higher quality of human capital, effects of globalization, urbanization and migration, and the Covid-19 pandemic. The key social drivers thus identified are highlighted in Table 4-2 below. Population growth, particularly the growth of the young, tech-savvy population such as generations Z and Alpha, will likely increase the demand for the adoption of technologies in the travel and tourism industry to respond to their consumption behaviors. Additionally, this means that human capital would also improve. As a result, the literacy rate would also improve, thereby ensuring sufficient human resources for the tourism industry.

[Table 4-2] Social drivers

Social Drivers	Evaluation Criteria			Total
	Policy-relevant	Urgency	Important	
Young population	5	4	5	14
Population growth	4	3	4	11
Covid-19 pandemic	4	4	4	12
Migration/urbanization	4	4	4	12
Better education/ higher education literacy rate	5	4	5	14
Human capital	5	4	5	14
Globalization	4	4	4	12
Religion	3	3	3	9
National standardization	3	3	3	9
Quality of life	4	4	4	12

3.8.2 Technological Drivers

It is undeniable that technologies have played significant roles in all industries including the travel and tourism industry. The identified key drivers were evaluated, as shown in Table 4-3. Therefore, it is necessary to pay close attention to the key technological drivers that could significantly affect tourism development. This would prepare the industry for the adoption of appropriate technologies, including emerging technologies, to accommodate the changing demands and needs of tourists. The adoption of technologies also contributes to improving visitors' experiences, enhancing competitive advantages and promoting the sustainability of tourism businesses and destinations. The key technological drivers requiring special attention include new technologies that are available in the markets, technologies used for marketing purposes (such as brand promotion and advertising) and social media platforms (such as Facebook, Twitter, Instagram, Blogs, etc.), technologies that support smart destinations and ensure cyber security, digitalization and digitization, and, finally, research and development of technology.

[Table 4-3] Technological drivers

Technological Drivers	Evaluation Criteria			Total
	Policy-relevant	Urgency	Important	
New technology	5	5	5	15
Research and development	5	4	5	14
Patents, intellectual properties, copyrights	4	4	4	12
Innovation and creativity	5	4	4	13
Smart city	4	4	4	12
Media/branding	4	4	5	14
Technology transfer	4	4	3	11
Digitalization/digitization	5	4	5	14
E-marketing	5	5	5	15
Platforms	5	5	5	15
Smart destination	4	4	5	13
Cyber security	5	4	4	13

3.8.3 Economic Drivers

Table 4-4 shows the key economic drivers that could bring major changes to the tourism industry. These key drivers include access to funding for technology adoption and transformation, higher value-added products and services, competition among high demand for technology sectors (such as business, industry, and military), and public-private partnerships. Additionally, regional and international integration is another key driver of the economic development of Cambodia. Consequently, these key drivers will influence the development of tourism and the adoption of technologies in Cambodia.

[Table 4-4] Economic drivers

Economic Drivers	Evaluation Criteria			Total
	Policy-relevant	Urgency	Important	
Global/regional integration	4	4	5	13
High return on investment in the tech business	4	4	4	12
High labour cost	4	4	4	12
Public-private partnerships	5	4	5	14
Privatization	4	3	4	11
Competition (tech, economy, military, business)	4	4	5	13
Access to funding	5	5	5	15
Value-added products/services	4	5	5	14

3.8.4 Environmental Drivers

Several environmental drivers will affect tourism development by 2030. As shown in the table below, the key drivers attracting the most attention include climate change, waste management problems, and the development of community-based tourism. These align very well with the global, regional, and national environmental priority issues. Although Cambodia is located in a strategic geographical location and possesses an abundance of natural resources and warm weather with great potential for tourism development, other drivers also need to be taken into consideration. Indeed, natural disasters and resource depletion resulting from climate change could have a negative effect on tourism development in the medium- and long-term perspectives.

[Table 4-5] Environmental drivers

Environment Drivers	Evaluation Criteria			Total
	Policy-relevant	Urgency	Important	
Climate change	5	5	5	15
Disaster	4	4	4	12
Natural resource depletion/abundance	4	4	4	12
Strategic geographical location	4	3	4	11
Community-based tourism/ecotourism	5	4	5	14
Waste management	5	5	5	15

3.8.5 Political Drivers

In order to design tourism products and services and adopt the technologies needed to support them, certain political drivers, which are also the main drivers, need to be understood because the tourism industry is very sensitive to political changes, including changes at the national, regional, and international levels. Among the various political drivers identified as having an impact on tourism development, four important drivers need to be prioritized above all others (See Table 4-6). These drivers include the supported government policy and the legal frameworks that are necessary for effectively and efficiently supporting the adoption of the technologies for the tourism industry and other related sectors. Another driver is the need to maintain peace and stability. This driver is naturally one of the most important drivers because tourism is very susceptible to political instability. The government's political agenda, i.e. the Whole-of-Government Approach to Tourism Development, could also have a significant effect on tourism development in the medium to long term. Finally, collaboration is required at the national, regional and international levels. This also includes collaboration between government institutions at different levels and collaboration between government institutions and the private sector and the third sector.

[Table 4-6] Political drivers

Political Drivers	Evaluation Criteria			Total
	Policy-relevant	Urgency	Important	
Government policy and legal framework	5	4	5	14
Tourism tax incentives	4	4	4	12
Economic sanctions	4	4	4	12
Political agenda	5	4	5	14
Political stability	5	4	5	14
Human rights	4	4	5	13
Collaboration	5	4	5	14
Covid 19	4	4	4	12

3.8.6 Opportunities and Threats

All key drivers of change can affect the future development of the tourism industry. That being the case, it is necessary to identify the opportunities and threats arising from these drivers. Consequently, the candidate products and services required to respond to or address these opportunities and threats were also initially identified and evaluated, as shown in Table 4-7.

[Table 4-7] Drivers, opportunities, threats, and candidates of products and services

Drivers			Products/Services
- Young population - Population growth - Covid-19 pandemic - Migration/urbanization - Better education/ higher education literacy rate - Digitalization/digitization - E-marketing - Platforms - Smart destination - Cyber security - Global/regional integration - High ROI in the tech business - Human capital - Globalization - Religion	- New technology - Research and development - Patents, intellectual properties, copyrights - Innovation & creativity - Smart city - Media/branding - Technology transfer - High labour costs - Public & private partnership - Privatization - Competition (tech, economy, military, business) - Access to funding - Value-added products/services	- Climate change - Natural disasters - Natural resource depletion/abundance - Geographical benefits - Community-based tourism/ecotourism - Waste management - Government policy & legal framework - Tourism tax incentives - Economic slowdown - Political agenda - Political stability - Human rights issues - Collaboration - Quality of life - National standardization	- Digital marketing services - Digital tourism platforms - Sustainable tourism services - Digital communication/e-communication - Destination management - Tourism security & safety services - Cyber security - Smart experience programs/smart destinations - Green tourism products - Integrated transportation - Digital transaction services - Research and innovation - Automated & contactless services - Virtual Tour Service
- Tourists might not travel to the destination	- Diverse impacts of climate change	- Fewer opportunities in other tourism services	



Drivers			Products/Services
- Rising inflation - Less income for stakeholders - Less use of all kinds of transportation - Limited access to VR/AR - Restrictions on travel - Senior citizens - People with disabilities - Geographical barriers - Insecurity - Environmental threats - Natural resource depletion - Pollution/waste and pollution - Complexity of new product development - Investment focused more on quick returns on investment - Fewer opportunities in other tourism services - Obsolete or outdated technologies - Uncompetitive - Limitations on knowledge	- Limited flexibility - Limited investment and support policy for sustainable services - Climate change - Lack of knowledge, skills, and responsibility - Natural resource depletion - Insecurity - Environmental threats - Natural resource depletion - Complexity of new product development - Privacy concerns - Limited Internet coverage area - High Internet prices - Reduced employment - Difficult to find information - Confusion - Can be time-consuming - Increased cost of information exchange	- Not competitive - Limitations on knowledge about smart systems - Complexity for operators/services providers - Experience with complexity and difficulty - Ineffective access - Loss of customers - Customer demand - Long distance - Pandemic - Fiercer competition - Quality of human resources - Reduced jobs/employment - Increase of expenditure on/investment in new - Difficulty of payment in local currency and exchange - Limited human resources - Despite the increasing number of people using digital transactions, knowledge about this is still limited.	

Drivers			Products/Services
- about smart systems - Complexity of operators/services providers - Experience with complexity and difficulty - Increase cost of operation - Ineffective accessibilities - Increase price - High fuel cost - Road and traffic congestion - Increased cost of transport - Lack of infrastructure does not meet the demand for modern transportation - Limited for local service providers - Benefits for specific groups - Limited level of support infrastructure development - Limited skills/human resources - Limited funds for research and development - Limitations in technology	- Long distance/ busier life styles - Demand for safety and security - Cybercrime - Safety concerns - Privacy concerns/security - Safety and security for tourists is still limited - Used for the wrong purposes - Pandemic - Changing behaviors - Real-life services and communication - Missed use of technologies - Lack of coordination at the destination level - More dependent on technologies - Changing behaviors - Some attractions have too many tourists, while others have too few	- Those who have limited knowledge about technology will not be able to get the benefits. - Hackers - Low demand - Requires highly skilled human resources - Language barriers - Difficult to correct errors or mistakes in transactions - Time-consuming transactions - High labour costs - Traditional marketing will lose its popularity - Competitive market - Traditional marketing will be time-consuming/unresponsive marketing techniques - International recognition - Limited number of users - Difficult to find direct service - Internet fraud (fake products/services/information) - Limited investment in tourism	

Drivers			Products/Services
- capability - Improved competitive advantages - Effective use of human resources - Competitive advantages for tourism destinations - Lack of data	- Limited e-platforms for facilitating visitor travel - Loss of visitor and public trust		
- Real-time tour and travel - Promotion of products through media in real time and development of video clips - An alternative tour in the future - Increased access and inclusiveness for those who have limited ability to travel - Advanced development of AR/VR technology - The young generation is more tech-savvy and uses more technology - State-of-the-art technology	- Free independent travelers - International tourists - More international travel - New technology development - Environmental consciousness - Population growth - More effective transportation - Diversify tourism products and services - More convenient - Reduce travel times - Reduce cost of travel - Linkage with international	- Tourists are more knowledgeable about the environment - More tourists are interested in the natural environment - Clean and green are new demand trends - Tourists are more knowledgeable about the environment - Sustainable Development Goals (SDGs) - Conservation of local resources - Increase demand for product quality, health, and reduce environmental pollution - Pro-environmental attitudes - Ecotourism	

Drivers			Products/Services
- available - Opportunity for new business development - Development of capacity to use new technologies - Research and development - Sharing economy - Increase market share - New demand trends (Clean and green) - Availability of new technologies in the market - Create new experiences - Updated information - Provide support to tourists - Response to tourists' need for smart systems - New technology adoption trend - Tendency to seek new experiences - Provide customer value with experiences and knowledge - Technology advancement	- transport - Sharing economy - Long-term development - Support for tourism sites - Tourists increase income - Improve access to products and services - Tourism more developed - Increase jobs - Environmental protection - Environmentally-friendly behaviors - Improve the trust of international tourists - Global trends - Population growth - Internet access - Flow of foreign resources - Contactless payment - Fast, convenient and real-time transactions - Convenient communication - Keeping people in contact/ providing updated	- Community-based tourism - Help communities to protect their resources - Help communities to improve their wellbeing - Contribute to SDGs - Maintain a clean environment - Reduce waste and pollution - Spread internet connections - The more advanced the technology, the more information exchange is needed - Technological development - New available technology and improvement - High service quality requirements - Tourism business competition - The more advanced technology, the more transactions are required - Relevant policy - Respect the law and morality - Security and safety awareness - Increase the number of tourists	

Drivers			Products/Services
- Population growth - New advanced technology - High-speed internet development - Urbanization - Changes in lifestyles - Technology development - Fast, convenient, attractive - International travel - More advanced technology - Easy to use more choices - Competitive services - Promotional support - Outreach/better and more efficient - New product development	- information - Speedy and effective communication - Improved image of destinations and businesses - Support for communication - Enhance service quality - Increasing number of new internet users - Increasing amount of new technologies available - Digital government - Increase competitive advantages of core products of attractions/destinations - Changing consumer behaviors - Decrease operation cost - Incentives for research and development	- More convenient for marketing - Reduce marketing cost - Foreign tourist attractions - Media branding - Effective marketing - Effective sales and increased income - Sustainable development - Database support - International tourists - Higher demand for diverse types of technology	

3.8.7 Candidate products and services

The initial identification and evaluation of the candidates for tourism products and services revealed twelve tourism products and services based on the opportunities and threats brought by diverse social, technological, environmental, economic, and political drivers (Table 4-8). The products and services deemed necessary for the tourism industry include digital marketing services, sustainable tourism services, digital communications (e-communication), destination management, and tourism security and safety services. This was followed by other supported products such as smart (tourism) experience products and programs, green tourism products, integrated tourism transportation services, digital transaction services, research and innovation, automated and contactless services, and virtual tours. These initial results were reassessed based on three main criteria, namely, economic impact, strategic importance, and success potential. The strategic products and services were identified accordingly.

[Table 4-8] Candidates of products/services for the tourism technology roadmap

Tourism product candidates	Average score*	Ranking
Digital marketing services	5	1
Sustainable tourism services	5	2
Digital communication/e-communication	5	2
Destination management	5	3
Tourism security and safety services	6	4
Smart experience (smart destination) products	6	6
Green tourism products	7	7
Integrated transportation services	7	8
Digital transaction services	7	9
Research and innovation	7	10
Automated & contactless services	9	11
Virtual Tour Services	9	12

* The original scores were computed based on the expert rating.

3.9 Technologies

- | | |
|--------------------------|--|
| 1) Touch screen | 16) Devices (smartphone, iPad, tablet, laptop, computer) |
| 2) Audio tool-headphones | 17) Social media (Facebook, YouTube, Instagram, TikTok) |
| 3) Mobile apps | 18) Online booking platforms |
| 4) Digital signature | 19) Property Management system (PMS) for tourism and hospitality |
| 5) Virtual reality | 20) Artificial Intelligence |
| 6) Augmented reality | 21) Cloud computing & high-speed performance |
| 7) Blockchain | 22) Image recognition technology |
| 8) Internet of Things | 23) Robotics |
| 9) Google map | 24) Big data |
| 10) GIS/GPS | 25) Censoring |
| 11) Analytical tools | 26) QR code |
| 12) Hologram | 27) Digital camera (including security camera) |
| 13) 3D printing | 28) Solar panel |
| 14) Internet | 29) Wind turbines |
| 15) Voice search | 30) Batteries |

3.10 Identification of Priority Strategic Products and Services

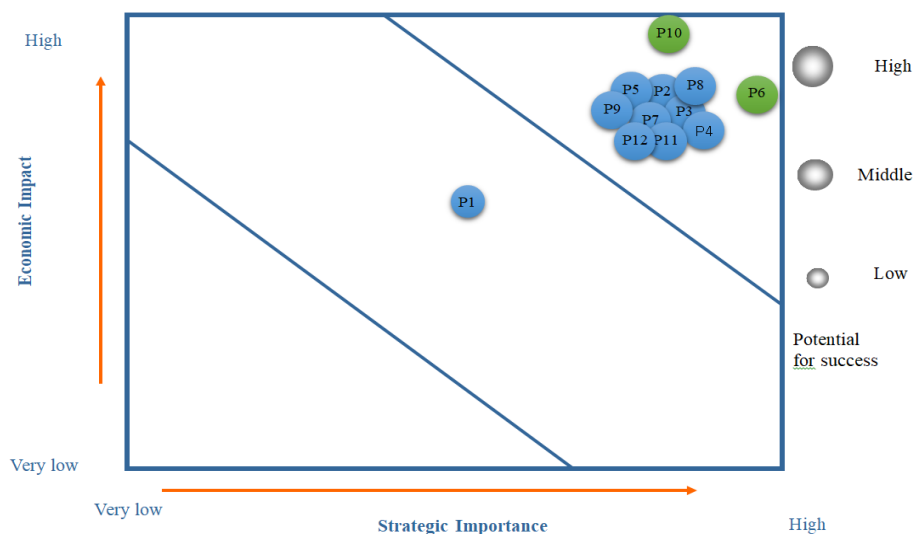
Twelve tourism products/services were evaluated based on a 5-point scale according to three criteria: economic impact, strategic importance, and success potential (See Table 4-9). Most of the products obtained 4 points for each criterion, for a total score of 12 points. Two products, digital transaction services and digital marketing services, obtained the highest score of 13, while only one virtual tour services received the lowest total score of 9, with 3 points for each criterion. Digital transaction services and digital marketing services were rated as the top priority products, whereas virtual tour services were considered to be the lowest priority. However, the variation in the score from 12 to 13 shows only a slight difference, thus suggesting that most of the products should be prioritized.

[Table 4-9] Evaluation of products/services

Products/service	Code	Evaluation criteria			Sum
		Economic impact	Strategic importance	Potential of success	
Virtual tour services	P1	3	3	3	9
Integrated transportation services	P2	4	4	4	12
Sustainable tourism services	P3	4	4	4	12
Green tourism products	P4	4	4	4	12
Smart experience programs	P5	4	4	4	12
Digital transaction services	P6	4	4	5	13
E-communication	P7	4	4	4	12
Tourism security and safety services	P8	4	4	4	12
Automated and contactless services	P9	4	4	4	12
Digital marketing services	P10	5	4	4	13
Research and innovation	P11	4	4	4	12
Destination management	P12	4	4	4	12

Based on the scores obtained in the evaluation (See Table 9-1), most of the products are in the upper right-hand corner (Figure 4-12), which indicates that they are strategic products. Only one product is in the middle area, which may also be considered a strategic product candidate.

[Figure 4-12] Portfolio analysis to identify strategic products and services



3.11 Technology Evaluation of Products/Services

Thirty selected technologies were rated based on their importance and feasibility in terms of 12 priorities. The rating was based on a scale ranging from 1 (very low) to 5 (very high). The evaluation was based on two criteria: importance and feasibility.

3.11.1 Virtual Tour Services

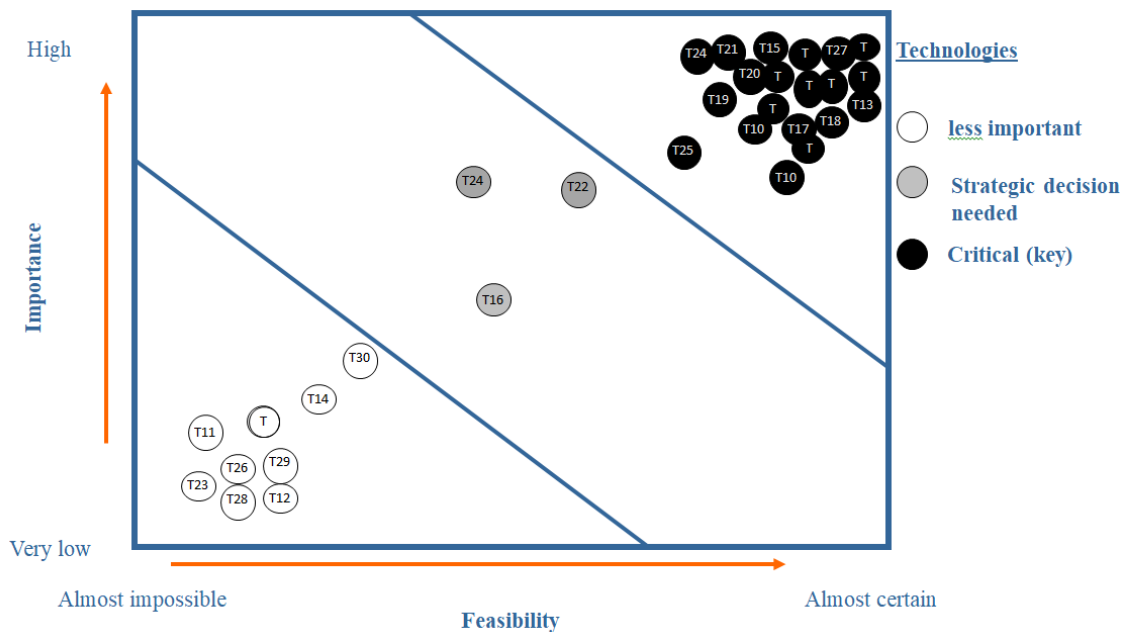
Most of the technologies were found to be important and feasible as virtual services in the future (See Table 4-10). Five technologies were rated least important and less feasible.

[Table 4-10] Evaluation of technologies based on two criteria for evaluating technologies for virtual tour services

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	5	5	10
PMS for tourism and hospitality	T3	1	1	2
Mobile apps	T4	5	5	10
Digital signature	T5	5	5	10
Virtual reality	T6	5	5	10
Augmented reality	T7	5	5	10
Blockchain	T8	5	5	10
Cloud computing	T9	5	5	10
Google map	T10	5	5	10
GIS/GPS	T11	2	2	10
Analytical tools	T12	3	3	6
Hologram	T13	5	5	10
3D printing	T14	1	1	2
Devices	T15	5	5	10
Social media	T16	4	4	8
Online booking platforms	T17	5	5	10
Voice search	T18	5	5	10
Internet	T19	5	5	10

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	4	3	7
Robotics	T23	1	1	2
Big data	T24	5	5	10
Censoring	T25	5	4	9
QR code	T26	2	2	4
Digital cameras	T27	5	5	10
Solar panels	T28	1	1	2
Wind turbines	T29	1	1	2
Batteries	T30	3	3	6

[Figure 4-13] Setting priorities to identify key technologies for virtual tour services



The majority of the key technologies were found to be critical for virtual tour services (Figure 4-13). Thus, these technologies are the most highly prioritized ones among all technology products and services.

3.11.2 Integrated Transportation Services

Most technologies were rated as most important for integrated transportation services, with a score of 5 and a total score of 10 (Table 4-11). While PMS for tourism and hospitality was given the lowest score, Hologram and 3D printing were not considered applicable to integrated transportation services.

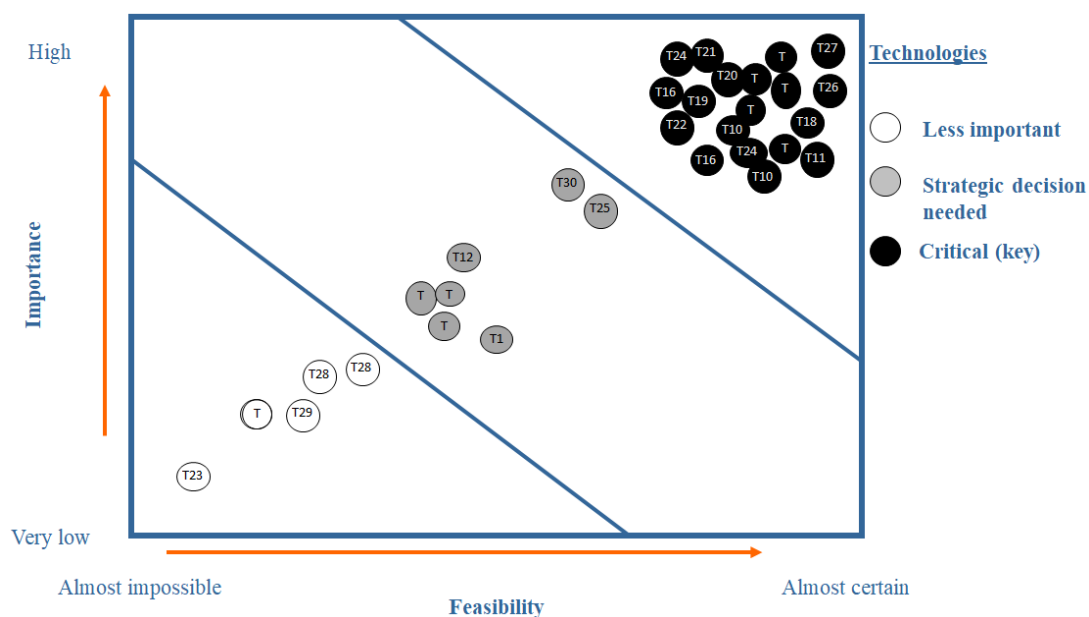
[Table 4-11] Evaluation of technologies based on two criteria for integrated transportation service

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	3	3	6
PMS for tourism and hospitality	T3	1	1	2
Mobile apps	T4	5	5	10
Digital signature	T5	5	5	10
Virtual reality	T6	3	3	10
Augmented reality	T7	3	3	10
Blockchain	T8	5	5	10
Cloud computing	T9	5	5	10
Google map	T10	5	5	10
GIS/GPS	T11	5	5	10
Analytical tools	T12	3	3	6
Hologram	T13	N/A	N/A	N/A
3D printing	T14	N/A	N/A	N/A
Devices	T15	5	5	10
Social media	T16	3	3	6
Online booking platforms	T17	5	5	10
Voice search	T18	5	5	10
Internet	T19	5	5	10
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	5	5	10
Robotics	T23	5	3	8
Big data	T24	5	5	10

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Censoring	T25	4	3	7
QR code	T26	5	5	10
Digital cameras	T27	5	5	10
Solar panels	T28	3	3	6
Wind turbines	T29	2	2	4
Batteries	T30	4	4	8

Over half of the key technologies were in the critical area (Figure 4-14). These technologies were evaluated as those with the highest priority for future integrated transportation services.

[Figure 4-14] Setting priorities to identify key technologies for integrated transportation services



3.11.3 Sustainable Tourism Services

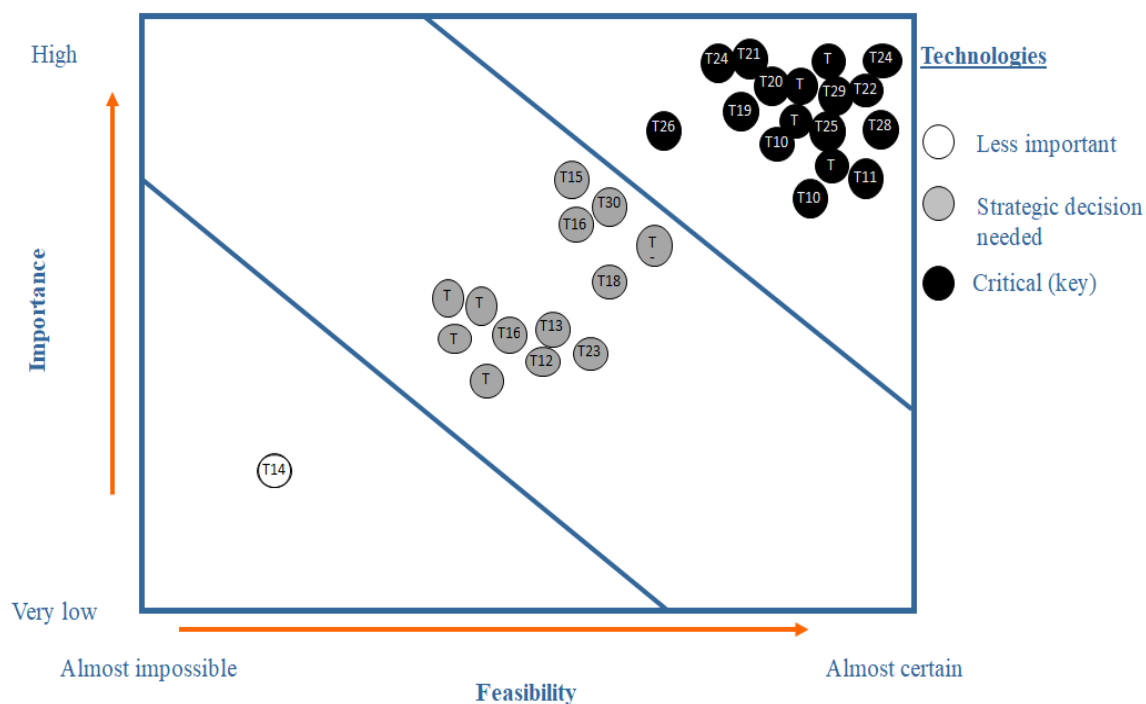
The results indicate that most of the technologies rate very highly in terms of their importance and feasibility for sustainable tourism services (See Table 10-3), whereas PMS, 3D printing, and robotics are considered the least important or feasible.

[Table 4-12] Evaluation of technologies based on two criteria for sustainable tourism services

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	5	5	10
PMS for tourism and hospitality	T3	1	1	2
Mobile apps	T4	5	5	10
Digital signature	T5	5	5	10
Virtual reality	T6	5	5	10
Augmented reality	T7	5	5	10
Blockchain	T8	5	5	10
Cloud computing	T9	5	5	10
Google map	T10	5	5	10
GIS/GPS	T11	2	2	10
Analytical tools	T12	3	3	6
Hologram	T13	5	5	10
3D printing	T14	1	1	2
Devices	T15	5	5	10
Social media	T16	4	4	8
Online booking platforms	T17	5	5	10
Voice search	T18	5	5	10
Internet	T19	5	5	10
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	4	3	7
Robotics	T23	1	1	2
Big data	T24	5	5	10
Censoring	T25	5	4	9
QR code	T26	4	5	9
Digital cameras	T27	4	5	9
Solar panels	T28	5	5	10
Wind turbines	T29	5	5	10
Batteries	T30	4	4	8

About two-thirds of the key technologies were categorized in the critical area (Figure 4-15). Thus, these technologies are considered to be of the highest priority for future sustainable tourism development.

[Figure 4-15] Setting priorities to identify key technologies for sustainable tourism products



3.11.4 Green Tourism Products

Most technologies were rated from the medium to high level in terms of their importance and feasibility for green tourism products (Table 4-13). Two technologies were considered to have low importance and feasibility regarding these products, i.e. big data and blockchain.

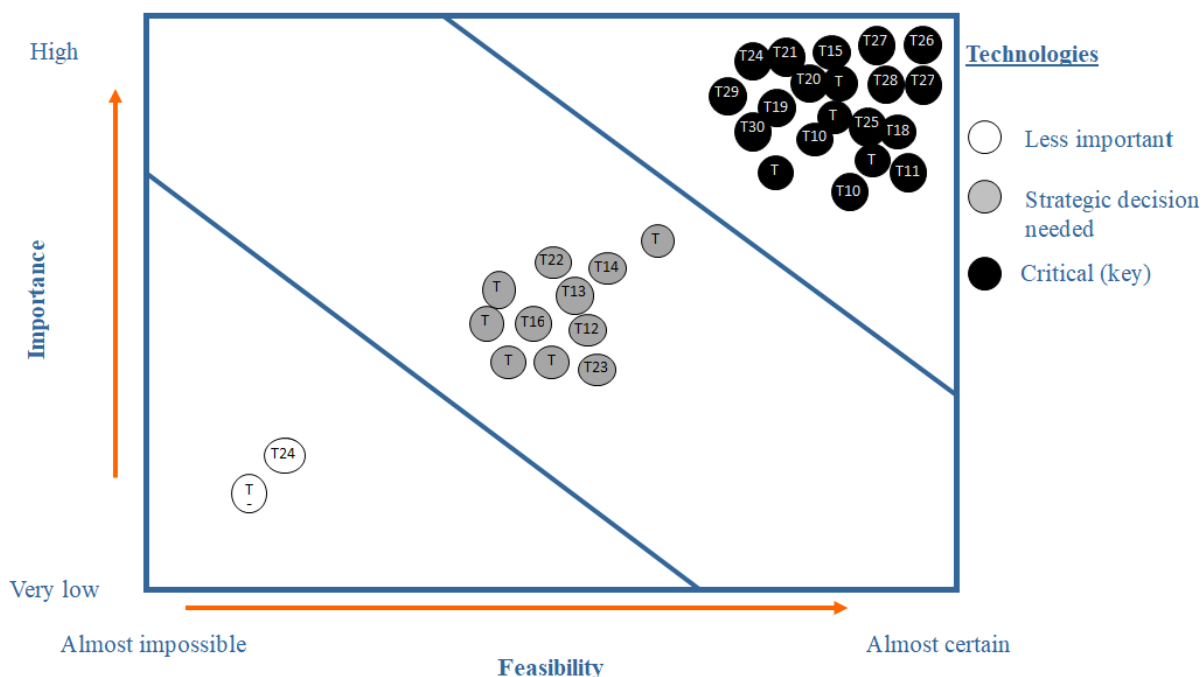
[Table 4-13] Evaluation of technologies based on two criteria for green tourism products

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	3	3	6

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
PMS for tourism and hospitality	T3	3	3	6
Mobile apps	T4	5	5	10
Digital signature	T5	5	5	10
Virtual reality	T6	3	3	10
Augmented reality	T7	3	3	10
Blockchain	T8	2	2	4
Cloud computing	T9	4	4	8
Google map	T10	5	5	10
GIS/GPS	T11	5	5	10
Analytical tools	T12	3	3	6
Hologram	T13	3	3	6
3D printing	T14	3	4	7
Devices	T15	5	5	10
Social media	T16	4	4	8
Online booking platforms	T17	5	5	10
Voice search	T18	3	4	7
Internet	T19	5	5	10
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	4	3	7
Robotics	T23	3	3	6
Big data	T24	2	2	4
Censoring	T25	5	5	10
QR code	T26	4	5	9
Digital cameras	T27	4	5	9
Solar panels	T28	5	5	10
Wind turbines	T29	5	5	10
Batteries	T30	5	5	10

Around two-thirds of the key technologies were categorized in the critical area (Figure 4-16). Thus, these technologies have been prioritized for future green tourism product development.

[Figure 4-16] Setting priorities to identify key technologies for green tourism products



3.11.5 Smart Experience Products

Most technologies were rated medium to high, while only around one-third were rated high to very high in terms of their importance and feasibility for smart experience products (Table 4-14). Only renewable energy technologies were rated to have low importance and feasibility for smart experience products.

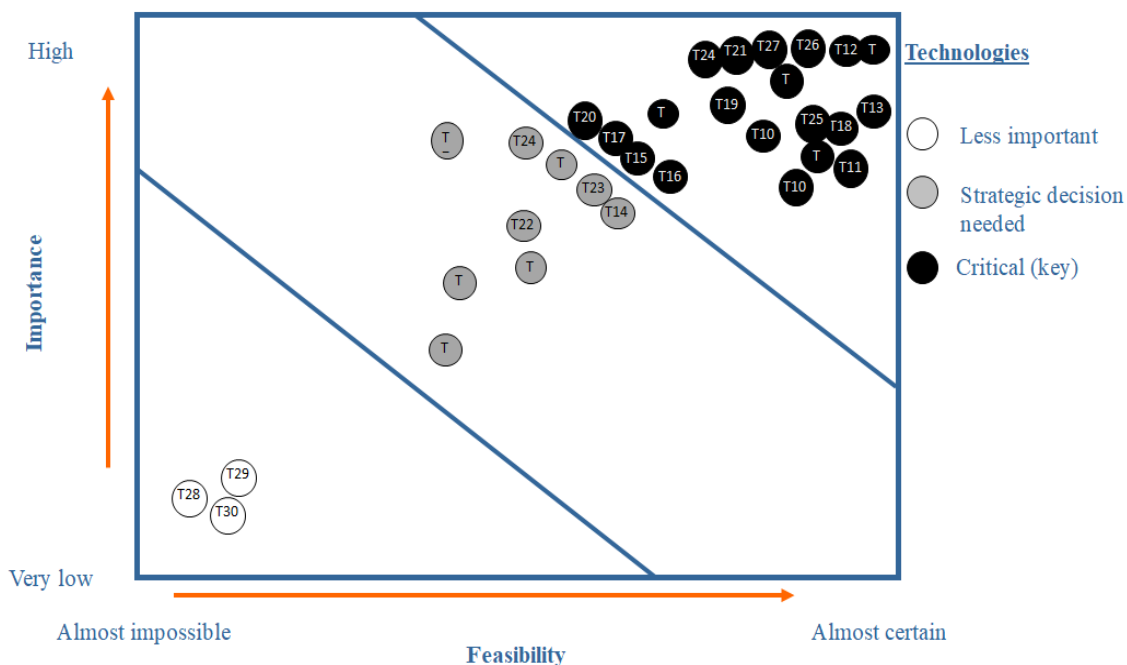
[Table 4-14] Evaluation of technologies based on two criteria for smart experience

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	5	5	6

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
PMS for tourism and hospitality	T3	3	3	6
Mobile apps	T4	5	5	10
Digital signature	T5	3	4	7
Virtual reality	T6	5	4	9
Augmented reality	T7	5	3	8
Blockchain	T8	3	3	6
Cloud computing	T9	4	4	8
Google map	T10	5	5	10
GIS/GPS	T11	5	5	10
Analytical tools	T12	5	5	10
Hologram	T13	5	5	10
3D printing	T14	4	4	8
Devices	T15	5	4	9
Social media	T16	4	5	9
Online booking platforms	T17	5	4	9
Voice search	T18	5	5	10
Internet	T19	5	5	10
Artificial Intelligence	T20	5	4	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	4	3	7
Robotics	T23	4	4	8
Big data	T24	4	4	10
Censoring	T25	5	5	9
QR code	T26	5	5	10
Digital cameras	T27	5	5	10
Solar panels	T28	2	2	4
Wind turbines	T29	1	1	2
Batteries	T30	2	2	4

Most of the key technologies were categorized in the critical area (Figure 4-17). Thus, these technologies will be prioritized for smart experience products.

[Figure 4-17] Setting priorities to identify key technologies for smart experience



3.11.6 Digital Transaction Services

The majority of the technologies were rated high to very high in terms of their importance and feasibility for digital transaction services (Table 4-15). Five technologies, including audio tool-headphones, virtual reality, augmented reality, social media, and voice search, are not considered applicable to digital transaction services.

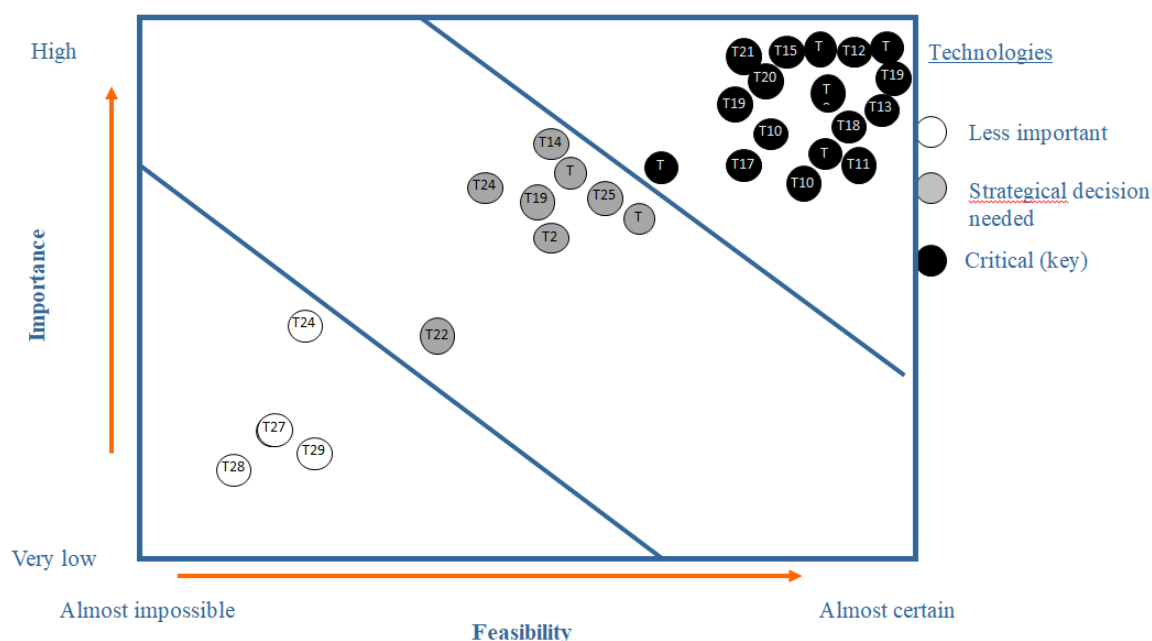
[Table 4-15] Evaluation of technologies based on two criteria for digital transaction services

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	N/A	N/A	N/A

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
PMS for tourism and hospitality	T3	5	5	10
Mobile apps	T4	4	4	8
Digital signature	T5	4	4	8
Virtual reality	T6	N/A	N/A	N/A
Augmented reality	T7	N/A	N/A	N/A
Blockchain	T8	4	4	8
Cloud computing	T9	5	4	9
Google map	T10	5	5	10
GIS/GPS	T11	5	5	10
Analytical tools	T12	5	5	10
Hologram	T13	5	5	10
3D printing	T14	4	4	8
Devices	T15	5	5	10
Social media	T16	N/A	N/A	N/A
Online Booking platforms	T17	5	5	10
Voice search	T18	N/A	N/A	N/A
Internet	T19	5	5	10
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	3	3	6
Robotics	T23	4	3	7
Big data	T24	5	3	10
Censoring	T25	4	4	9
QR code	T26	5	5	10
Digital cameras	T27	2	2	4
Solar panels	T28	2	2	4
Wind turbines	T29	1	1	2
Batteries	T30	4	4	8

Around half of the key technologies were categorized in the critical area (Figure 4-18). Thus, these technologies are the most prioritized for the development of future digital transaction services.

[Figure 4-18] Setting priorities to identify key technologies for digital transaction services



3.11.7 Digital Communication

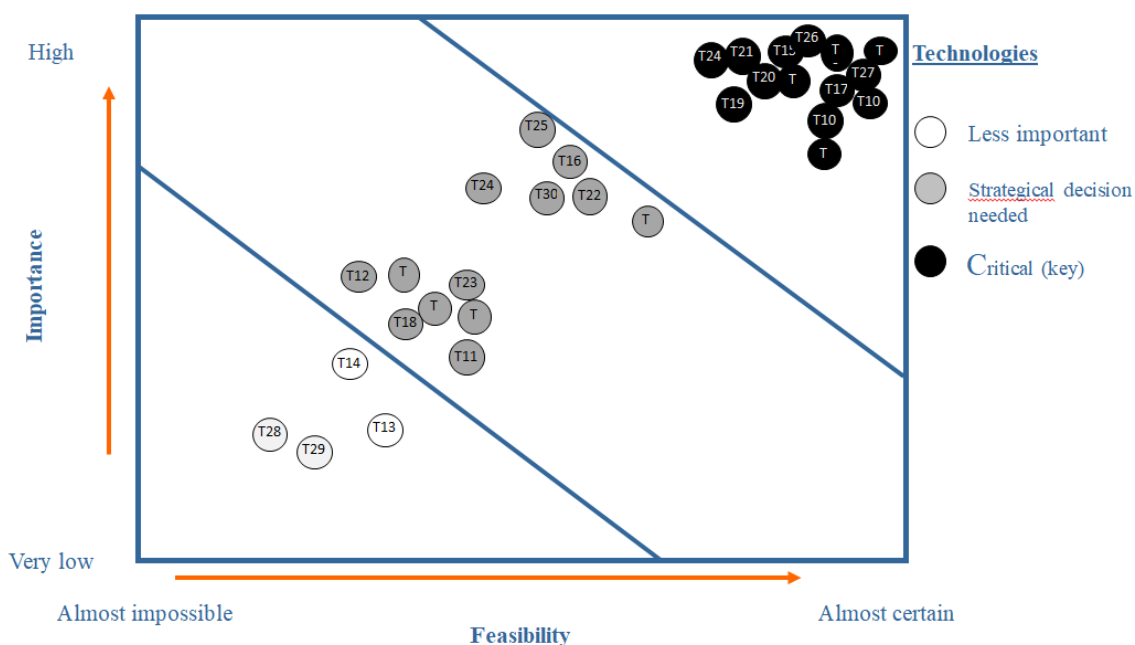
Around one-third of the technologies were rated as being of medium importance and feasibility for e-communication, while the rest were rated at high and very high levels in terms of their importance and feasibility (Table 4-16). Smartphones, laptops, computers, mobile apps, and online booking were the five top-rated technologies in terms of importance and feasibility. In contrast, PMS for tourism and hospitality was not considered important or feasible for digital communication services.

[Table 4-16] Evaluation of technologies based on two criteria for digital communication

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphone	T2	5	5	10
PMS for tourism and hospitality	T3	N/A	N/A	N/A
Mobile apps	T4	5	5	10
Digital signature	T5	4	4	8
Virtual reality	T6	3	3	6
Augmented reality	T7	3	3	6
Blockchain	T8	3	3	5
Cloud computing	T9	5	4	9
Google map	T10	5	5	10
GIS/GPS	T11	3	3	6
Analytical tools	T12	3	3	6
Hologram	T13	2	3	5
3D printing	T14	3	3	6
Devices	T15	5	5	10
Social media	T16	4	4	8
Online booking platforms	T17	5	5	10
Voice search	T18	3	3	6
Internet	T19	5	5	10
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	4	3	7
Robotics	T23	3	3	6
Big data	T24	5	5	10
Censoring	T25	4	4	8
QR code	T26	5	5	10
Digital cameras	T27	5	5	10
Solar panels	T28	2	2	4
Wind turbines	T29	1	1	2
Batteries	T30	4	4	8

About half of the key technologies were categorized in the critical areas (Figure 4-19). Thus, these technologies are considered to be of the highest priority for future digital communication services.

[Figure 4-19] Setting priorities to identify key technologies for digital communication



3.11.8 Tourism Security and Safety Services

Most technologies were rated as having high and very high levels of importance and feasibility for tourism security and safety services (See Table 4-17). While holograms received 2 points for medium, and 3D printing 2 points for low level, renewable energy technologies were rated very low in terms of their importance and feasibility for tourism security and safety services.

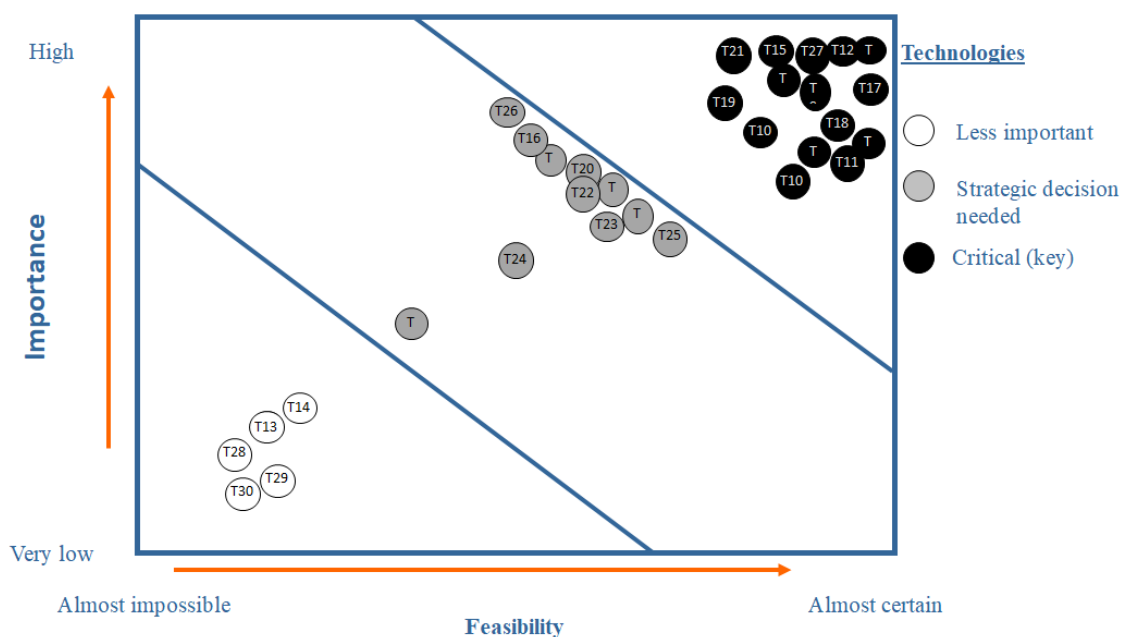
[Table 4-17] Evaluation of technologies based on two criteria for tourism security and safety service

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	5	5	10

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
PMS for tourism and hospitality	T3	5	5	10
Mobile apps	T4	5	5	10
Digital signature	T5	4	4	8
Virtual reality	T6	4	4	8
Augmented reality	T7	4	4	8
Blockchain	T8	3	3	6
Cloud computing	T9	5	5	10
Google map	T10	5	5	10
GIS/GPS	T11	5	5	10
Analytical tools	T12	5	5	10
Hologram	T13	2	2	4
3D printing	T14	2	2	4
Devices	T15	5	5	10
Social media	T16	4	4	8
Online booking platforms	T17	5	5	10
Voice search	T18	5	5	10
Internet	T19	5	5	10
Artificial Intelligence	T20	4	4	8
Internet of Things	T21	5	5	10
Image recognition technology	T22	4	3	7
Robotics	T23	4	4	8
Big data	T24	3	4	7
Censoring	T25	4	4	8
QR code	T26	3	5	8
Digital cameras	T27	5	5	10
Solar panels	T28	1	1	2
Wind turbines	T29	1	1	2
Batteries	T30	1	1	2

Around half of the key technologies were categorized in the critical area (Figure 4-20). Thus, these technologies are the most prioritized ones for future improvements of tourism security and safety services.

[Figure 4-20] Setting priorities to identify key technologies for tourism security and safety services



3.11.9 Automated and Contactless Services

Seven technologies were rated to have medium levels of importance and feasibility, while the rest were rated to have high and very high levels, with ten technologies receiving 5 points each in terms of their importance and feasibility for automated and contactless services (See Table 4-18). Only 3D printing and renewable energy technologies received a very low score.

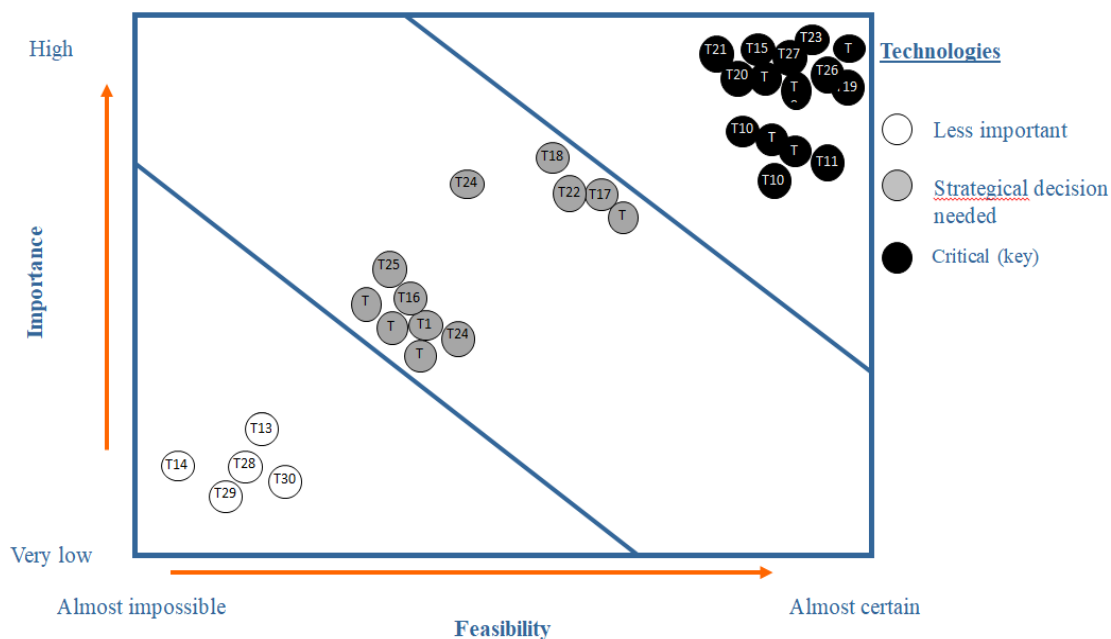
[Table 4-18] Evaluation of technologies based on two criteria for automated and contactless services

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	5	5	10

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
PMS for tourism and hospitality	T3	5	5	10
Mobile apps	T4	5	5	10
Digital signature	T5	4	4	8
Virtual reality	T6	3	3	6
Augmented reality	T7	3	3	6
Blockchain	T8	3	3	6
Cloud computing	T9	5	5	10
Google map	T10	5	5	10
GIS/GPS	T11	5	5	10
Analytical tools	T12	3	3	6
Hologram	T13	2	2	4
3D printing	T14	1	1	2
Devices	T15	5	5	10
Social media	T16	3	3	6
Online booking platforms	T117	4	4	8
Voice search	T18	4	4	8
Internet	T19	5	5	10
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	4	4	10
Robotics	T23	5	5	10
Big data	T24	3	3	10
Censoring	T25	3	3	6
QR code	T26	5	5	10
Digital cameras	T27	5	5	10
Solar panels	T28	1	1	2
Wind turbines	T29	1	1	2
Batteries	T30	2	2	4

Around half of the key technologies were categorized in the critical area (Figure 4-21). Thus, these technologies are the most highly prioritized ones for the future development of automated and contactless services.

[Figure 4-21] Setting priorities to identify key technologies for automated and contactless service



3.11.10 Digital Marketing Service

More than one-third of the technologies were rated very highly in terms of their importance and feasibility (Table 4-19). Digital signature and analytical tools were rated below the medium level for importance or feasibility, while two of the renewable energy technologies were considered very low in terms of their importance and feasibility for digital marketing services.

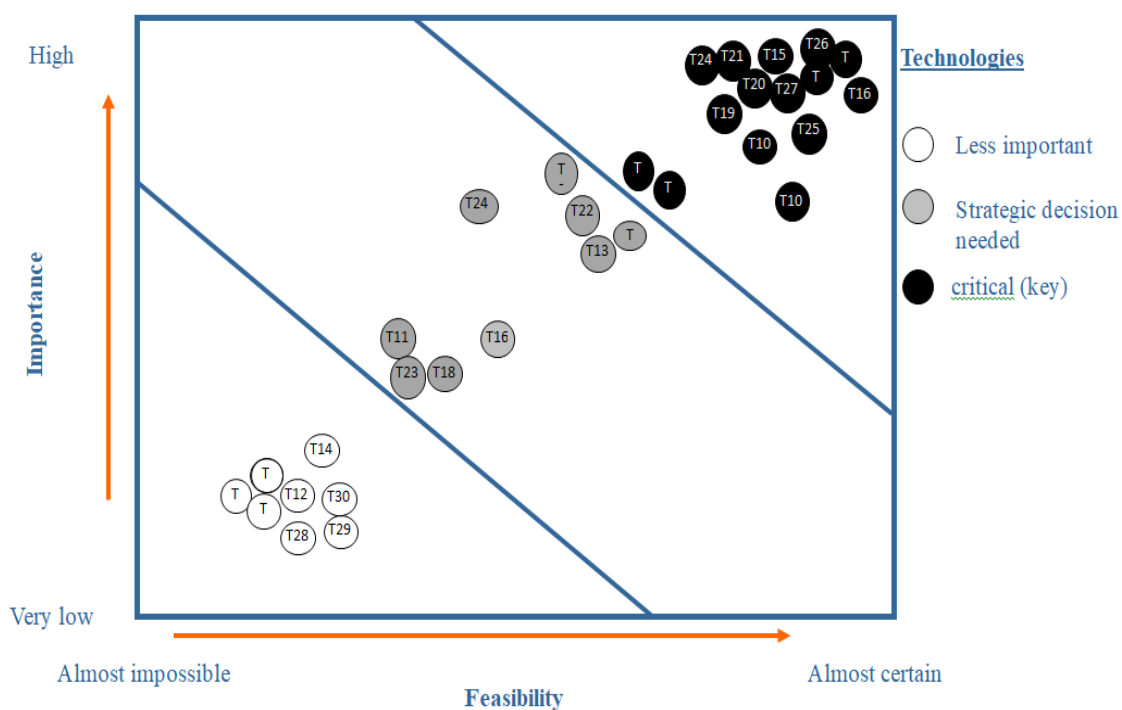
[Table 4-19] Evaluation of technologies based on two criteria for digital marketing services

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	4	4	8

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
PMS for tourism and hospitality	T3	2	2	4
Mobile apps	T4	5	5	10
Digital signature	T5	2	2	4
Virtual reality	T6	5	4	9
Augmented reality	T7	5	4	9
Blockchain	T8	2	2	4
Cloud computing	T9	4	4	8
Google map	T10	5	5	10
GIS/GPS	T11	3	3	6
Analytical tools	T12	2	2	4
Hologram	T13	4	4	8
3D printing	T14	5	5	10
Devices	T15	5	5	10
Social media	T16	5	5	10
Online booking platforms	T117	3	3	6
Voice search	T18	3	3	6
Internet	T19	5	5	10
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	3	3	7
Robotics	T23	3	3	6
Big data	T24	4	4	10
Censoring	T25	4	4	8
QR code	T26	5	5	10
Digital cameras	T27	5	5	10
Solar panels	T28	1	1	2
Wind turbines	T29	1	1	2
Batteries	T30	2	2	4

Around half of the key technologies were categorized in the critical area (Figure 4-22). Thus, these technologies are the most highly prioritized for the future development of digital marketing services.

[Figure 4-22] Setting priorities to identify key technologies for digital marketing services



3.11.11 Research and Innovation

Over one-third of the technologies were rated very highly in terms of their importance and feasibility for research and innovation (Table 4-20). While digital signature, voice search, and robotics were rated very low, holograms and 3D printing were not considered important or feasible for research and innovation.

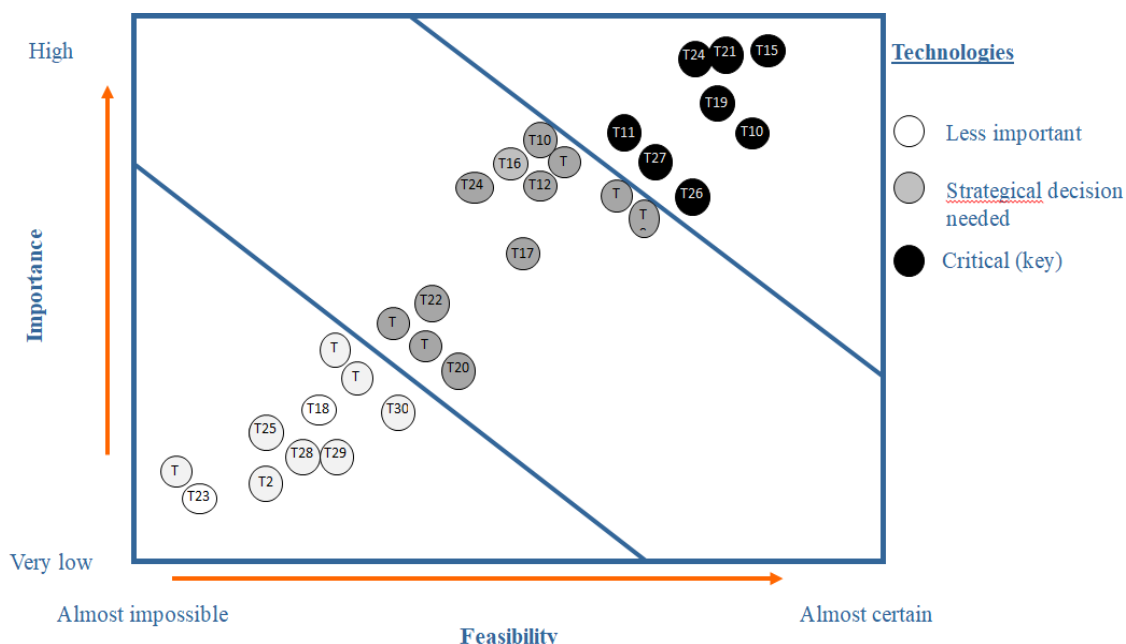
[Table 4-20] Evaluation of technologies based on two criteria for research and innovation

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	3	5	8
Audio tool-headphones	T2	2	2	4

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
PMS for tourism and hospitality	T3	3	3	6
Mobile apps	T4	4	4	8
Digital signature	T5	1	5	6
Virtual reality	T6	2	3	5
Augmented reality	T7	2	3	5
Blockchain	T8	3	3	6
Cloud computing	T9	4	4	8
Google map	T10	3	5	8
GIS/GPS	T11	4	5	9
Analytical tools	T12	4	4	8
Hologram	T13	N/A	N/A	N/A
3D printing	T14	N/A	N/A	N/A
Devices	T15	5	5	10
Social media	T16	4	4	8
Online booking platforms	T17	3	4	7
Voice search	T18	1	3	5
Internet	T19	5	5	10
Artificial Intelligence	T20	3	3	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	3	3	7
Robotics	T23	1	1	2
Big data	T24	5	5	10
Censoring	T25	2	2	4
QR code	T26	4	5	9
Digital cameras	T27	4	5	9
Solar panels	T28	2	2	4
Wind turbines	T29	2	2	4
Batteries	T30	3	3	6

Only one-third of the key technologies were categorized in the critical area, while the majority were categorized in the “strategic decision required” area (Figure 4-23). Thus, more careful consideration will be required when choosing technologies for future research and innovation in the tourism industry. In other words, the future research and innovation for the tourism industry may not demand many technologies.

[Figure 4-23] Setting priorities to identify key technologies for research and innovation



3.11.12 Destination Management

Around half of the technologies received the top score of 5 for their importance and feasibility for destination management (Table 4-21). No technologies were rated below the medium level in terms of their importance and feasibility for destination management.

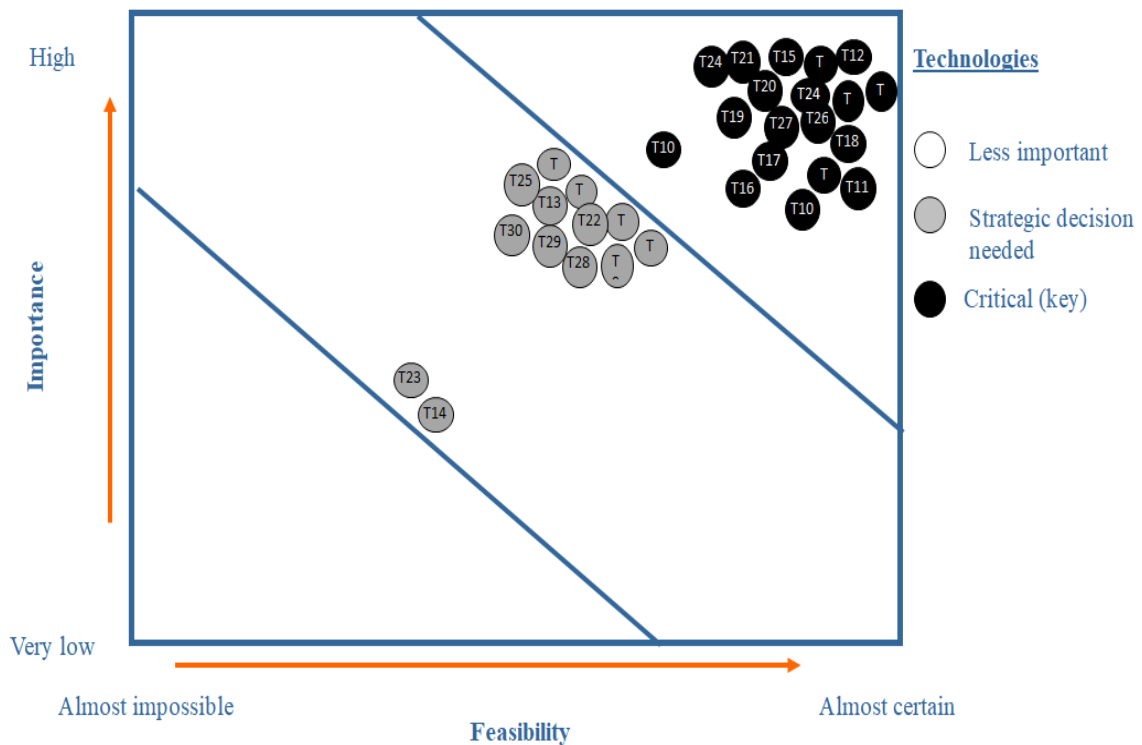
[Table 4-21] Evaluation of technologies based on two criteria for destination management

Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
Touch screen	T1	5	5	10
Audio tool-headphones	T2	5	5	10

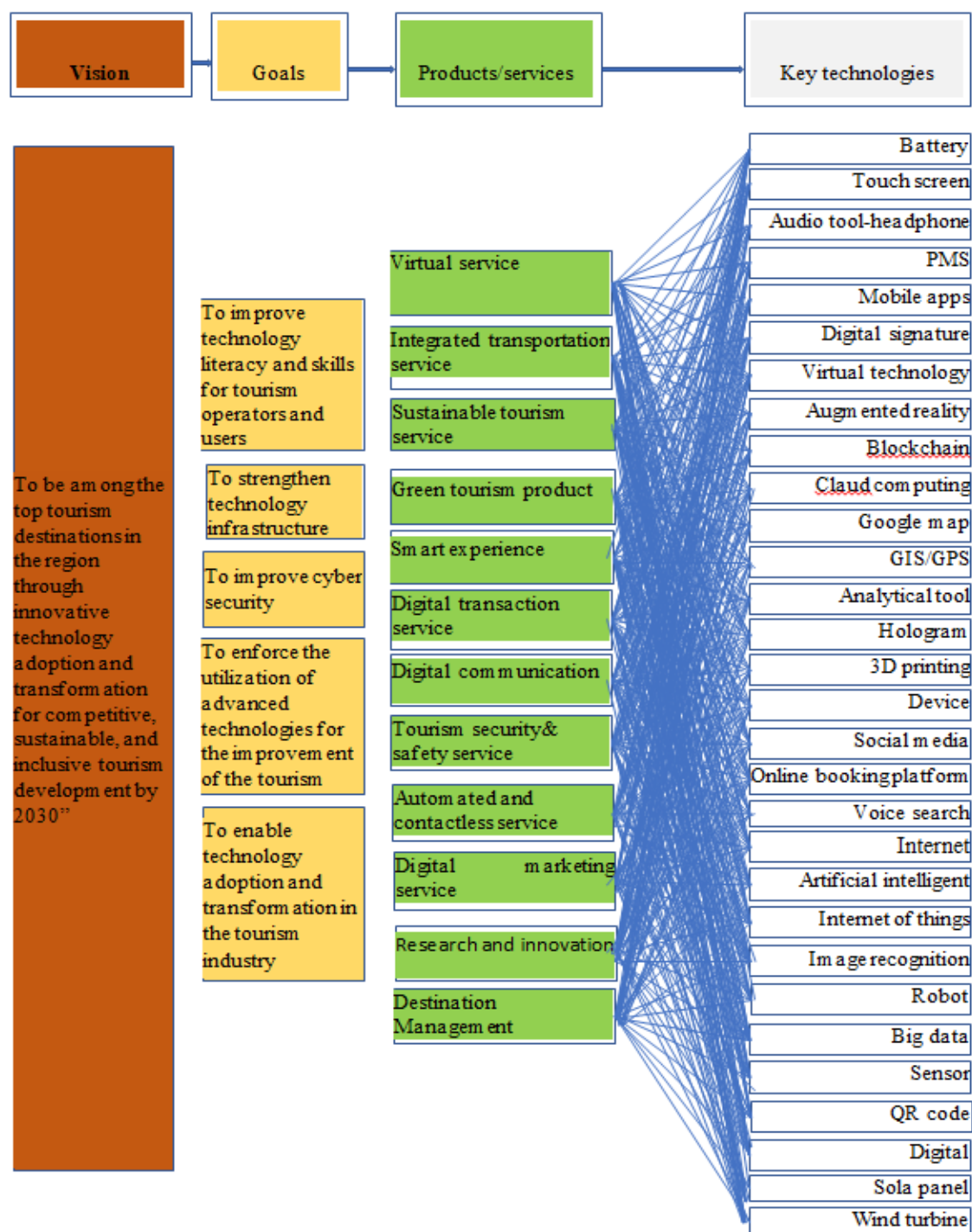
Technologies	Code	Evaluation criteria		SUM
		Importance	Feasibility	
PMS for tourism and hospitality	T3	4	5	8
Mobile apps	T4	4	4	8
Digital signature	T5	3	3	6
Virtual reality	T6	5	5	10
Augmented reality	T7	5	5	10
Blockchain	T8	5	5	10
Cloud computing	T9	4	4	8
Google map	T10	5	5	10
GIS/GPS	T11	5	5	4
Analytical tools	T12	4	5	9
Hologram	T13	4	4	8
3D printing	T14	3	3	6
Devices	T15	5	5	10
Social media	T16	5	5	10
Online booking platforms	T117	5	5	10
Voice search	T18	5	5	10
Internet	T19	5	5	10
Artificial Intelligence	T20	5	5	10
Internet of Things	T21	5	5	10
Image recognition technology	T22	3	4	7
Robotics	T23	3	3	6
Big data	T24	5	5	10
Censoring	T25	4	4	8
QR code	T26	5	5	10
Digital cameras	T27	5	5	5
Solar panels	T28	4	4	8
Wind turbines	T29	4	4	8
Batteries	T30	4	4	8

More than two-thirds of the key technologies were categorized in the critical area (Figure 4-24). Thus, these are considered the most highly prioritized technologies for future tourist destination management.

[Figure 4-24] Setting priorities to identify key technologies for destination management



3.12 Diagram Linking Vision, Goals, Products, Services and Key Technologies



3.13 Finalization of Key Technologies for Products/Services

3.13.1 Finalization of Key Technologies for Virtual Services

Virtual Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IH : L	IH : L	IH : L
		Facebook	Users	IL : L	IL : L	IL : L
		Instagram	Users	IL : L	IL : L	IL : L
		Twitter	Users	IL : L	IL : L	IL : L
		Blog	Users	IL : L	IL : L	IL : L
		YouTube	Users	IL : L	IL : L	IL : L
		TikTok	Users	IL : L	IL : L	IL : L
	Apps	Mobile apps	Volume	IH : L, R	IH : L, R	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital/security cameras)		IH : L	IH : L	IH : L
	Network	Internet	Speed	IH : L	IH : L	IH : L
		Internet of Things	Volume	IH : L, IC, O, R	IH : L, IC, O, R	IH : L, IC, O, R
		Blockchain	Volume	IH : L	IH : L	IH : L
		Cloud computing	Volume	IH : L	IH : L	IH : L
	Biometric Security and Image recognition	Digital signature	Volume	IH : L, R	IH : L, R	IH : L, R
		Image recognition technologies	Volume	IH : L	IH : L	IH : L

Virtual Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Digital Map	QR code	Volume		IL : L	IL : L	IL : L
	Google map	Number		IH : L	IH : L	IH : L
	GIS/GPS	Volume		IH : L	IH : L	IH : L
Artificial Intelligence	Virtual reality	Number		IH : L, R	IH : L, R	IH : L, R
	Augmented reality	Number		IH : L	IH : L	IH : L
	AI	Number		IH : L	IH : L	IH : L
	Voice search	Volume		IH : L	IH : L	IH : L
	Robotics	Volume		IL : L	IL : L	IL : L
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IL : L	IL : L	IL : L
	Big data	Volume		IH : L	IH : L	IH : L
	Property management system for tourism and hospitality	Volume		IL : L	IL : L	IL : L
Audio and Visual	Hologram	Volume		IL : L	IL : L	IL : L
	3D printing	Volume		IL : L	IL : L	IL : L
	Audio tool-headphones	Volume		IH : L	IH : L	IH : L
	Touchscreen	Volume		IH : L	IH : L	IH : L
Renewable Energy	Solar panels	Volume		IL : L	IL : L	IL : L
	Wind turbines	Volume		IL : L	IL : L	IL : L
	Batteries	Volume		IL : L	IL : L	IL : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.2 Finalization of Key Technologies for Integrated Transportation Services

Integrated Transportation Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IH : L, IC, R	IH : L, IC, R	IH : L, IC, R
		Facebook	Users	IM : L	IM : L	IM : L
		Instagram	Users	IM : L	IM : L	IM : L
		Twitter	Users	IM : L	IM : L	IM : L
		Blog	Users	IM : L	IM : L	IM : L
		YouTube	Users	IM : L	IM : L	IM : L
		TikTok	Users	IM : L	IM : L	IM : L
	Apps	Mobile Apps	Volume	IH : L, O, R	IH : L, R	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital /security cameras)		IH : L	IH : L	IH : L
	Network	Internet	Speed	IH : L	IH : L	IH : L
		Internet of Things	Volume	IH : L	IH : L	IH : L
		Blockchain	Volume	IH : L	IH : L	IH : L
		Cloud computing	Volume	IM : L, O	IH : L, IC	IH : R
	Biometric Security and Image Recognition	Digital signature	Volume	IH : L, O	IH : L	IH : L, R
		Image recognition technologies	Volume	IH : L	IH : L	IH : L

Integrated Transportation Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Digital Map	QR code	Volume		IH : L	IH : L	IH : L
	Google map	Number		IH : L	IH : L	IH : L
	GIS/GPS	Volume		IH : L	IH : L, IC	IH : L, IC
Artificial Intelligence	Virtual reality	Number		IH : L, O	IH : L, O	IH : L, R
	Augmented reality	Number		IM : L, O	IH : L, O	IH : L, R
	AI	Number		IH : L	IH : L	IH : L
	Voice search	Volume		IL : L	IM : L	IM : L
	Robotics	Volume		IH : L	IH : L	IH : L
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IM: L, O	IH : L, O	IH : L, O
	Big data	Volume		IH : L	IH : L	IH : L
	Property management system for tourism and hospitality	Volume		IH : L	IH : L	IH : L
Audio and Visual	Hologram	Volume		IL : L	IL : L	IL : L
	3D printing	Volume		IL : L	IL : L	IL : L
	Audio tool-headphones	Volume		IH : L	IH : L	IH : L
	Touchscreen	Volume		IH : L	IH : L	IH : L, R
Renewable Energy	Solar panels	Volume		IL : L	IM : L	IH : L
	Wind turbines	Volume		IL : L	IM : L	IH : L
	Batteries	Volume		IL : L	IM : L	IH : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.3 Finalization of Key Technologies for Sustainable Tourism Services

Sustainable Tourism Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IM : L, IC, R	IM : I, IC, R	IM : I, IC, R
		Facebook	Users	IH : L	IH : L	IH : L
		Instagram	Users	IH : L	IH : L	IH : L
		Twitter	Users	IH : L	IH : L	IH : L
		Blog	Users	IH : L	IH : L	IH : L
		YouTube	Users	IH : L	IH : L	IH : L
		TikTok	Users	IH : L	IH : L	IH : L
	Apps	Mobile apps	Volume	IH : L, O, R	IH : L, O, R	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital /security cameras)		IH : L	IH : L	IH : L
	Network	Internet	Speed	IH : L	IH : L	IH : L
		Internet of Things	Volume	IH : L	IH : L, R	IH : L, R
		Blockchain	Volume	IL : L	IM : L	IH : L
		Cloud computing	Volume	IH : L	IH : L	IH : L
	Biometric Security and Image Recognition	Digital signature	Volume	IM : L	IM : L	IM : L
		Image recognition technologies	Volume	IM : L	IM : L	IM : L
		QR code	Volume	IH : L	IH : L	IH : L

Sustainable Tourism Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Digital Map	Google map	Number		IH : L	IH : L	IH : L
	GIS/GPS	Volume		IM : L	IM : L	IH : L
Artificial Intelligence	Virtual reality	Number		IH : L	IH : L	IH : L
	Augmented reality	Number		IH : L	IH : L	IH : L
	AI	Number		IL : L	IM : L	IH : L
	Voice search	Volume		IH : L	IH : L	IH : L
	Robotics	Volume		IM : L, R	IM : L, R	IM : L, R
Sensing	Sensors	Volume		IH : L	IH : L, R	IH : L, R
Machine Learning	Analytical tools	Volume		IM : L	IM : L	IH : L, R
	Big data	Volume		IH : L	IH : L	IH : L
	Property management system for tourism and hospitality	Volume		IH : L	IH : L	IH : L
Audio and Visual	Hologram	Volume		IL : L, O	IM : L, O	IM : L, O
	3D printing	Volume		IH : L	IH : L	IH : L
	Audio tool-headphones	Volume		IM : L	IM : L	IM : L
	Touchscreen	Volume		IH : L	IH : L	IH : L
Renewable Energy	Solar panels	Volume		IH : L	IH : L	IH : L
	Wind turbines	Volume		IH : L	IH : L	IH : L
	Batteries	Volume		IH : L	IH : L	IH : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.4 Finalization of Key Technologies for Green Tourism Products

Green Tourism Products	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IL : L, R	IL : L, R	IL : L, R
		Facebook	Users	IM : L	IM: L	IM : L
		Instagram	Users	IM: L	IM: L	IM : L
		Twitter	Users	IL: L	IL : L	IL : L
		Blog	Users	IH : L, R	IH : L, R	IH : L, R
		YouTube	Users	IH : L	IH : L	IH : L
		TikTok	Users	IM : L	IM : L	IM : L
	Apps	Mobile apps	Volume	IH : L, R	IH : L, R	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital /security cameras)		IH : L	IH : L	IH : L
	Network	Internet	Speed	IH : L, R	IH : L, R	IH : L, R
		Internet of Things	Volume	IH : L	IH : L, R	IH : L, R
		Blockchain	Volume	IH : L	IH : L	IH : L
		Cloud computing	Volume	IH : L	IH : L	IH : L
	Biometric	Digital signature	Volume	IL : L	IL : L	IL : L

Green Tourism Products	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Security and Image recognition	Image recognition technologies	Volume		IH : L	IH : L	IH : L
	QR code	Volume		IH : L	IH : L	IH : L
Digital Map	Google map	Number		IH: L, IC	IH: L, IC	IH: L, IC
	GIS/GPS	Volume		IM : L	IM : L	IM : L
Artificial Intelligence	Virtual reality	Number		IL : L	IM : L	IH : L
	Augmented reality	Number		IL : L	IM : L	IH : L
	AI	Number		IM : L	IH : L	IH : L
	Voice search	Volume		IH : L	IH : L	IH : L
	Robotics	Volume		IL : L	IM : L	IH : L
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IH : L	IH : L	IH : L
	Big data	Volume		IL : L	IL : L	IL : L
	Property management System	Volume		IH : L	IH : L	IH : L
Audio and Visual	Hologram	Volume		IH : L	IH : L	IH : L
	3D printing	Volume		IH : L	IH : L	IH : L
	Audio tool-headphones	Volume		IL : L	IL : L	IL : L
	Touchscreen	Volume		IH : L	IH : L, R	IH : L, R
Renewable Energy	Solar panels	Volume		IH : L	IH : L	IH : L
	Wind turbines	Volume		IH : L	IH : L	IH : L
	Batteries	Volume		IH : L	IH : L	IH : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.5 Finalization of Key Technologies for Smart Services

Smart Experience	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IH : L, O, R	IH : L, O, R	IH : L, IC, O, R
		Facebook	Users	IH : L	IH : L	IH : L
		Instagram	Users	IH : L	IH : L	IH : L
		Twitter	Users	IH : L	IH : L	IH : L
		Blog	Users	IH : L, O	IH : L, O, R	IH : L, R
		YouTube	Users	IH : L, O	IH : L, O, R	IH : L, R
		TikTok	Users	IH : L	IH : L	IH : L
	Apps	Mobile apps	Volume	IH : L, O, R	IH : L, O, R	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital /security cameras)		IH : L	IH : L	IH : L
	Network	Internet	Speed	IH : L	IH : L	IH : L
		Internet of Things	Volume	IH : L	IH : L, IC, R	IH : L, IC, R
		Blockchain	Volume	IH : L	IH : L	IH : L
		Cloud computing	Volume	IH : L	IH : L	IH : L
	Biometric Security and Image Recognition	Digital Signature	Volume	IM : L	IH : L	IH : L
		Image recognition technologies	Volume	IH : L	IH : L	IH : L
		QR code	Volume	IH : L	IH : L	IH : L

Smart Experience	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Digital Map	Google map	Number		IH : L	IH : L	IH : L
	GIS/GPS	Volume		IH : L	IH : L	IH : L, R
Artificial Intelligence	Virtual reality	Number		IM : L	IH : L	IH : L
	Augmented reality	Number		IM : L	IH : L	IH : L
	AI	Number		IM : L	IH : L	IH : L
	Voice search	Volume		IH : L	IH : L	IH : L
	Robotics	Volume		IH : L	IH : L, R	IH : L, R
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IM : L	IH : L	IH : L
	Big data	Volume		IM : L	IM : L	IM : L
	Property management system	Volume		IH : L, O, R	IH : L, O, R	IH : L, O, R
Audio and Visual	Hologram	Volume		IH : L, O	IH : L, R	IH : L, R
	3D printing	Volume		IH : L	IH : L	IH : L
	Audio tool-headphones	Volume		IM : L, R	IH : L, R	IH : L, R
	Touchscreen	Volume		IH : L, R	IH : L, R	IH : L, R
Renewable Energy	Solar panels	Volume		IL : L	IL : L	IL : L
	Wind turbines	Volume		IL : L	IL : L	IL : L
	Batteries	Volume		IL : L	IL : L	IL : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.6 Finalization of Key Technologies for Digital Transaction Services

Product: Digital Transaction Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IH : L, IC, O	IH : L, IC, O	IH : L, IC, R
		Facebook	Users	IL : L	IL : L	IL : L
		Instagram	Users	IL : L	IL : L	IL : L
		Twitter	Users	IL : L	IL : L	IL : L
		Blog	Users	IL : L	IL : L	IL : L
		YouTube	Users	IL : L	IL : L	IL : L
		TikTok	Users	IL : L	IL : L	IL : L
	Apps	Mobile apps	Volume	IH : L	IH : L, R	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital /security cameras)	Number	IH : L	IH : L	IH : L
	Network	Internet	Speed	IH : L	IH : L	IH : L
		Internet of Things	Volume	IH : L	IH : L, R	IH : L, R
		Blockchain	Volume	IH : L	IH : L	IH : L
		Cloud computing	Volume	IH : L	IH : L	IH : L
	Biometric Security and Image Recognition	Digital signature	Volume	IH : L	IH : L	IH : L
		Image recognition technologies	Volume	IH : L	IH : L	IH : L
		QR code	Volume	IH : L	IH : L	IH : L
	Digital Map	Google map	Number	IL : L	IL : L	IL : L

Product: Digital Transaction Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
	GIS/GPS	Volume		IL : L	IL : L	IL : L
Artificial Intelligence	Virtual reality	Number		IL : L	IL : L	IL : L
	Augmented reality	Number		IL : L	IL : L	IL : L
	AI	Number		IH : L	IH : L	IH : L
	Voice search	Volume		IL : L	IL : L	IL : L
	Robotics	Volume		IL : L	IM : L	IH : L
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IH : L	IH : L	IH : L
	Big data	Volume		IM : L	IM : L	IM : L
	Property management system	Volume		IH : L	IH : L	IH : L
Audio and Visual	Hologram	Volume		IL : L	IL : L	IL : L
	3D printing	Volume		IL : L	IL : L	IL : L
	Audio tool-headphones	Volume		IL : L	IL : L	IL : L
	Touchscreen	Volume		IH : L	IH : L	IH : L
Renewable Energy	Solar panels	Volume		IL : L	IL : L	IL : L
	Wind turbines	Volume		IL : L	IL : L	IL : L
	Batteries	Volume		IL : L	IL : L	IL : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.7 Finalization of Key Technologies for Digital Communication

Digital Communication	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume		IH : L	IH : L
		Facebook	Users		IH : L	IH : L
		Instagram	Users		IH : L	IH : L
		Twitter	Users		IH : L	IH : L
		Blog	Users		IH : L	IH : L
		YouTube	Users		IH : L	IH : L
		TikTok	Users		IH : L	IH : L
	Apps	Mobile apps	Volume		IH : L	IH: L, O, R
	Devices	Smartphone	Volume		IH : L	IH : L
		I-pad/tablet	Volume		IH : L	IH : L
		Computer/laptop	Volume		IH : L	IH : L
		Cameras (digital /security cameras)	Number			
	Network	Internet	Speed		IH : L	IH : L
		Internet of Things	Volume		IH : L	IH : L
		Blockchain	Volume		IH : L	IH : L
		Cloud computing	Volume		IH: L	IH : L
	Biometric Security and Image Recognition	Digital signature	Volume		IH : L	IH : L
		Image recognition technologies	Volume		IH : L	IH : L
		QR code	Volume		IH : L	IH : L
	Digital Map	Google map	Number		IH : L	IH : L

Digital Communication	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
	GIS/GPS	Volume		IH : L	IH : L	IH : L
Artificial Intelligence	Virtual reality	Number		IM : L	IH : L	IH : L
	Augmented reality	Number		IM : L	IH : L	IH : L
	AI	Number		IH : L	IH : L	IH : L
	Voice search	Volume		IH : L	IH : L	IH : L
	Robotics	Volume		IL : L	IM : L	IH : L
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IL : L	IM : L	IH : L
	Big data	Volume		IH : L	IH : L	IH : L
	Property management system	Volume		ILM: L	IM : L	IH : L
Audio and Visual	Hologram	Volume		IL : L	IL : L	IL : L
	3D printing	Volume		IL : L	IL : L	IL : L
	Audio tool-headphones	Volume		IH : L	IH : L	IH : L
	Touchscreen	Volume		IH : L	IH : L	IH : L, R
Renewable Energy	Solar panels	Volume		IM : L	IM : L	IM : L
	Wind turbines	Volume		IM : L	IM : L	IM : L
	Batteries	Volume		IM : L	IM : L	IM : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.8 Finalization of Key Technologies for Tourism Security and Safety Services

Tourism Security and Safety Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IH : L	IH : L, R	IH : L, R
		Facebook	Users	IL : L	IL : L	IL : L
		Instagram	Users	IL : L	IL : L	IL : L
		Twitter	Users	IL : L	IL : L	IL : L
		Blog	Users	IL : L	IL : L	IL : L
		YouTube	Users	IL : L	IL : L	IL : L
		TikTok	Users	IL : L	IL : L	IL : L
	Apps	Mobile apps	Volume	IH : L, R	IH : L, R	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital /security cameras)		IH : L	IH : L	IH : L
	Network	Internet	Speed	IH : L, R	IH : L, R	IH : L, R
		Internet of Things	Volume	IH : L	IH : L	IH : L
		Blockchain	Volume	IH : L	IH : L	IH : L
		Cloud Computing	Volume	IH : L	IH : L	IH : L
	Biometric Security and Image Recognition	Digital signature	Volume	IH : L	IH : L	IH : L
		Image recognition technologies	Volume	IH : L	IH : L	IH : L
		QR code	Volume	IH : L	IH : L	IH : L

Tourism Security and Safety Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Digital Map	Google map	Number		IH : L	IH : L	IH : L
	GIS/GPS	Volume		IH : L	IH : L	IH : L
Artificial Intelligence	Virtual reality	Number		IL : L	IM : L	IH : L
	Augmented reality	Number		IL : L	IM : L	IH : L
	AI	Number		IH : L	IH : L	IH : L
	Voice search	Volume		IH : L	IH : L	IH : L
	Robotics	Volume		IL : L	IM : L	IH : L
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IH : L	IH : L	IH : L
	Big data	Volume		IM : L	IM : L	IH : L
	Property management system	Volume		IH : L	IH : L	IH : L
Audio and Visual	Hologram	Volume		IL : L	IL : L	IL : L
	3D printing	Volume		IL : L	IL : L	IL : L
	Audio tool-headphones	Volume		IH : L	IH : L	IH : L
	Touchscreen	Volume		IH : L	IH : L	IH : L
Renewable Energy	Solar panels	Volume		IH : L	IH : L	IH : L
	Wind turbines	Volume		IH : L	IH : L	IH : L
	Batteries	Volume		IH : L	IH : L	IH : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.9 12.9 Finalization of Key Technologies for Automated and Contactless Services

Automated and Contactless Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IH : L	IH : L	IH : L, R
		Facebook	Users	IL : L	IL : L	IL : L
		Instagram	Users	IL : L	IL : L	IL : L
		Twitter	Users	IL : L	IL : L	IL : L
		Blog	Users	IL : L	IL : L	IL : L
		YouTube	Users	IL : L	IL : L	IL : L
		TikTok	Users	IL : L	IL : L	IL : L
	Apps	Mobile apps	Volume	IH : L	IH : L, R	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital/security cameras)		IH : L	IH : L	IH : L
	Network	Internet	Speed	IH ; L	IH : L	IH : L
		Internet of Things	Volume	IH ; L	IH : L, R	IH : L, R
		Blockchain	Volume	IH: L	IH : L	IH : L
		Cloud computing	Volume	IM : L	IH : L	IH : L
	Biometric Security and Image Recognition	Digital signature	Volume	IH : L	IH : L	IH : L
		Image recognition technologies	Volume	IH : L	IH : L	IH : L
		QR Code	Volume	IH : L	IH : L	IH : L

Automated and Contactless Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Digital Map	Google map	Number		IH : L	IH : L	IH : L
	GIS/GPS	Volume		IH : L	IH : L	IH : L
Artificial Intelligence	Virtual reality	Number		IL : L	IM : L	IH : L
	Augmented reality	Number		IL : L	IM : L	IH : L
	AI	Number		IH : L	IH : L	IH : L
	Voice search	Volume		IH : L	IH : L	IH : L
	Robotics	Volume		IH : L	IH : L, R	IH : L, R
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IL : L	IL : L, R	IM : L, R
	Big data	Volume		IL : L	IL : L	IL : L
	Property management system	Volume		IM: L, O, R	IH : L, O, R	IH : L, R
Audio and Visual	Hologram	Volume		IL : L	IL : L	IL : L
	3D printing	Volume		IL : L	IL : L	IL : L
	Audio tool-headphones	Volume		IH : L	IH : L	IH : L
	Touchscreen	Volume		IH : L, O	IH : O, R	IH : R
Renewable Energy	Solar panels	Volume		IM : L	IM : L	IH : L
	Wind turbines	Volume		IM : L	IM : L	IH : L
	Batteries	Volume		IM : L	IM : L	IH : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.10 Finalization of Key Technologies for Digital Marketing Services

Digital Marketing Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume	IH ; L	IH : L, R	IH : L, R
		Facebook	Users	IH : L	IH : L	IH : L
		Instagram	Users	IH : L	IH : L	IH : L
		Twitter	Users	IH : L	IH : L	IH : L
		Blog	Users	IH : L	IH : L	IH : L
		YouTube	Users	IH : L	IH : L	IH : L
		TikTok	Users	IH : L	IH : L	IH : L
	Apps	Mobile apps	Volume	IH : L, R	IH : L, R	IH : L, R
	Devices	Smartphone	Volume	IH ; L	IH : L	IH : L
		I-pad/tablet	Volume	IH ; L	IH : L	IH : L
		Computer/laptop	Volume	IH ; L	IH : L	IH : L
		Cameras (digital /security cameras)		IH ; L	IH : L	IH : L
	Network	Internet	Speed	IH : L	IH/L	IH : L
		Internet of Things	Volume	IH : L	IH : L	IH : L
		Blockchain	Volume	IH ; L	IH : L	IH : L
		Cloud computing	Volume	IH ; L	IH : L	IH : L
	Biometric Security and Image Recognition	Digital signature	Volume	IL : L	IM : L, R	IH: L, R
		Image recognition technologies	Volume	IH : L	IH : L	IH : L
		QR code	Volume	IH : L	IH : L	IH : L

Digital Marketing Services	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Digital Map	Google map	Number		IM : L	IH : L	IH : L
	GIS/GPS	Volume		IM : L	IH : L	IH : L
Artificial Intelligence	Virtual reality	Number		IM : L	IH : L	IH : L
	Augmented reality	Number		IM : L	IH : L	IH : L
	AI	Number		IH : L	IH : L	IH : L
	Voice search	Volume		IL : L	IM : L	IM : L
	Robotics	Volume		IL : L	IM : L	IM : L
Sensing	Sensors	Volume		IH : L	IH : L	IH : L
Machine Learning	Analytical tools	Volume		IH ; L	IH : L	IH : L
	Big data	Volume		IH : L	IH : L	IH : L
	Property management system	Volume		IH : L, R	IH: L, R	IH: L, R
Audio and Visual	Hologram	Volume		IH : L	IH : L	IH : L
	3D printing	Volume		IH : L	IH : L	IH : L
	Audio tool-headphones	Volume		IH ; L	IH : L	IH : L
	Touchscreen	Volume		IH : L, R	IH : L, R	IH : L, R
Renewable Energy	Solar panels	Volume		IL : L	IM : L	IH : L
	Wind turbines	Volume		IL : L	IM : L	IH : L
	Batteries	Volume		IL : L	IM : L	IH : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.11 Finalization of Key Technologies for Research and Innovation

Research and Innovation	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
Technology Areas	Platform	Online booking platforms	Volume			
		Facebook	Users	IH : L	IH : L	IH : L
		Instagram	Users	IH : L	IH : L	IH : L
		Twitter	Users	IH : L	IH : L	IH : L
		Blog	Users	IH : L	IH : L	IH : L
		YouTube	Users	IH : L	IH : L	IH : L
		TikTok	Users	IH : L	IH : L	IH : L
	Apps	Mobile apps	Volume	IH : L	IH : L	IH : L, R
	Devices	Smartphone	Volume	IH : L	IH : L	IH : L
		I-pad/tablet	Volume	IH : L	IH : L	IH : L
		Computer/laptop	Volume	IH : L	IH : L	IH : L
		Cameras (digital/security cameras)		IH : L	IH : L	IH : L
	Network	Internet	Speed	IH : L	IH : L	IH : L
		Internet of Things	Volume	IH : L	IH : L	IH : L
		Blockchain	Volume	IH : L	IH : L	IH : L
		Cloud computing	Volume	IM : L	IM : L	IH : L
	Biometric Security and Image Recognition	Digital signature	Volume	IL : L	IL : L	IL : L
		Image recognition technologies	Volume	IH : L	IH : L	IH : L
		QR code	Volume	IH : L	IH : L	IH : L

Research and Innovation	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1-3 years)	Mid term (4-7 years)	Long term (8-10 years)
	Digital Map	Google map	Number		IH : L	IH : L
		GIS/GPS	Volume		IH : L	IH : L
	Artificial Intelligence	Virtual reality	Number		IL : L	IL : L
		Augmented reality	Number		IL : L	IL : L
		AI	Number		IH : L	IH : L
		Voice search	Volume		IM : L	IM : L
		Robotics	Volume		IL : L	IM : L
	Sensing	Sensors	Volume		IH : L	IH : L
	Machine Learning	Analytical tools	Volume		IH : L	IH : L, R
		Big data	Volume		IH : L	IH : L
		Property management system	Volume		IL : L	IL : L
	Audio and Visual	Hologram	Volume		IL : L	IL : L
		3D printing	Volume		IL : L	IL : L
		Audio tool-headphones	Volume		IH : L	IH : L
		Touchscreen	Volume		IH : L	IH : L, R
	Renewable Energy	Solar panels	Volume		IL : L	IL : L
		Wind turbines	Volume		IL : L	IL : L
		Batteries	Volume		IL : L	IL : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.13.12 Finalization of Key Technologies for Destination Management

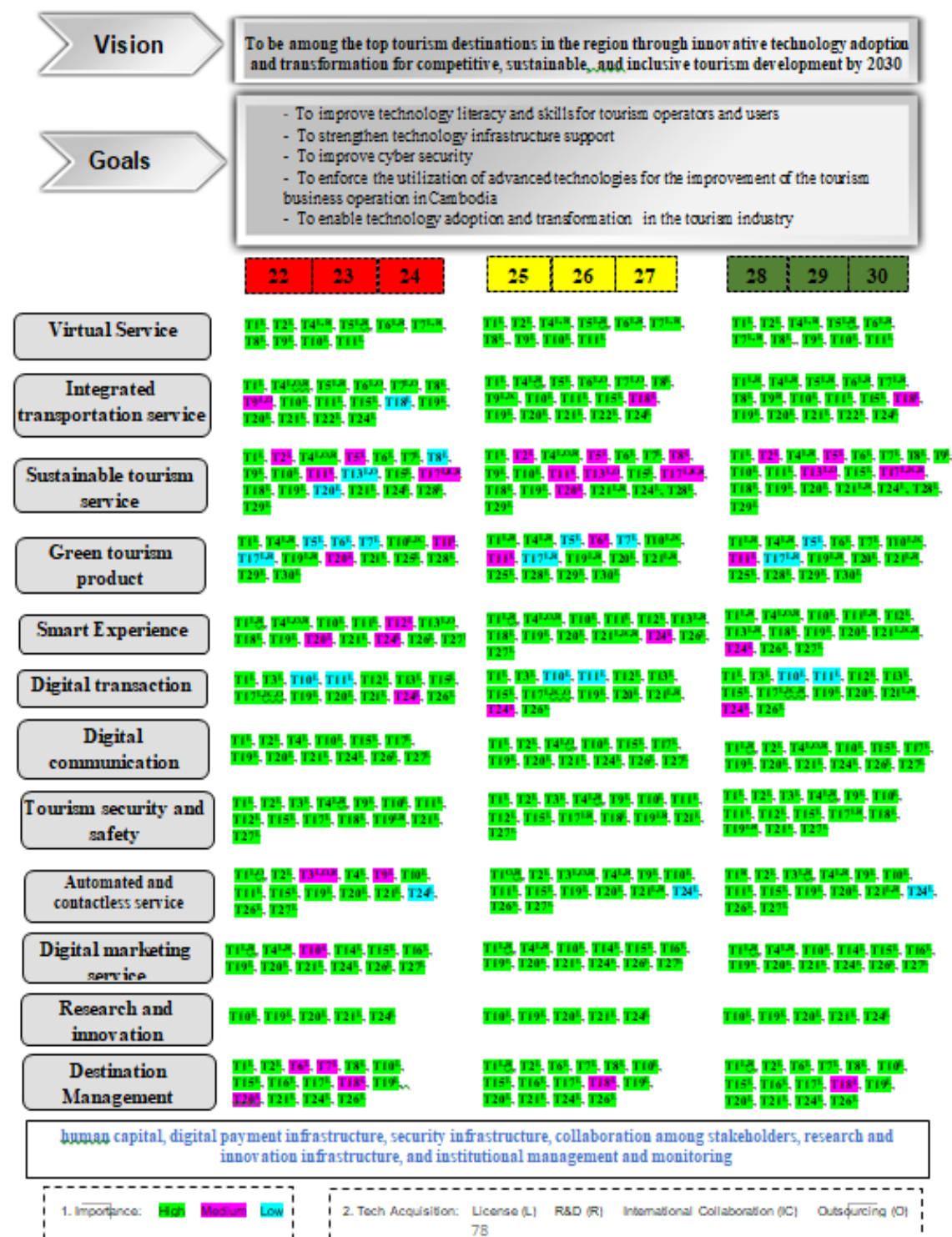
Destination Management	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1~3 years)	Mid term (4~7 years)	Long term (8~10 years)
Technology Areas	Platform	Online booking platforms	Volume		IH : L	IH : L
		Facebook	Users		IH : L	IH : L
		Instagram	Users		IH : L	IH : L
		Twitter	Users		IH : L	IH : L
		Blog	Users		IH : L	IH : L
		YouTube	Users		IH : L	IH : L
		TikTok	Users		IH : L	IH : L
	Apps	Mobile apps	Volume		IH : L	IH : L, R
	Devices	Smartphone	Volume		IH : L	IH : L
		I-pad/tablet	Volume		IH : L	IH : L
		Computer/laptop	Volume		IH : L	IH : L
		Cameras (digital/security cameras)			IH : L	IH : L
	Network	Internet	Speed		IH : L	IH : L
		Internet of Things	Volume		IH : L	IH : L
		Blockchain	Volume		IH : L	IH : L
		Cloud computing	Volume		IH : L	IH : L
	Biometric Security and Image Recognition	Digital signature	Volume		IL : L	IM : L
		Image recognition technologies	Volume		IH : L	IH : L
		QR code	Volume		IH : L	IH : L

Destination Management	Technology Solution	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1~3 years)	Mid term (4~7 years)	Long term (8~10 years)
	Digital Map	Google map	Number		IH : L	IH : L
		GIS/GPS	Volume		IH : L	IH : L
	Artificial Intelligence	Virtual reality	Number		IM : L	IH : L
		Augmented reality	Number		IM : L	IH : L
		AI	Number		IM : L	IH : L
		Voice search	Volume		IM : L	IM : L
		Robotics	Volume		IL : L	IM : L
	Sensing	Sensors	Volume		IH : L	IH : L
	Machine Learning	Analytical tools	Volume		IH : L	IH : L
		Big data	Volume		IH : L	IH : L
		Property management system	Volume		IH : L	IH : L, R
	Audio and Visual	Hologram	Volume		IH : L	IH : L
		3D printing	Volume		IH : L	IH : L
		Audio tool-headphones	Volume		IH : L	IH : L
		Touchscreen	Volume		IH : L	IH : L, R
	Renewable Energy	Solar panels	Volume		IH : L	IH : L
		Wind turbines	Volume		IM : L	IM : L
		Batteries	Volume		IH : L	IH : L

Importance: High (IH), Medium (IM), Low (IL)

Tech Acquisition strategy: License (L), International Collaboration (IC), Outsource (O), R&D (R)

3.14 Charting Tourism Roadmap (Macro)



3.15 Conclusion and Recommendation

Technologies play essential roles in society, economy, politics, environment, and technologies. Furthermore, certain technologies became even more important during the Covid-19 pandemic. Tourism is regarded as a sector that is quick to adopt new technology, and certain technologies have contributed to the management and recovery of the tourism industry during and after the Covid-19 crisis. Meanwhile, the Royal Government of Cambodia (RGC) is paying a lot of attention to (the role of?) technology in the development of the country. The RGC has initiated various technology policies and roadmaps for 2030 in different sectors. As a crucial sector in Cambodia's economic development, the RGC has paid particular attention to tourism, and encouraged the development of the "Tourism Technology Roadmap 2030".

The vision of the Tourism Technology Roadmap 2030 is to transform Cambodia into one of the top tourism destinations in the region through the adoption of innovative technologies and to develop a competitive, sustainable, and inclusive tourism industry by 2030. As such, it has the five following five goals: to improve the technology literacy and skills of tourism operators and users; to strengthen support for the country's technology infrastructure; to improve cyber security; to enforce the utilization of advanced technologies to improve the operation of the tourism business in Cambodia; and to promote the adoption of technologies that will transform the tourism industry.

A series of drivers has been identified and examined in terms of society, economy, politics, environment, and technology, and on the basis of these drivers, prioritized tourism products and services and key technologies have been identified.

The roadmap prioritizes the following twelve tourism products and services: virtual tour services, integrated transportation services, green tourism products, digital transaction services, digital communication, tourism security and safety services, automated and contactless services, digital marketing services, research and innovation, and destination management. However, virtual tour services are regarded as having the lowest priority, while digital transaction services and digital marketing services are the most prioritized ones. Thus, digital transaction and marketing services

are highly prioritized and strongly recommended as the focus of tourism development.

Thirty key technologies have been identified to support the development of the prioritized tourism products and services. The key technologies are rated based on their importance and the feasibility of each product and service, and are then identified as 'critical', 'less important', and 'strategic decision needed'. These technologies are classified into such areas as platforms, apps, devices, networks, biometric security and image recognition, digital map, artificial intelligence, sensing, machine learning, audio and visual, and renewable energy. Since the roadmap is divided into three periods: short-term, medium-term, and long-term, the products/services and technologies are prioritized according to those periods.

Supporting infrastructure should be provided to support the implementation of this roadmap. The infrastructures include human resources, digital payment infrastructure, security infrastructure, collaboration among stakeholders, research and innovation infrastructure, and institutional management and monitoring.

References

- Alford, P., & Jones, R. (2020) The lone digital tourism entrepreneur: knowledge acquisition and collaborative transfer, *Tourism Management*, pp. 81, 1–12. doi:10.1016/j.tourman.2020.104139.
- Alaei, A., Becken, S., & Stantic, B. (2017) Sentiment analysis in tourism: Capitalising on big data. *Journal of Travel Research*, 58(2), 1-17. doi:10.1177/0047287517747753.
- Altaş, A. (2020) Digital transformation in restaurants. In A. B. Waluyo, & L. Tan (Eds.), *Mobile Computing and Technology Applications in Tourism and Hospitality* (pp. 1-23). Doi:10.4018/978-1-7998-6904-7.ch001.
- American Marketing Association. (2022) *What is Digital Marketing?* Retrieved from <https://www.ama.org/pages/what-is-digital-marketing/>
- Anil, B. (2018) Smart E-Communication through Smart Phones. In B. Anil, *Handbook of Research on Mobile Devices and Smart Gadgets in K-12 Education* (pp. 240-251). Pennsylvania: IGI Global.
- Balasubramanian, S., Sethi, J. S., Ajayan, S., & Paris, C. M. (2022) An enabling framework for blockchain in tourism. *Information Technology & Tourism*, 24, 165–179. doi:10.1007/s40558-022-00229-6.
- Benyon, D., Quigley, A., O’Keefe, B., & Riva, G. (2014) Present and digital tourism, *AI & Soc*, 521–529 . doi:10.1007/s00146-013-0493-8.
- Buhalis, D. (2000) Tourism and Information Technologies: Past, Present, and Future, *Tourism Recreation Research*, 25(1), 41–58. doi:10.1080/02508281.2000.11014899.
- Buhalis, D., & Inversini, A. (2014) Tourism Branding, Identity, Reputation Co-creation, and Word-of-Mouth in the Age of Social Media, In M. M. Mariani, R. Baggio, D. Buhalis, & C. Longhi, *Tourism Management, Marketing, and Development Volume I: The Importance of Networks and ICTs* (pp. 15-40). New York: Palgrave Macmillan.
- Buhalis, D., & Law, R. (2008) Progress in information technology and tourism management: 20 years on and ten years after the Internet—The state of e-Tourism research, *Tourism Management*, 29(4), 609-623. doi:10.1016/j.tourman.2008.01.005.
- Buhalis, D., & Amaranggana, A. (2015) Smart Tourism Destinations Enhancing Tourism Experience through the Personalisation of Services. In A. Inversini, & L. Tussyadiah, *Information and Communication Technologies in Tourism* (pp. 377–389). York: Springer International Publishing.
- Cox, S. A., & Perkins, J. S. (2008) The Role of E-Collaboration Systems in Knowledge Management. In S. A. Cox, & J. S. Perkins, *Encyclopedia of E-Collaboration* (pp. 527-533). Pennsylvania: IGI Global.

- Digital innovation of cultural tourism: Handbook for tourist destinations and cultural heritage institutions* (2022) Nova Gorica: Arctur d.o.o.
- Drosos, D., Chalikias, M., Skodoulis, M., Kalantonis, P., & Papagrigoriou, A. (2017) The Strategic Role of Information Technology in Tourism: The Case of Global Distribution Systems, In V. Katsoni, A. Upadhya, & A. Stratigea, *Tourism, Culture and Heritage in Smart Economy* (pp. 207-220). Cham: Springer Nature. doi:10.1007/978-3-319-47732-9_15.
- Furqan, A., Som, A. P., & Hussin, R. (2010) Promoting green tourism for future sustainability, *Theoretical and Empirical Researches in Urban Management*, 5(8), 64-74. doi:10.2307/24872610.
- Gössling, S. (2021) Tourism, Technology, and ICT: a critical review of affordances and concessions, *Journal of Sustainable Tourism*, 29(5), 733–750. doi:10.1080/09669582.2021.1873353.
- Gretzel, U. (2022) The Smart DMO: A new step in the digital transformation of destination management organizations, *European Journal of Tourism Research*, 30(3002), 1–12. Retrieved from <https://ejtr.vumk.eu/index.php/about/article/view/2589/508>.
- Ivars-Baidal, J. A., Celdrán-Bernabeu, M. A., Mazón, J.-N., & Perles-Ivars, Á. F. (2019) Smart destinations and the evolution of ICTs: a new scenario for destination management? *Current Issues in Tourism*, 22(3), 1581-1600. doi:10.1080/13683500.2017.1388771.
- Ivanov, S., & Webster, C. (2021) Willingness-to-pay for robot delivered tourism and hospitality services: an exploratory study, *International Journal of Contemporary Hospitality Management*, 33(11), 3926-3955. doi:10.1108/IJCHM-09-2020-1078.
- Jamal, S., & Habib, M. A. (2022) Smartphone and daily travel: How the use of smartphone applications affects travel decisions, *Sustainable Cities and Society*, 53, 1-9. doi:10.1016/j.scs.2019.101939.
- Koo, C., Gretzel, U., Hunter, W. C., & Chung, N. (2015) The Role of IT in Tourism, *Asia Pacific Journal of Information Systems*, 25(1), 99-102. doi:10.14329/apjis.2015.25.1.099.
- Leav, P. (2019) *Impact of social media advertising value on youth purchase intention: a case study of consumer goods*, Bachelor's Thesis (Royal University of Phnom Penh).
- Maurer, C. (2022) DigitalMarketing in Tourism. In Z. Xiang, M. Fuchs, U. Gretzel, & W. Höpken, *Handbook of e-Tourism* (p. 1311), 1334: Springer Nature Switzerland. doi:10.1007/978-3-030-48652-5_82,
- Mauro, A. D., Greco, M., & Grimaldi, M. (2015) What is big data? A consensual definition and a review of key research topics, *International Conference on Integrated Information*, (pp. 97-104). doi:10.1063/1.4907823.
- Mikhailov, S., & Kashevnik, A. (2020) Tourist behaviour analysis based digital pattern of life and approach and case study, *Future Internet*, 12(10), 1–16. doi:10.3390/fi12100165.

- Ministry of Industry, Science, Technology, and Innovation (2002) *Cambodia digital education roadmap: Enhancing education with technology*, Phnom Penh: Ministry of Industry, Science, Technology, and Innovation.
- Ministry of Industry, Science, Technology, and Innovation (2022) *AgriTechRoadmap*, Phnom Penh: Ministry of Industry, Science, Technology, and Innovation.
- Ministry of Industry, Science, Technology, and Innovation (2022) *HealthTech Roadmap*, Phnom Penh: Ministry of Industry, Science, Technology, and Innovation.
- Ministry of Industry, Science, Technology, and Innovation (2022) *Cambodia's science, technology, and innovation*, Phnom Penh: Ministry of Industry, Science, Technology, and Innovation.
- Ministry of Post and Telecommunications. (2022) *Cambodia digital government policy 2022-2035*. Retrieved from https://asset.cambodia.gov.kh/mptc/media/Cambodia_Digital_Government_Policy_2022_2035_English.pdf.
- Ministry of Tourism (2019) *Tourism Statistics Report 2019*, Phnom Penh: Ministry of Tourism.
- Ministry of Tourism (2021) *Roadmap for Cambodia tourism recovery during and post Covid-19*, Phnom Penh: Ministry of Tourism.
- Popkova, E. G., & Sergi, B. S. (2019) *Digital Economy: Complexity and Variety vs. Rationality*, Cham: Springer Nature.
- Rodrigue, J.-P. (2020) *The Geography of Transport Systems*, New York: Routledge.
- Royal Government of Cambodia (2021) *Tourism Development Master Plan Siem Reap 2021-2035*. Retrieved 12 15, 2021, from https://data.opendevelopmentmekong.net/en/library_record/tourism-development-master-plan-siem-reap-2021-2035.
- Supreme National Economic Council (2021) *Cambodia's digital economy and society policy framework 2021-2035*. Retrieved from https://asset.cambodia.gov.kh/mptc/media/Cambodia_Digital_Government_Policy_2022_2035_English.pdf
- Samala, N., Katkam, B. S., Bellamkonda, R. S., & Rodriguez, R. V. (2022) Impact of AI and robotics in the tourism sector: A critical insight, *Journal of Tourism Future*, 8(1), 73-87. doi:10.1108/JTF-07-2019-0065.
- Song, H. (2012) Tourism supply chain management, In S. Page (Ed.), *Advances in Tourism* (1 ed.). New York, NY: Routledge.
- Talwar, S., Kaur, P., Nunkoo, R., & Dhir, A. (2022) Digitalization and sustainability: virtual reality tourism in a post pandemic world, *Journal of Sustainable Tourism*, 1-28. doi:10.1080/09669582.2022.2029870.
- Tang, R., Huang, C., Zhao, X., & Tang, Y. (2022) Research on smart tourism-oriented sensor network construction and information service mode, *Sensors*, 22(24), 1-17. doi:10.3390/s222410008.

- Thakur, A. (2022) Sensor-based technology in the hospitality industry. In A. B. Waluyo, & L. Tan (Eds.), *Mobile computing and technology applications in tourism and hospitality* (pp. 24-34), Doi: 10.4018/978-1-7998-6904-7.ch004.
- Türkmenbaş, Z. (2022) Artificial intelligence in the delivery of mobile tourism services. In A. B. Waluyo, & L. Tan (Eds.), *Mobile computing and technology applications in tourism and hospitality* (pp. 162-187), doi: 10.4018/978-1-7998-6904-7.ch008.
- Tourism 4.0 (2022, 7 September) *Digital innovation of cultural heritage: a handbook for tourist destinations and cultural heritage institutions*. Retrieved from <https://tourism4-0.org/>.
- UNWTO (1996) *Tourist Safety and Security: Practical Measures for Destinations*. Madrid: UNWTO.
- UNWTO. (2022). *Policy and destination management*. Retrieved from <https://www.unwto.org/policy-destination-management>
- UNWTO (2022). *Sustainable tourism development*. Retrieved from <https://www.unwto.org/sustainable-development#:~:text=%22Tourism%20that%20takes%20full%20account,the%20environment%20and%20host%20communities%22>.
- Vireak, T. (2021) *Determinants of customer satisfaction with online shopping: a case of generation Y*, Bachelor's Thesis (Royal University of Phnom Penh).
- Wise, N., & Heidari, H. (2019) Developing Smart Tourism Destinations with the Internet of Things. In M. S. Rahimi, & M. Thelwall (Eds.), *Big Data and Innovation in Tourism, Travel, and Hospitality—Managerial Approaches, Techniques, and Applications* (pp. 21–29). Singapore: Springer.
- Xiang, Z., & Fesenmaier, D. R. (2017) Big Data Analytics, Tourism Design and Smart Tourism. In Z. Xiang, & D. R. Fesenmaier, *Analytics in Smart Tourism Design: Concepts and Methods* (pp. 299-307). Cham: Springer. doi:10.1007/978-3-319-44263-1_17.
- Yung, R., & Khoo-Lattimore, C. (2017) New realities: a systematic literature review on virtual reality and augmented reality in tourism research, *Current Issues in Tourism*. doi:10.1080/13683500.2017.1417359

Appendices

Appendix I

Guiding questions for an interview with an expert/key informant
Project title: Cambodian Tourism Technology Roadmap

We, the research team from the Department of Tourism of the Royal University of Phnom Penh, are conducting a research project on the development of a Tourism Technology Roadmap 2030 in Cambodia for the Ministry of Industry, Science, Technology & Innovation (MISTI). Your participation is essential for providing the necessary inputs for this project. We would like to thank you very much for your kind cooperation.

ផ្នែកទី១៖ ព័ត៌មានផ្ទាល់ខ្លួន (Part I: Socio-demographic data)

ឈ្មោះ (Name):..... ភេទ (Gender):.....

ការអប់រំ Education:..... តួនាទី Position:.....

វិស័យ Sector:..... បទពិសោធន៍ការងារ Working experience in the sector:years

ផ្នែកII ៖ ការប្រើប្រាស់បច្ចេកវិទ្យាក្នុងឧស្សាហកម្មទេសចរណ៍

Part II: Technology adoption in the tourism industry in general

- តាមយោបល់របស់អ្នក ទេសចរណ៍ឌីជីថលមានន័យបែបណា?

In your opinion, please tell us what digital tourism means to you.

.....

.....

.....

.....

- តើអ្នកគិតថាបច្ចេកវិទ្យាមានតួនាទីអ្វីខ្លះក្នុងការអភិវឌ្ឍឧស្សាហកម្មទេសចរណ៍? (In general and in the Cambodian context)

.....

.....

.....

.....

3. តើអ្នកគិតថាការប្រើប្រាស់បច្ចេកវិទ្យាមានការប្រែប្រួលដូចម្តេចខ្លះនៅក្នុងឧស្សាហកម្ម ទេសចរណ៍ក្នុងរយៈពេល២០ឆ្នាំចុងក្រោយ?

What do you think about the changes that have occurred with the technologies (including digital devices, digital platforms, supporting infrastructure, etc.) used in the tourism industry in Cambodia over the last 20 years?

.....

.....

.....

.....

4. តើការប្រែប្រួលនៃបច្ចេកវិទ្យាបានជះឥទ្ធិពលដូចម្តេចខ្លះលើការអភិវឌ្ឍឧស្សាហកម្ម ទេសចរណ៍ក្នុងប្រទេសកម្ពុជា?

How have these changes affected tourism development in Cambodia?

.....

.....

.....

.....

5. តើការប្រើប្រាស់បច្ចេកវិទ្យាពេលបច្ចុប្បន្នក្នុងឧស្សាហកម្មទេសចរណ៍ក្នុងប្រទេសកម្ពុជា មានស្ថានភាពបែបណាដែរ?

What is the current status of technology (including digital devices, digital platforms, supporting infrastructure, etc.) adoption in the tourism industry in Cambodia?

.....

.....

.....

.....

ផ្នែក III៖ បច្ចេកវិទ្យាប្រើក្នុងវិស័យនៃអាជីវកម្មរបស់អ្នក

Part III: Technology used in the sector

6. តើអាជីវកម្មរបស់អ្នកដំណើរការដូចម្តេចដែរ?

How is your business performance?

a. មុន Covid-19

Before the Covid-19 Pandemic

-
-
-
- b. កំឡុងពេល Covid-19 (២០២០-២០២១)
During the Covid-19 Pandemic (between 2020 and 2021)
-
-
-
- c. បច្ចុប្បន្ន (ចាប់ពីដើមឆ្នាំ២០២២)
Currently (since early 2022)
-
-
-
7. តើបច្ចេកវិទ្យាអ្វីខ្លះត្រូវប្រើប្រាស់ក្នុងវិស័យនៃអាជីវកម្មរបស់អ្នក?
Which technologies are used in your business sector?
- a. មុន Covid-19
Before the Covid-19 Pandemic
-
-
-
- b. កំឡុងពេល Covid-19 (២០២០-២០២១)
During the Covid-19 Pandemic (between 2020 and 2021)
-
-
-
- c. បច្ចុប្បន្ន (ចាប់ពីដើមឆ្នាំ២០២២)
Currently (since early 2022)
-
-
-
8. តើបច្ចេកវិទ្យាអ្វីខ្លះត្រូវបានប្រើប្រាស់ក្នុងអាជីវកម្មរបស់អ្នក? ក្នុងគោលបំណងអ្វី?
Which technologies are used in your business and for what purposes?
-
-
-
-

9. តាមការយល់ឃើញរបស់អ្នក តើកត្តាចម្បងៗអ្វីខ្លះដែលអាជីវកម្មរបស់និងវិស័យនៃ អាជីវកម្មរបស់អ្នកប្រឈម ដើម្បីទទួលយកបច្ចេកវិទ្យាថ្មីៗ?

From your point of view, what are the main challenges facing your business and your business sector regarding the adoption of new technologies?

.....

.....

.....

.....

ផ្នែក IV: ទំនោរបច្ចេកវិទ្យាក្នុងវិស័យទេសចរណ៍

Part IV: Technology trends in tourism

10. តាមយោបល់របស់អ្នក បច្ចេកវិទ្យាសំខាន់ៗអ្វីខ្លះ

ដែលត្រូវការសម្រាប់ជួយដល់អាជីវកម្ម របស់អ្នកក្នុងពេល១០ឆ្នាំខាងមុខ?

In your opinion, what are the key technologies required to support your business sector in the next 10 years?

.....

.....

.....

.....

11. តើអ្នកគិតដូចម្តេចចំពោះទេសចរណ៍ឌីជីថលក្នុងពេល១០ឆ្នាំចុងក្រោយនេះ?

What do you think about the development of digital tourism in Cambodia over the next 10 years?

.....

.....

.....

.....

12. តើអ្វីខ្លះដែលអ្នកគួរធ្វើពីពេលបច្ចុប្បន្នដើម្បីឱ្យលទ្ធភាព ទទួលយកបច្ចេកវិទ្យាថ្មីៗបាននៅ ក្នុងឆ្នាំ២០៣០?

What should be done from now on to encourage the sector to adopt those key technologies by 2030?

.....

.....

.....

.....

សូមអរគុណសម្រាប់ការចូលរួមរបស់អ្នក!

Thank you very much for your kind cooperation!

Appendix 2

Matrix Aligning the Vision of the Tourism Technology Roadmap 2030 with the Vision of Cambodia's Digital Economy and Society Policy Framework 2021-2035 (DESPF)

TsmTech Roamap's Vision DESPF's Vision	To become one of the top tourism destinations in the region through innovative technology adaptation and transformation for the development of a competitive, sustainable and inclusive tourism industry 2030					
	Top tourism destinations in the region	Innovative technology adoption and transformation	Competitive	Inclusive	Sustainable	By 2030
To construct a vibrant digital economy and society by laying foundations to promote digital adoption and transformation of all social actors, including the state, the general public, and businesses, so as to accelerate new economic growth and promote social welfare in the “new normal”	Partial Support	Full Support	Full Support	Full Support	Full Support	Partial Support

Appendix 3

Matrix Aligning the Goals of the Tourism Technology Roadmap 2030 with the Goals of Cambodia's Digital Economy and Society Policy Framework 2021-2035 (DESPF)

DESPF's Goals \ TsmTech Roamap's Goals	To improve the technology literacy and skills of tourism operators and users	To strengthen technology infrastructure support	To improve cyber security	To enforce the utilization of advanced technologies in order to improve tourism business operations in Cambodia.	To enable technology adoption and transformation in the tourism industry.
Develop infrastructures		Full Support	Partial Support	Partial Support	Partial Support
Build reliability and confidence in the digital system	Partial Support	Partial Support	Full Support	Partial Support	Partial Support
Foster digital citizens	Full Support	Partial Support	Partial Support	Partial Support	Partial Support
Build digital government	Partial Support	Partial Support	Partial Support	Full Support	Partial Support
Enable digital businesses	Full Support	Partial Support	Partial Support	Partial Support	Full Support

Appendix 4

Matrix Aligning the Goals with the Vision of the Tourism Technology Roadmap 2030

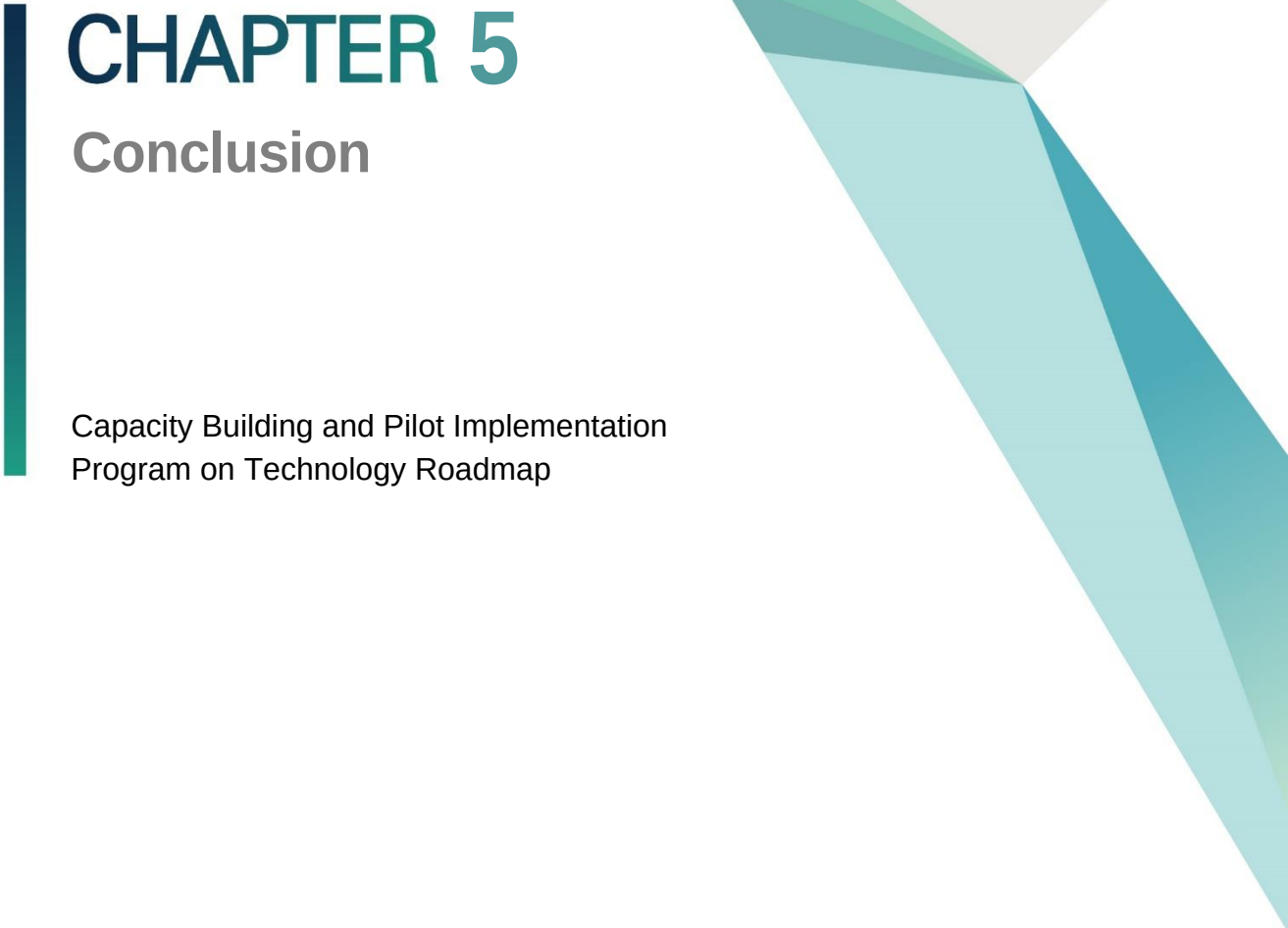
TsmTech Roadmap's Goals	TsmTech Roadmap's Vision To become one of the top tourism destinations in the region through innovative technology adoption and transformation for the development of a competitive, sustainable and inclusive tourism industry by 2030".					
	Top tourism destinations in the region	Innovative technology adoption and transformation	Competitive	Inclusive	Sustainable	2030
Improve the technology literacy and skills of tourism operators and users	Full Support	Full Support	Full Support	Full Support	Full Support	Full Support
Strengthen technology infrastructure support	Full Support	Full Support	Full Support	Full Support	Full Support	Full Support
Improve cyber security	Full Support	Full Support	Full Support	Full Support	Full Support	Full Support
Enforce the utilization of advanced technologies in order to improve tourism business operations in Cambodia	Full Support	Full Support	Full Support	Full Support	Full Support	Full Support
Enable technology adoption and transformation in the tourism industry.	Full Support	Full Support	Fully Support	Fully Support	Fully Support	Fully Support

Appendix 5

Meanings/definitions of Prioritized Products and Services

Products/services	Meanings/definitions
Virtual tour service	AI-based applications that use information communication systems to allow real-time experience of and interaction with a place within a real-world environment from a distance.
Integrated transportation service	A combination of different transportation modes designed to maximize travel effectiveness and efficiency by improving comfort and convenience, reducing cost and time, and improving access and safety for travelers (Rodrigue, 2020).
Sustainable tourism service	Tourism development that fully takes into consideration present and future economic, social, and environmental impacts while simultaneously accommodating the needs of the tourism industry, visitors, and local communities (UNWTO, 2022).
Green tourism product	Various tourism related activities that are environmentally-friendly or any environmentally-friendly tourism service or product offered to visitors (Furqan, Som, & Hussin, 2010).
Smart experience	Use of technology to facilitate the (active) experiences of tourists (Buhalis & Amaranggana, 2015).
Digital transaction service	An action or series of actions aimed at creating, issuing, and circulating digital financial assets (Popkova & Sergi, 2019).
Digital communication	Interchangeable use of “ Digital communication ” services, including the use of any type of digital communication or electronic communication, such as email, SMS text messages or other types of electronic or digital communication (Cox & Perkins, 2008; Anil, 2018).
Tourism security and safety service	Offering high-quality tourism experiences by guaranteeing a safe and secure environment for visitors to a tourism destination is essential (UNWTO, 1996).
Automated and contactless service	Any transactions that provide services and/or products, either

Products/services	Meanings/definitions
	wirelessly or without contact with persons.
Digital marketing service	Use of digital technologies or social platforms to promote a brand or reach customers (American Marketing Association, 2022), this term is often used interchangeably with other terms such as online marketing, internet marketing or e-marketing (Maurer, 2022).
Research and innovation	Use of technology to create a meaningful contribution and leveraging of innovation for tourism development.
Destination Management	the promotion of a brand or reaching the customers by using digital technologies or social platforms (American Marketing Association, 2022), this term often use interchangeably with other terms such as online marketing, internet marketing or e-marketing (Maurer, 2022).

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and green, also pointing downwards.

CHAPTER 5

Conclusion

Capacity Building and Pilot Implementation
Program on Technology Roadmap

Chapter 5. Conclusion

1. Outcomes and Expected Benefits

1.1 Outcomes (Announcement ceremony of the National Science and Technology Roadmap 2021)

- The announcement ceremony of the National Science and Technology Roadmap 2021 was held online on June 23, 2022. The consulting program for the establishment of the 1st Technology Roadmap 2021 was adopted as Cambodia's "National Science and Technology Roadmap." The three technology roadmap sectors, namely, agriculture, health, and education, were promoted last year. The consulting program was implemented to contribute to the achievement of the Cambodia Vision 2050.
- The announcement ceremony organized by the MISTI was attended online and offline by more than 120 experts from Korea and Cambodia. Insights were shared on the background, progress and strategy of the roadmaps for agriculture, health, and education technologies. STEPI President Mun Mi-ok delivered the opening remarks.

STEPI President Mun said that she was delighted to witness the achievements that had been made based on Korea's experience in the field of science and technology development and expressed her hope that the roadmap would greatly contribute to the economic and social development of Cambodia.

- Key discussion points

- (Agriculture): With the focus on strategies for improving agricultural productivity in Cambodia, plans will be made to construct the infrastructure required to achieve qualitative improvements.
- (Health): Efforts will be made to build up a corpus of data using ICT technology.
- (Education): Strategies for making the Internet more widely accessible to ordinary citizens and improving students' learning outcomes will be developed.

[Figure 5-1] Technology launch event program draft



Dissemination and Launching Event

"EduTech, HealthTech, and AgriTech Roadmaps"

Concept

The Royal Government of Cambodia endorsed the National Policy on Science Technology, and Innovation 2020-2030 in December 2019. In the following year, the Ministry of Industry, Science, Technology, & Innovation (MISTI) was established by the Royal Kram NS/RKM/0320/009 dated on 26 March 2020 with role and functioning made through sub decree dated on 06 April 2020. In the same year, National Council of Science, Technology & Innovation (NCSTI) was also established by the Royal Decree NS/RKT/1020/1073 dated on 10 October 2020 to enhance STI development in Cambodia. MISTI has been mandated as a subordinate body to the Royal Government of Cambodia to oversee this sector in the purpose of boosting the development of industry, small and medium enterprises including potable water.

With regard to the highest considerations of the government and to succeed in the mission with significant and fruitful outcomes, and together with National Strategic Development Plan 2019-2023 and Cambodia's STI Roadmap 2030, under the direction of NCSTI, MISTI has framed and developed "EduTech, HealthTech, and AgriTech Roadmaps", with the support from the Science Technology Policy Institute (STePI) of South Korea, which are crucial blueprints for stakeholders, and strategic policy documents in realizing the government's vision to transform Cambodia to an upper-middle income country by 2030 and to a high-income country by 2050. These 3 roadmaps specifically focus on providing strategic directions for the development of key emerging technologies in Education, Health and Agriculture.

To encourage the line ministries, institutions, and key actors to implement officially, effectively, inclusively, and accountably, MISTI holds this launching event to disseminate this roadmap for official use.

Objectives

- To disseminate and roll out the "EduTech, HealthTech, and AgriTech Roadmaps"
- To promote readiness to the relevant ministries, institutions, and stakeholders to develop action plans in compliance with advancement of key technologies as discussed in each roadmap.

Expected Outcomes

- Handover officially the "EduTech, HealthTech, and AgriTech Roadmaps"
- Readiness for the collaboration with MISTI in implementing the STI activities and monitoring and evaluating the implementation under the guidance of NCSTI.

Agenda Dissemination and Launching Event "EduTech, HealthTech, and AgriTech Roadmap"

Phnom Penh, Thursday, 23 June 2022

09:00 – 10:30 (Phnom Penh Time) 1 hour and 30mm duration

Time	Agenda	Key Speakers/Moderators
08:30-09:00	Arrival of distinguished Guests	All
09:00-09:05	National Anthem	Ms. Taing Chanreaksamey
09:05-09:15	Welcoming remarks	Dr. Mi Ock Mun President, STePI, South Korea
09:15-09:25	Official-launching Speech	H.E. Kitti Settha Pandita CHAM Prasidh Senior Minister, Chairman of NCSTI and Minister of MISTI
09:25-09:35	Background of "Tech Roadmap"	H.E. Dr. Hul Seingheng Director General, General Department of STI, MISTI and Head of NCSTI Secretariat
09:35-09:50	Brief on EduTech, HealthTech, and AgriTech Roadmap	Dr. LY Sokny Director, Department STI Cooperation Member of steering committee
09:50-09:55	EduTech Roadmap	H.E. Dr. San Vathana Under Secretary of State, MoEYS and Chairman of EduTech Roadmap Committee
09:55-10:00	HealthTech Roadmap	H.E. Dr. Hok Kimcheng Director General General Department of Health, MoH and Chairman of HealthTech Roadmap
10:00-10:05	AgriTech Roadmap	Mr. Prak Cheatho Deputy Director General General Directorate of Agriculture, MAFF and Chairman of AgriTech Roadmap
10:05-10:20	Comments on the implementation of "EduTech, HealthTech, and AgriTech Roadmap"	Distinguished Guests
10:20-10:30	The announcement of official launching of "EduTech, HealthTech, and AgriTech Roadmap" and Closing Remark	H.E. Kitti Settha Pandita CHAM Prasidh Senior Minister, Chairman of NCSTI and Minister of MISTI

The End

Zoom ID: 850 4549 9741
Passcode: 668870

Please Note: This agenda is subject to change without prior notice.



Please scan me for materials

[Figure 5-2] Participants in the technology launch event



1.2 Expected Benefits

■ Expected outcomes

- Consultation on the preparation of an NTRM for strategic technologies as the Cambodian government's STI policy implementation strategy, and contribution to enhancing Cambodia's national innovation capacity by helping the NTRM sanctioned as the official government plan.
- Contribution to the creation of effective outcomes through strategic investment with scarce R&D resources, in particular.

2. Dissemination and Future Plans

2.1 Dissemination

■ Press release of the Kyosu Shinmun (July 12, 2022)

인쇄하기

인쇄

KCS&K > 뉴스 (news/pressrelease/section_code=5121) > 학술연구 (news/academic/section_code=5212)

과학기술정책 ODA 사업, 캄보디아 국가과학기술로드맵으로 채택

김승우 | 2022.07.12 10:46

대한민국의 선진 과학기술정책이 개발도상국 국가과학기술혁신을 선도하는 역할을 했다.

과학기술정책연구원(이하 과기정책연)이 과기정책연(STEP), 원장 문미숙)은 ODA 사업인 「2021년도 국제과학기술협력사업(K-Innovation Partnership Program)」의 일환으로 캄보디아 공무원 및 현지전문가들을 대상으로 한 「기술로드맵 자문사업」이 캄보디아 「국가과학기술로드맵」으로 채택되었다고 밝혔다.

지난해 1년 간 진행되었던 「기술로드맵 자문사업」은 ▲농업 ▲보건 ▲교육 분야를 캄보디아 전략기술분야로 선정, 관련한 정책 로드맵 구축을 위해 진행됐다.

먼저, 농업 분야는 캄보디아 농업생산성과 품질개선을 목표로 설정했으며, 보건 분야는 ICT 기술을 활용한 정보에 입각하여 결정을 내릴 수 있도록 지원하는 것을 로드맵 구축의 목표로 삼았다.

마지막으로 교육 분야는 컴퓨터를 통해 학교와 집에서 인터넷에 대한 접근성을 향상시키고, 학습결과를 개선하는 것을 목표로 로드맵을 구축했다.

캄보디아 국가과학기술로드맵으로 채택이 된 두 사업은 캄보디아 과학기술혁신(Science Technology and Innovation, STI) 정책을 수립하기 위해 지난 2018년부터 2019년까지 수행된 1단계 정책자문 사업의 후속 사업이다.

두 자문사업은 2050년 캄보디아 국가발전 목표 달성을 위해 경제사회적 목표 이행계획 아래 진행되었다.



캄보디아 국가과학기술로드맵 공표식(캄보디아 산업과학기술혁신부 공무원)

특히, 지난 1단계 정책자문 사업은 캄보디아 정부 최초의 국가과학기술 기본계획인 「National STI Policy 2020-2030」이 수립을 하는 성과도 있었다.

* National STI Policy 2020-2030 : 주요 정책목표로 성별균형을 고려하면서 직업 윤리를 갖춘 과학기술혁신 인력을 개발 및 육성하고, 효율적이고 효과적이며 국가연구개발과 해외 기술 흡수 역량 강화 등이 있음

이와 함께 2022년에는 ▲디지털 ▲관광 ▲에너지 분야가 추가되어 캄보디아의 과학기술 혁신 역량을 한 단계 더 발전시킬 예정이다.

한편, 캄보디아는 지난 6월 23일(목), 국가과학기술로드맵을 공식적으로 채택하는 행사를 개최했다.

캄보디아 산업과학기술혁신부(Ministry of Industry, Science, Technology and Innovation, MISTI) 주최로 진행된 이번 공표식은 한국과 캄보디아 관련 전문가 약 120여 명이 온-오프라인 참석하여, 농업, 보건, 교육 분야의 각 로드맵에 대한 배경, 기술 및 전략 등을 공유하기도 했다.

온라인으로 참석한 문미숙 과기정책연구원장은 축사를 통해 "한국의 과학기술 로드맵 시행 경험을 전하고자 노력한 우리측 연구자 포함 국내 전문가들의 노력이 오늘 캄보디아 국가과학기술로드맵을 선연하게 되어 매우 기쁘게 생각한다"라며 "국가과학기술로드맵이 캄

■ Press release of the Electronic Times (Sep. 22, 2022)²⁶

[테크코리아 우리가 이끈다]과학기술정책연구원

발행일 : 2022-09-22 16:00 | 지면 : 2022-09-23 | 10면



«지난 6월 23일 캄보디아는 국가과학기술로드맵 공표식을 진행했다. 캄보디아 국가과학기술로드맵은 과기정책연의 기술로드맵 자문사업에 힘입어 탄생했다.»

과학기술정책연구원(원장 문미숙)은 우리나라 선진 과학기술 정책을 기반으로 개발도상국 국가 과학기술혁신(STI)을 선도, 테크코리아 위상을 높이고 있다.

2014년 ODA 사업인 「국제과학기술협력사업(K-이노베이션 사업)」을 발족해 개발도상국 지속가능 발전 자원을 9년째 이어오고 있다.

에티오피아를 첫 대상으로 삼아 현재까지 아시아와 아프리카, 중남미 각 13개국에서 36개 K-이노베이션 사업을 수행해 오고 있다. 올해의 경우 가나, 라오스, 인도네시아, 아제르바이잔, 캄보디아, 배루 6개국에서 사업을 수행하고 있다.

국가 과학기술 마스터플랜 수립 등 정책 개선은 물론 국가혁신체계(NIS) 전반에 정책 자문을 하고 있다. 이는 우리나라 국가 위상을 높이고 향후 국내기업 해외 진출에도 큰 도움이 된다.

K-이노베이션 사업 대표 성과사례는 캄보디아 정책 자문이다. 특히 농업과 보건, 교육 분야를 캄보디아 전략기술 분야로 선정한 「기술로드맵 자문사업」이 올해 캄보디아 「국가과학기술로드맵」으로 이어지기도 했다.

기술로드맵 자문사업은 농업 품질 기반 시설 확충, 정보통신기술(ICT)을 활용한 보건 분야 결정 지원, 인터넷 접근성 향상을 통한 학습 결과 개선 등 내용을 로드맵 구축 목표로 삼았다.

과기정책연은 이에 앞서 1단계 정책 자문사업을 진행, 캄보디아 정부 최초 국가 과학기술 기본계획인 「국가 STI 정책 2020-2030」이 수립되는 성과를 이뤘다. 올해에도 디지털, 관광, 에너지 분야에서 캄보디아 STI 역량을 한 단계 발전시킬 예정이다.

문미숙 과기정책연구원장은 "우리 과학기술 발전 경험을 전한 과기정책연 연구자 등의 노력이 구체적인 성과를 보여 기쁘다"며 "우리 기업이 K-이노베이션 사업으로 공평한 개발도상국 발전 초석이 되기를 바란다"고 말했다.

²⁶ <https://www.etnews.com/20220915000222>, search date: Nov. 15, 2022

2.2 Future Plans

- Completion of the 2nd Cambodia NTRM final report
 - To complete the final report by mid-December 2022, after revising and supplementing the 2nd draft of Cambodia's NTRM for the three areas of digital, tourism, and energy.
 - To prepare the English-language version of the report on the 2nd year NTRM before translating it and publishing the Khmer version, as in the first year.
- Holding of a proclamation ceremony for Cambodia's NTRM 2030 (early 2023)
 - To submit the final draft of Cambodia's NTRM, to be prepared in two phases for a total of six technology fields, to the National Council of Science, Technology and Innovation (NCSTI) for its approval as a government plan
 - To hold the proclamation ceremony of Cambodia NTRM 2030 after receiving the final approval as a government proposal.
 - To use it as a guideline for the Cambodian government's R&D budget allocation and science and technology plans following its proclamation to not only the science and technology community but also the public.
- To launch the 3rd stage project (2023-2025) – “Consulting on the establishment of the national R&D management system and the capacity building of R&D entities” for the efficient implementation of Cambodia's NTRM
 - The Cambodian government requested the Korean side to carry out “Consulting on the establishment of the national R&D management system and the capacity building of R&D entities” as the 3rd stage program for the efficient implementation of Cambodia's NTRM, which is being prepared by the Cambodian government in the 2nd stage project (2020-2022).
- To provide policy consulting on the “conceptual design and construction plan of the national R&D management system”* after considering Cambodia's current capacity and development stage, in order to efficiently utilize its limited R&D investment resources and enhance the country's innovation outcomes.
- To prepare a “plan for establishing a digital-based Cambodian national R&D management system” and identify potential follow-up programs for its implementation based on the results mentioned in the above paragraph.

REFERENCES

Jungwon Lee and Jieun Kim(2020) “2020 K-Innovation ODA Program with Cambodia”, *Science and Technology Policy Institute*, pp. 27-69.

Jungwon Lee and Jeehye Min(2021) “2021 K-Innovation ODA Program with Cambodia”, *Science and Technology Policy Institute*, pp. 37-257.

Press release of the Kyosu Shinmun (July 12, 2022) “STEPI’s ODA program adopted as Cambodia’s NTRM” 」。 .

Press release of the Electronic Times (Sep. 22, 2022) “STEPI says “We lead Tech Korea!”

The website of the Embassy of the Republic of Korea in Cambodia,

https://overseas.mofa.go.kr/kh-ko/brd/m_3104/view.do?seq=1345597, search date

The website of the MISTI, Cambodia:

<https://www.misti.gov.kh/public/file/202108261629990117.pdf>, search date

WHO COVID-19 Dashboard Website, <https://covid19.who.int/region/wpro/country/kh>

APPENDIX

Development of a National Technology Roadmap

2022. 8. 3.

Hyun Yim

KISTEP 한국과학기술기획평가원
Korea Institute of S&T Evaluation and Planning

Outline

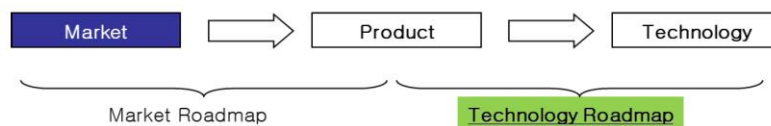
- I. Technology Roadmapping**
- II. Korea's NTRM Cases**
- III. NTRM Procedure**
- IV. Summary**

I. Technology Roadmapping

Announcement ceremony of the National Science and Technology Roadmap

Definition of Technology Roadmapping

- **Technology Roadmapping** is a needs-driven technology planning process to help identify, select, and develop technology alternatives to satisfy a set of product needs



- **A technology roadmap** is the document that is generated by the technology roadmapping process
- Technology roadmapping
 - is a learning process
 - is a consensus building process

2021

What is a Technology Roadmap for?

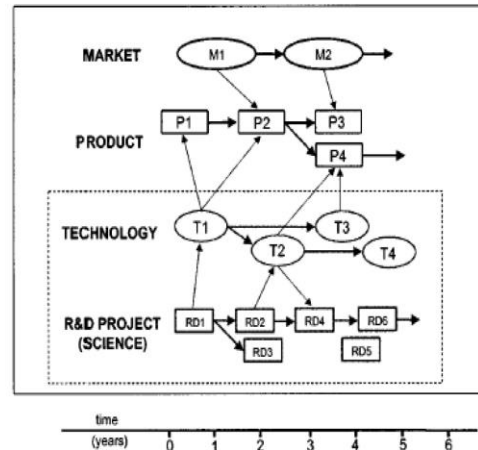
- In a complicated and uncertain world TRM **provides a way to identify, evaluate, and select strategic technology alternatives** that can be used to achieve desired objectives

very complex and uncertain



The elements of a Technology Roadmap

- An Technology Roadmap consists of **nodes and links**
- These elements can have *quantitative* and *qualitative* attributes.
- Construction of a technology roadmap requires
 - identification of the nodes and their attributes,
 - connecting the nodes with links, and
 - specifying the link attributes



Technology Roadmap is Not a List !!!

A generalized Roadmap architecture

4 Dimensions (“when”, “why”, “what”, “how”)

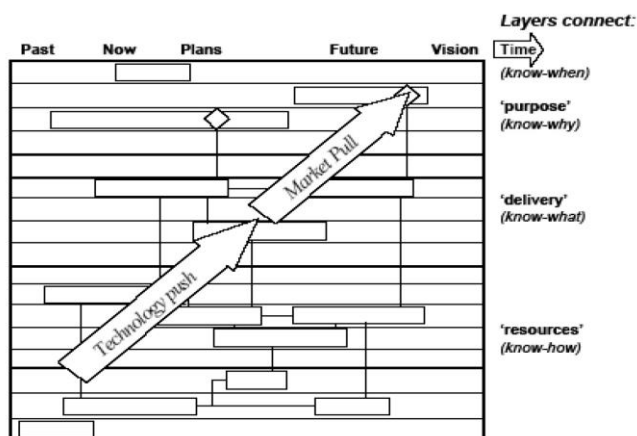
Roadmap Components

Markets/Customers/Competitors/
Environment/Industry/
Business/Trends/Drivers/Threats/
Objectives/Milestones/Strategy

Products/Services/Applications/
Capabilities/Performance/
Features/Components/Families/
Processes/Systems/Platforms/
Opportunities/Requirements/Risks

Technology/Competencies/
Knowledge/Development

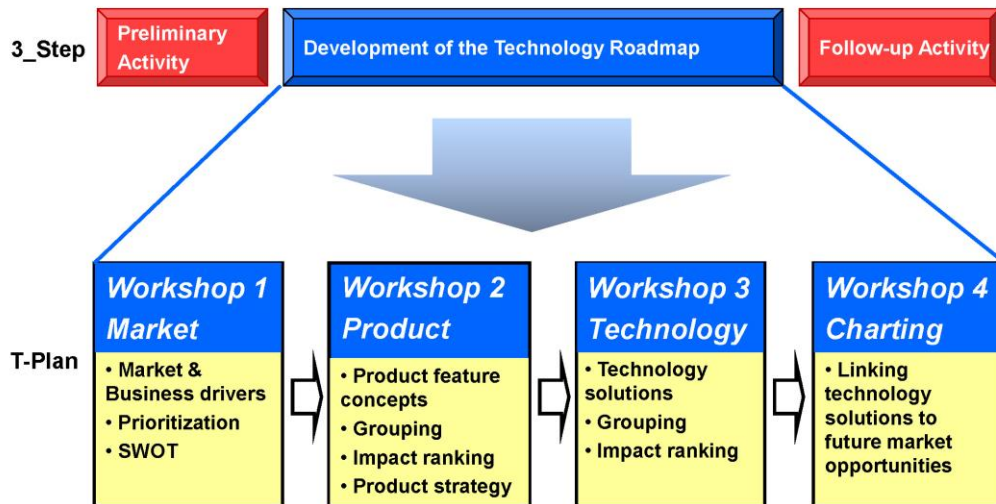
Skills/Partnerships/Suppliers/
Facilities/Infrastructure/
Organisation/Standards/Science/
Finance/Research



Phaal, 2003

Roadmapping is Planning for the future and Links Future to Present

Technology Roadmapping Process



STEMI

Roadmap Development via workshop

Workshop is a collaborative tool to develop the roadmap!



- A workshop allows the participants to:
 - exchange information, views and insights,
 - identify points of agreement, disagreement and uncertainty
 - create new shared understandings

II. Korea's NTRM Cases

Korea's National Technology Roadmap (2002)

Background

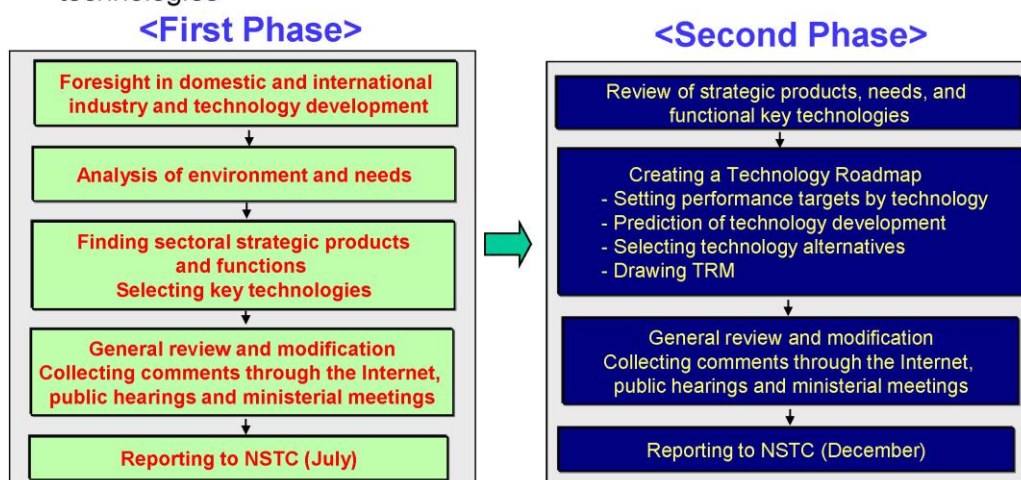
- Required to focus the strategically chosen R&D areas for optimal utilization of limited R&D resources
- Necessity to Improve our national technological capabilities from 'Imitation stage' to 'Innovation stage'
- Rise of necessity for a comprehensive and coherent approach to resource allocation

Purpose

- To build the National Technology Roadmap (NTRM) for the need of national strategy and economic growth toward 2012
- To follow top-down vision-driven approach to the identification of key technologies

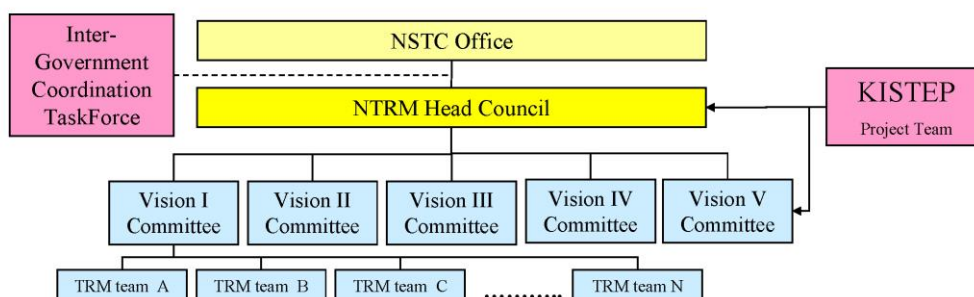
Process

- Two-phase process
- In the first phase strategic products/functions and key technologies are identified
- In the second phase TRMs are developed for strategic products and key technologies



Organizational Structure

- NTRM were developed for 5 visions
 - Macro Roadmaps for strategic product/functions and Micro Roadmaps for key technologies
 - Incorporate existing TRMs into NTRM with necessary modifications
 - Total 74 TRM teams were set up and each team was composed of around 10 technology experts



**Analysis of Industrial Need ➡ 5 visions ➡ 13 Directions ➡
49 Strategic Product/Functions ➡ 99 Key Technologies ➡ NTRM**

- ❖ Incorporate existing TRMs into NTRM with necessary modifications
- ❖ Handle basic S&T separately from NTRM based on bottom-up approach

- ❖ **NTRM include Macro Roadmaps for strategic product/functions and detailed TRM for chosen key technologies**

VISION 1 : Building an information-knowledge-intelligence society
VISION 2 : Aiming at Bio-Healthtopia
VISION 3 : Advancing the E2 Frontier
VISION 4 : Upgrading the Value of Major Industries of Korea Today
VISION 5 : Improving National Safety and Prestige

VISION 2 : Aiming at Bio-Healthtopia

VISION 3 : Advancing the E2 Frontier

VISION 4 : Upgrading the Value of Major Industries of Korea Today

VISION 5 : Improving National Safety and Prestige

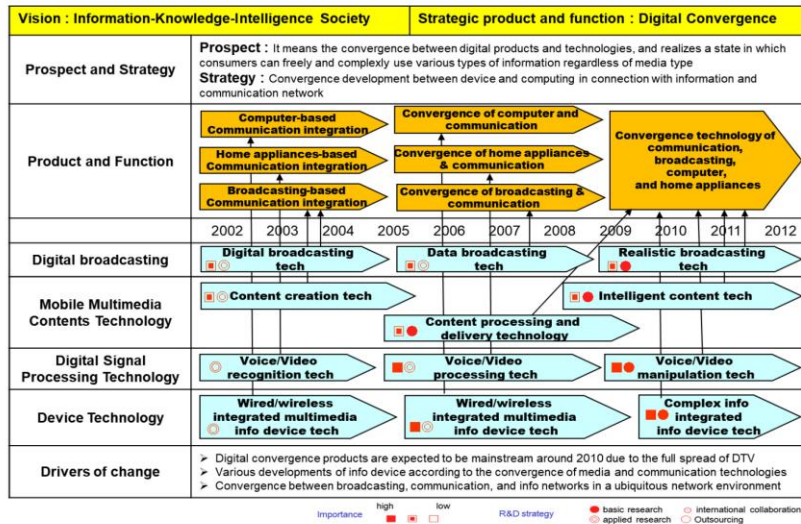
Meeting a variety of human needs in all areas of life by making IT service more intelligent, mobile, and user-friendly

Meeting a variety of human needs in all areas of life by making IT service more intelligent, mobile, and user-friendly

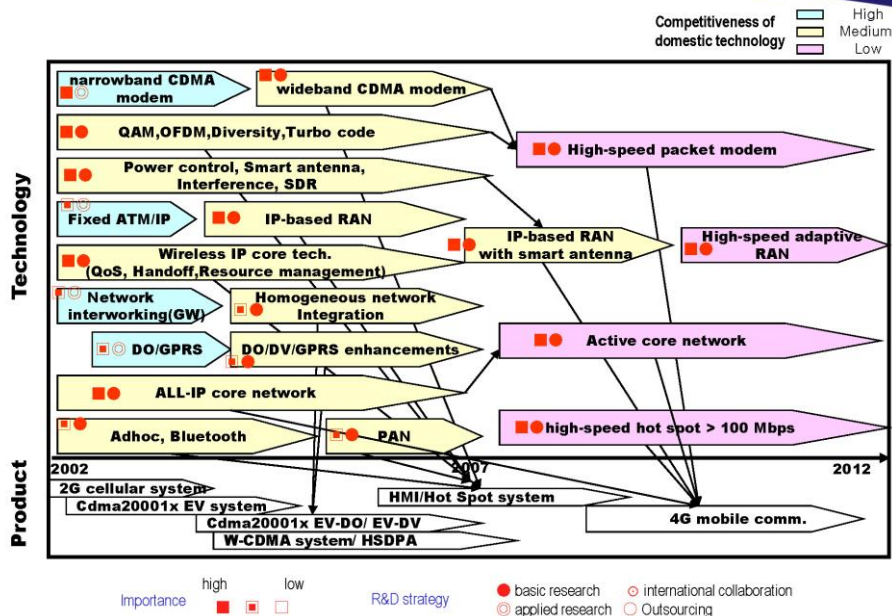


Macro TRM : Digital Convergence

A macro roadmap of digital convergence derived from Korea in 2002



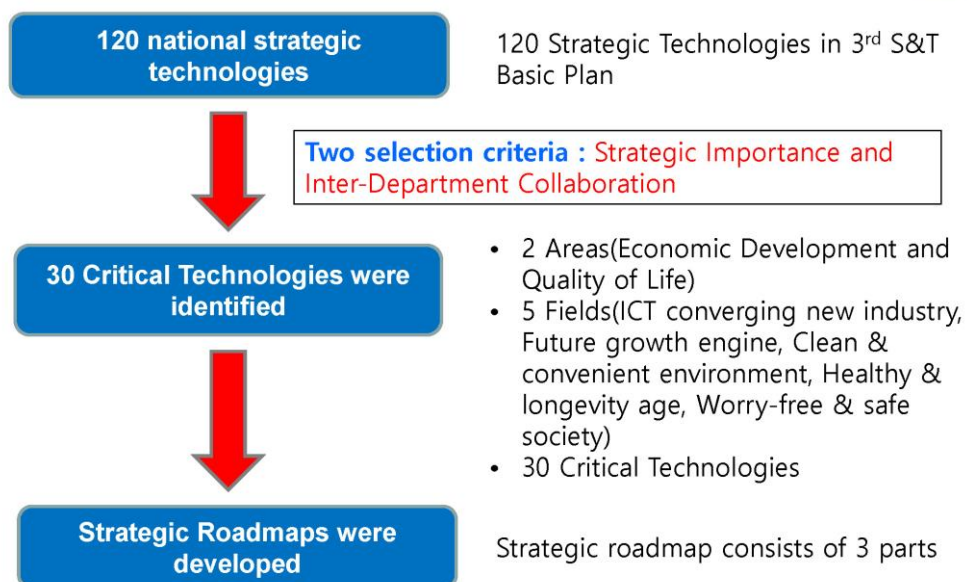
Micro TRM : High-speed Wireless Multimedia/ 4G Mobile Communication



Impacts

- Enhance the national R&D investment with the optimal utilization of limited R&D resources
- Promote coordination and synergy among industry, academia, and government in technology development
- Present guidelines for the selection and evaluation of government R&D programs
- Incorporated in “National S&T Basic Plan(2002-2006)” which play a role as a guideline for national R&D
- Play a benchmarking role of non-governmental sector’s S&T plan

Korea’ s National Critical S&T Strategic Roadmap (2014)

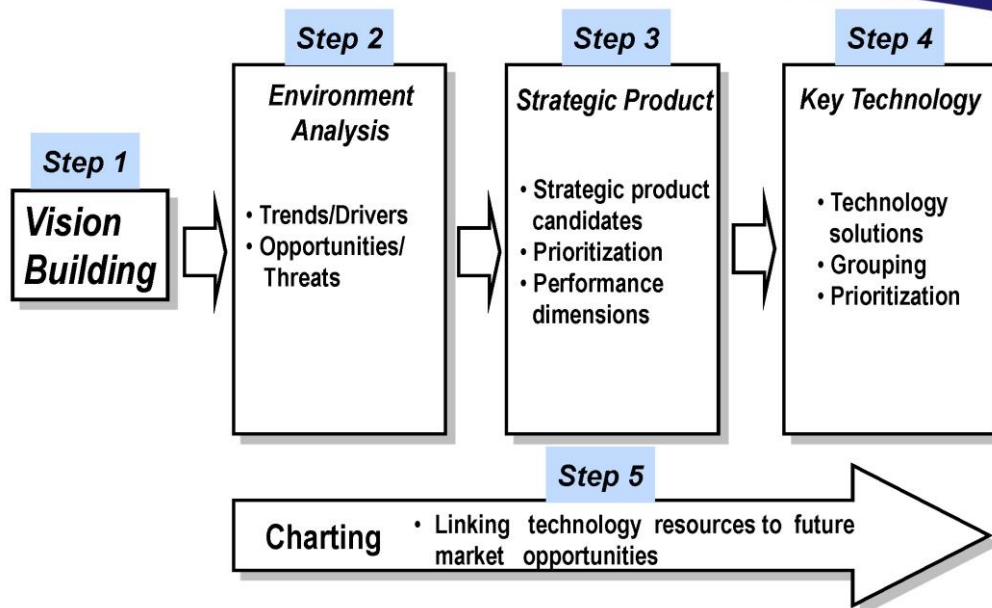


Structure of Strategic Roadmap



III. NTRM Procedure

NTRM Procedure



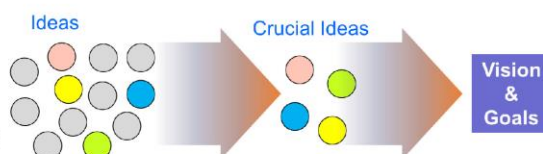
Step 1 : Vision Building

■ Objective

- Build a vision and goals

■ Detailed Process

- Conduct a vision workshop
 - ✓ Draw a desired future
 - (Example) Participants are asked to transport themselves into an imaginary world ten years in the future and draw key words which might be important for the desired future
 - ✓ Draw a vision and goals which are necessary to achieve the desired future
 - (Example) Each participant writes his or her own ideas to post-it and placed it on the paper
 - The group discusses to adjust..., and also to possibly add new ideas
 - The most crucial ideas can be filtered out of the mass of reflections, for example by voting



[Worksheet] Visioning

Develop the vision and goals for each sector; **agriculture, education, and health**

Vision	■ <i>Building a sustainable and carbon-neutral society</i>
Goals	■ <i>to cut greenhouse gas emissions by 20% by 2030 and 50% by 2050</i> ■ ■ ■

Step 2 : Environment Analysis

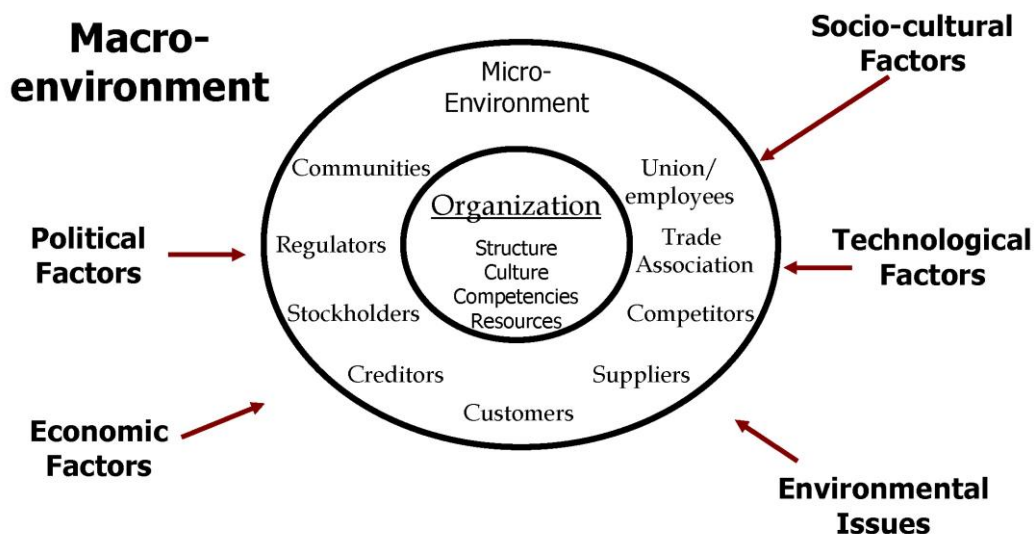
■ Objective

- Collect and analyze the drivers

■ Detailed Process

- Collects and analyzes the drivers
 - ✓ Global trends/issues are gathered from literatures, web sites and by consultation
 - ✓ Identify the drivers that may have an impact on the desired future using STEEP analysis in the workshop
- Conduct a survey among experts with various backgrounds to evaluate the drivers (**optional**)
 - ✓ (Example)Rate the extent of the impact the driver will have
 ➤ **1 = No impact, 2 = Small impact, 3 = Moderate impact, 4 = Extensive impact**
 - ✓ (Example)Add the driver which is not included in the list, but might be considered to be important for the desired future

Macro Environment: STEEP Analysis



Examples of possible Drivers in the Energy sector in Ukraine

- **Social:** ageing population, increasing consumerism, growing ecological awareness ...
- **Technological:** clean coal power techs, nuclear, solar PV, energy efficiency techs, CHP, biomass ...
- **Economic:** global oil and gas prices, national energy pricing, growing economy (GDP) ...
- **Environmental:** Kyoto Protocol, Chernobyl legacy, energy conservation ...
- **Political:** security of energy supply (relations with Russia), industry political lobbying, laws on conservation ...

[Worksheet] Identification of Drivers

STEEP	Drivers
Social	• <i>Ageing population</i>
Technological	
Economic	• <i>Global oil and gas prices</i>
Environmental	• <i>Climate change</i> • <i>Environmental pollution</i>
Political	

Step 3 : Strategic Products/Services (1/2)

■ Objective

- Identify candidates of strategic products/services
- Identification of criteria for assessing and choosing strategic products/services
- Identification of strategic products/services with priority

■ Detailed Process

- Identify candidates of strategic products to fulfill a vision and goals
 - ✓ (Example) Participants are asked to draw opportunities and threats after considering the drivers identified in Step 2
 - ✓ (Example) Identify products/services to utilize the opportunities and/or to avoid the threats
- Adjustment : Collected candidates of products/services should be adjusted by committee. In this step, candidate with unclear definition will be removed, similar candidates will be merged and, depth and wideness of each candidate will be corrected to the same level

Step 3 : Strategic Products/Services (2/2)

■ Detailed Process

- Each candidate of strategic product/service will be assessed based on the criteria by participants in the workshop or survey
 - ✓ (Example) Participants are asked to draw criteria that are important to assess the candidates
 - ✓ (Example) Participants are asked to assess the candidates in 5-point scale for parameters "economic impact", "strategic importance", and "potential for success"
- Final decision of strategic products/services will be made by committee
 - ✓ The results of voting should not be accepted automatically as the final outcome of prioritization. They should be thoroughly discussed in an expert group to confirm the results of voting and to identify possible pitfalls

[Worksheet] Opportunity/Threat

Drivers	Opportunity/Threat
A. <i>Environmental pollution</i>	<i>Restrictions on the use of fossil fuels, Use of renewable energy</i>
B. <i>Climate change</i>	...
C.	
D.	
E.	
F.	
G.	
H.	
I.	
J.	
K.	
...	

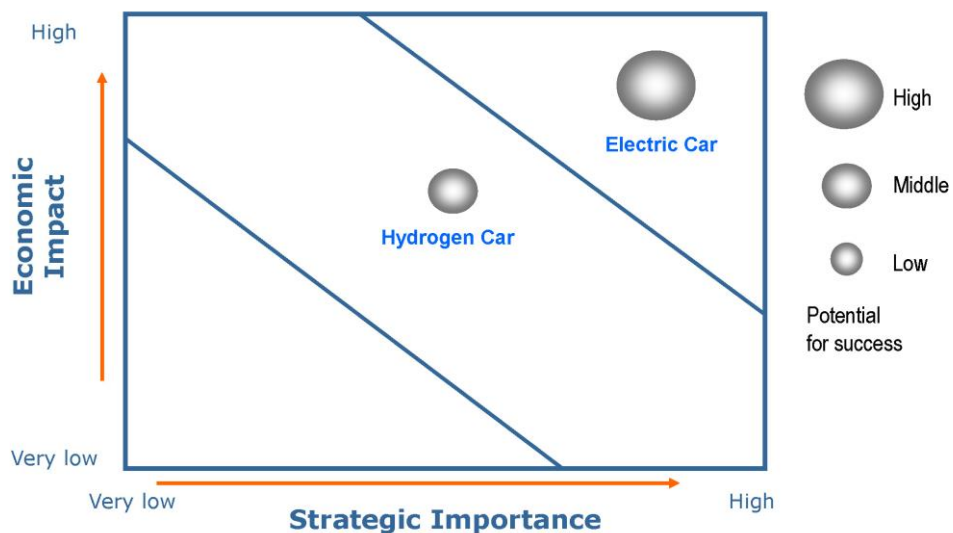
[Worksheet] Identification of Products/Services

	Opportunities	Threats
Opportunities & Threats	<i>Use of renewable energy, ...</i>	<i>Restrictions on the use of fossil fuels, ...</i>
Products/ Services	● <i>Electric car</i> ● <i>Hydrogen car</i> ● <i>...</i>	

[Worksheet] Evaluation of Products/Services

Products/Services	Code	Evaluation Criteria			Sum
		Economic Impact	Strategic Importance	Potential for Success	
<i>Electric car</i>	<i>P1</i>	<i>4</i>	<i>4</i>	<i>3</i>	<i>11</i>
<i>Hydrogen car</i>	<i>P2</i>	<i>3</i>	<i>3</i>	<i>2</i>	<i>8</i>
...	<i>P3</i>				

[Worksheet] Prioritization : Strategic Products/Services



Step 4 : Key Technologies (1/2)

■ Objective

- Identify product performance dimensions
- Identify technologies corresponding to performance dimensions
- Develop the technology tree
- Identify criteria for assessing and choosing key technologies
- Identify key technologies with priority for which roadmap will be deployed

■ Detailed Process

- Identify performance dimensions that are important for product performance, and that technology can deliver
 - ✓ The 'Performance' provides a means of linking market, product and technology
- Explore the related technologies to meet the performance dimensions
- Arrange the technologies and develop the technology tree
 - ✓ Similar technologies will be merged and technologies will be sorted into the appropriate groups

Step 4 : Key Technologies (2/2)

■ Detailed Process

- Identify criteria for assessing and choosing key technologies
 - ✓ (Example)Participants are asked to draw criteria that are important to assess the technologies
 - ✓ (Example)Participants are asked to assess the technologies in 5-point scale(very low-low-medium-high-very high) for parameters "attractiveness" and "feasibility"
- Final decision of key technologies will be made by committee
 - ✓ The results of voting should not be accepted automatically as the final outcome of prioritization. They should be thoroughly discussed in an expert group to confirm the results of voting and to identify possible pitfalls

Performance Dimensions

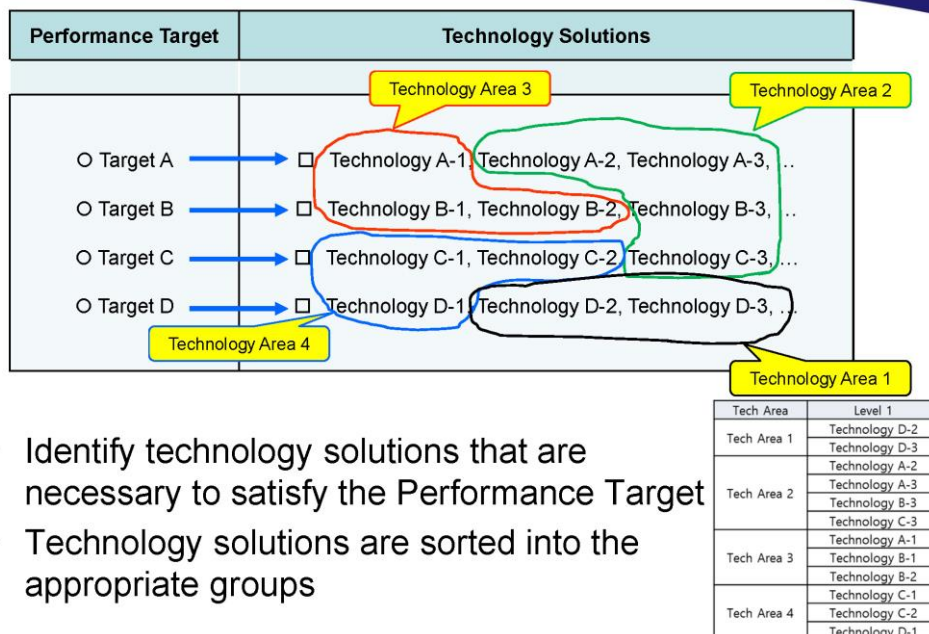
- These are aspects of product performance which are (or may be) important to the customer, and which technology can deliver
- Examples: size; speed; quality; reliability

- **Brainstorm performance dimensions**
- **Sort into appropriate groups***
- **Note 'key' dimensions**

* optional

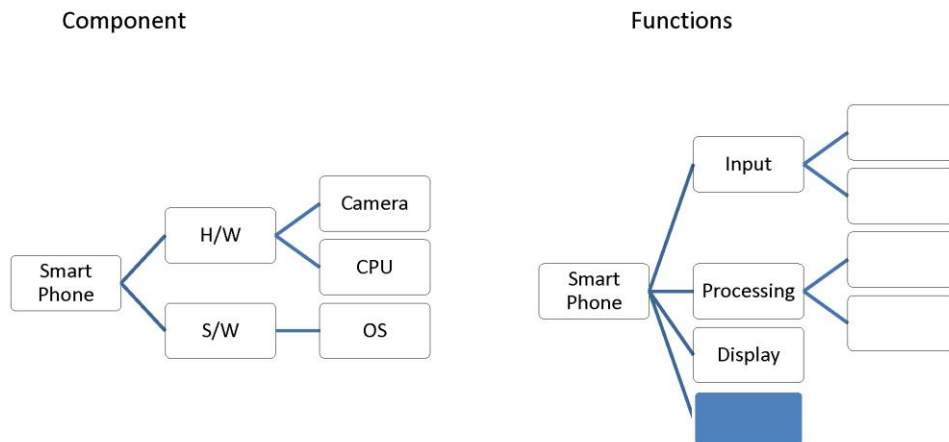
- Consider quantitative and qualitative dimensions
- May include market, product or technology 'views'
- 'Key' dimensions are those which stand out as having high potential value to the customer or attractiveness to [Company]

Technology Solutions



- Identify technology solutions that are necessary to satisfy the Performance Target
- Technology solutions are sorted into the appropriate groups

Technology Tree



[Worksheet] Identification of Performance dimension & Technology solutions

Product : Electric Car

Performance Dimension	Performance Target	Technology solutions	Technology Area	Tech Level 1
Mileage per charge (km/charge)	1,000 km/charge by 2030	Lithium-ion batteries, All-solid-state batteries, Lithium-air batteries	Battery	Lithium-ion batteries
				All-solid-state batteries
				Lithium-air batteries
...	...			

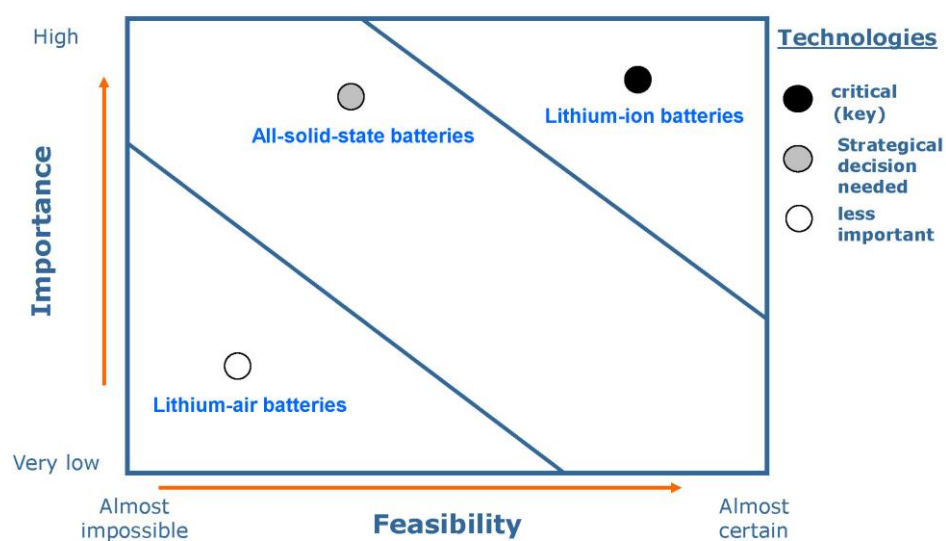
[Worksheet] Evaluation of Technologies

Product :

Technologies	Code	Evaluation Criteria		Sum
		Importance	Feasibility	
<i>Lithium-ion batteries</i>	<i>T1</i>	<i>4</i>	<i>4</i>	<i>8</i>
<i>All-solid-state batteries</i>	<i>T2</i>	<i>4</i>	<i>2</i>	<i>6</i>
<i>Lithium-air batteries</i>	<i>T3</i>	<i>1</i>	<i>1</i>	<i>2</i>

[Worksheet] Prioritization : Key Technologies

Product :



[Worksheet] Analysis of Technology Solutions

Product :

Technology areas	Technology solutions	Technology Performance Indicator	Current Technology Level	Technical performance goals and technology acquisition measures for key technologies		
				Short term (1~3 years)	Mid term (4~7 years)	Long term (8~10 years)
Battery	Lithium-ion batteries	Energy Density (Wh/kg)	100	200(■, M)	300(■, M)	
	All-solid-state batteries	Energy Density (Wh/kg)	100	300(■, M)	400(■, M)	600(■, C)
	Lithium-air batteries					

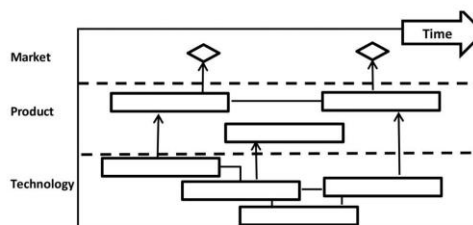
1. Importance High ↔ Low
 ■ ■ □ □

2. Tech Acquisition Strategy
 ● License ○ R&D
 ○ International Collaboration
 ○ Outsourcing

Step 5 : Charting

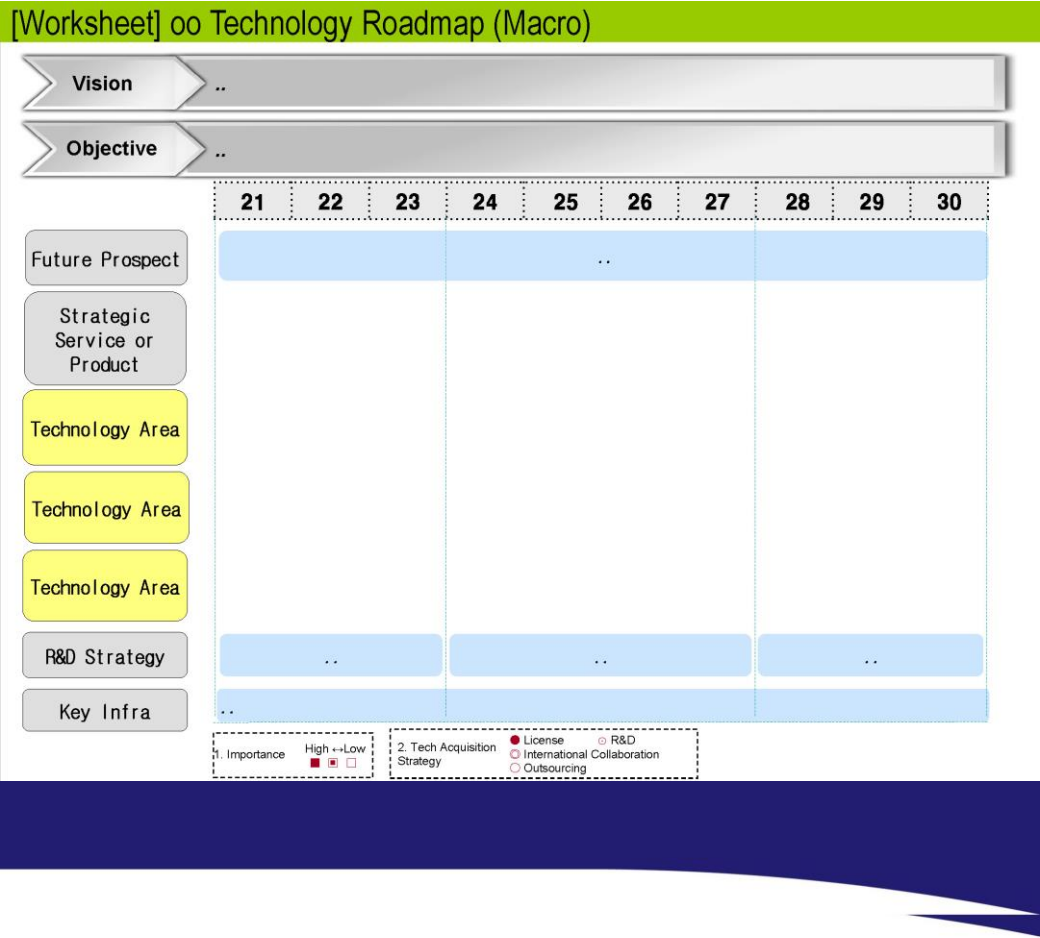
Objective

- Deployment of key technology areas by time frame
- Linkage of technologies with products/markets



Detailed Process

- Chart technology areas, key technologies, target, and technology elements
- Draw linkages between market, product and technology elements
 - ✓ A roadmap is created in multiple iterations over the views and abstraction levels
 - ✓ The iterations ensure feedback and increasing focus during the roadmap creation



IV. Summary

Summary

- ◆ Technology Roadmapping is one of Technology Planning activities and a useful tool to identify the business opportunities through normative approach for desirable futures.
- ◆ Technology roadmapping is a learning experience, and that a flexible approach is required, adapted to the particular circumstances being considered.
- ◆ The National Technology Roadmap (NTRM) is expected to provide a strategic direction for innovation in S&T and to increase the networking and consensus- building in S&T community

Thank You!!!

Hyun Yim
hyim@kistep.re.kr